

Code Logic Reference

Important functions, methods, branching logic, invariants, and failure behavior.

Eurskem AI · Agentic Workflow Platform

Generated 2026-08-24

Code revision 8bd16ca4a2a3

Derived from the live repository

Source of truth. Inventories and symbol tables are generated from the current checkout. Narrative explains observed structures and does not replace executable contracts.

Reading logic safely

The branch score is a lightweight AST count, not a quality verdict. High scores identify orchestration, validation, parsing, and fallback code that deserves narrow changes and strong regression tests. Read each unit with callers, state mutations, external effects, and error translation.

Logic unit	Location	Branch score	Contract
plan_workspace	app/workflow/ chat_workspace_planner.py:773	74	<code>objective: str, *, experience_id: str None=None, skill_name: str None=None, selected_workflow: str None=None, preferred_output: OutputKind='auto', has_attachments: bool=False, attachment_categories: list[str] None=None, collection_id: str None=None, retrieval_profile_id: str None=None, rag_agent_id: str None=None, document_ids: list[str] None=None, integration_connection: str None=None, integration_tool: str None=None, previous_run_id: str None=None, previous_result: Any=None</code> No summary declared.
ProposalEvidenceFactoryAgent.run	app/nodes/ proposal_evidence_factor.py:374	62	<code>self, state: dict[str, Any], resolved_config: dict[str, Any]</code> Run the result.
lifespan	app/main.py:104	56	<code>app: FastAPI</code> Compute the lifespan.
RetrievalService._retrieve_impl	app/retrieval/ service.py:211	49	<code>self, request: RetrievalQuery, *, owner_scope_id: str, request_id: str, llm: Any None=None</code> Retrieve the impl.
inspect_python	scripts/ build_documentation_pdf.py:234	48	<code>path: Path, model: Model</code> No summary declared.
_probe_services	app/runtime/ preflight.py:2356	46	<code>spec: WorkflowSpec, services: dict[str, Any], report: WorkflowPreflightReport, *, require_run_history: bool, owner_scope_id: str None=None</code> Probe the services.
_make_runtime_fn	app/runtime/ compiler.py:63	45	<code>instance, bus: RunEventBus None, services: dict</code> Wrap a NodeType instance so it conforms to LangGraph's 'state -> partial state' contract.
_wire_edges	app/runtime/ compiler.py:428	43	<code>graph: StateGraph, edges: list[EdgeSpec], hitl_ids: set[str], multi_router_ids: set[str] None=None</code> Add edges, return the set of source node ids (used to compute terminals).

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IngestionJobRunner.run	app/ingestion/jobs.py:115	42	<pre>self, job: IngestionJob, paths: list[Path], *, metadata: dict[str, Any] None=None</pre> Run the result.
_apply	app/nodes/data_transform.py:299	39	<pre>op: TransformOperation, context: dict[str, Any]</pre> Apply the result.
_validate_template_output_path	app/runtime/preflight.py:1635	38	<pre>spec: WorkflowSpec, current_node: NodeSpec, reference: str, report: WorkflowPreflightReport, path: str, forward: dict[str, set[str]], guaranteed: dict[str, set[str]], optional: bool=False</pre> Validate the template output path.
WorkflowSpec.validate_graph_references	app/runtime/schema.py:406	38	<pre>self</pre> Validate the graph references.
RAGService.query	app/rag/service.py:147	37	<pre>self, *, owner_scope_id: str, rag_agent_id: str, query: str, runtime_filters: dict[str, Any] MetadataFilterGroup None=None, document_ids: list[str] None=None, runtime_context: dict[str, Any] None=None, llm: Any None=None</pre> Query the result.
BoundedToolResearchService.research	app/research/deep_research.py:241	37	<pre>self, *, brief: ResearchBrief, instructions: str, max_duration_seconds: float=1800.0, max_iterations: int=15, max_cost_per_call_usd: float=15.0</pre> Compute the research.
validate_field_constraints	app/nodes/workflow_input.py:454	36	<pre>bindings: list[InputFieldBinding], data: dict[str, Any]</pre> Business-friendly validation beyond shape (§20/21 — never a raw Pydantic path like VALIDATION_SCHEMA_FIELD_002): text length/pattern, email/url/phone format, percentage range, and the semantic checks a compound preset's own object shape can't express on its own (date_range ordering, an allowed unit/
extract_identifiers	app/evidence/identifiers.py:173	34	<pre>url: str None=None, *, metadata: dict[str, Any] None=None, text: str None=None</pre> Best-effort offline identifier extraction.
_check	app/nodes/consistency_checker.py:57	34	<pre>graph: ProposalGraph</pre> Check the result.
analyze_chat_output	app/workflow/chat_output_compatibility.py:40	34	<pre>spec: WorkflowSpec</pre> Describe whether a workflow has a meaningful normalized Chat result.
Settings.validate_production_security	app/config.py:442	33	<pre>self</pre> Refuse to boot production with development security settings.
start_ingestion	app/api/knowledge.py:386	32	<pre>collection_id: str, request: Request, files: list[UploadFile]=File(...), parser_profile_id: str None=Form(None),</pre>

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			<pre> parser_profile_version: int None=Form(None), chunking_profile_id: str None=Form(None), chunking_profile_version: int None=Form(None), embedding_profile_id: str None=Form(None), embedding_profile_version: int None=Form(None), embedding_model: str None=Form(None), metadata_json: str=Form('{}'), user: CurrentUser=Depends(require_permission('k nowledge:write')) Start the ingestion. </pre>
build_call_coverage_matrix	app/proposal_graph/coverage.py:88	32	<pre> graph: ProposalGraph </pre> Build the call coverage matrix.
_select	scripts/openrouter_health_check.py:667	32	<pre> catalog: list[CatalogEntry], args: argparse.Namespace </pre> Returns (selected, excluded) — excluded rows are reported, never silently dropped.
evaluate_condition	app/runtime/rules.py:634	31	<pre> condition: Condition, context: dict[str, Any] </pre> Compute the evaluate condition.
_explain	app/api/builder.py:739	29	<pre> type_name: str, output: dict[str, Any] </pre> Turn a node's output into the "why did this happen?" view (§24, §47).
CitationRegistryBuilder.run	app/nodes/citation_registry_builder.py:152	29	<pre> self, state: dict[str, Any], resolved_config: dict[str, Any] </pre> Run the result.
build_rows	scripts/build_workflow_qa_matrix.py:127	28	<pre> root: Path, preflight_reports: dict[str, dict[str, Any]] </pre> No summary declared.
run	scripts/verify_knowledge_studio.py:474	28	<pre> base: str, user: str, password: str, keep: bool </pre> Run the result.
_validate_graph	app/runtime/preflight.py:1460	27	<pre> spec: WorkflowSpec, report: WorkflowPreflightReport </pre> Validate the graph.
_ScholarlyCandidateDiscovery.run	app/nodes/evidence_agent.py:179	26	<pre> self, state: dict[str, Any], resolved_config: dict[str, Any] </pre> Run the result.
_chunk_from_object	app/retrieval/strategies.py:16	26	<pre> obj: Any, *, score_kind: str, rank: int </pre> Chunk the from object.
_register_builders	app/runtime/logic_preflight.py:52	26	<pre> </pre> Populate the typed-output builders lazily.
_validate_router_edges	app/runtime/preflight.py:828	26	<pre> spec: WorkflowSpec, report: WorkflowPreflightReport </pre> Validate the router edges.
InternalProjectEvidenceRetrieverAgent.run	app/nodes/internal_project_evidence_retriever.py:153	25	<pre> self, state: dict[str, Any], resolved_config: dict[str, Any] </pre> Run the result.

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resume_workflow_durable	app/runtime/hitl.py:109	25	<pre>run_id: str, decision: dict[str, Any], *, services: dict[str, Any] None=None, session_id: str None=None, actor: str='system'</pre> Resume in memory when possible, otherwise rebuild from Mongo.
_validate_start_fields	app/runtime/preflight.py:1365	25	<pre>spec: WorkflowSpec, report: WorkflowPreflightReport</pre> Validates one Start node's own field/file_field list.
_HorizonDocxBuilder._append_inline	app/tools/docx_proposal_rendering.py:1413	25	<pre>self, paragraph, container, style: _InlineStyle None=None</pre> Append the inline.
_adjacency	app/nodes/about_synthesis.py:77	24	<pre>type_name -> {"upstream": [...], "downstream": [...], "example": str None}, mined once from every workflow YAML on disk.</pre>
PriorProjectRetrieverAgent.run	app/nodes/prior_project_retriever.py:163	24	<pre>self, state: dict[str, Any], resolved_config: dict[str, Any]</pre> Run the result.
ProposalSubmissionGate.run	app/nodes/proposal_submission_gate.py:125	24	<pre>self, state: dict[str, Any], resolved_config: dict[str, Any]</pre> Run the result.
RAGAgent.run	app/nodes/rag.py:178	24	<pre>self, state, resolved_config: dict[str, Any]</pre> Run the result.
MinIOEvidenceIngestion.run	app/nodes/minio_evidence_ingestion.py:126	23	<pre>self, state: dict[str, Any], resolved_config: dict[str, Any]</pre> Run the result.
ProposalEvidenceFactoryAgent_quality_report	app/nodes/proposal_evidence_factory.py:1235	23	<pre>*, policy: EvidencePolicy, verified_claims: list[VerifiedClaim], citation_registry: list[CitationRegistryEntry], links: list[ClaimEvidenceLink], audit: list[SearchAuditRecord], rejected: list[RejectedCandidate], quantitative_registry: list[dict[str, Any]]</pre> Internal helper for the quality report step.
_check_conditional_join_reachability	app/runtime/logic_preflight.py:873	23	<pre>spec: WorkflowSpec, issue: Callable[... , None]</pre> Two conditional dispatches into one shared step form an unsatisfiable join.
_HorizonDocxBuilder._render_block	app/tools/docx_proposal_rendering.py:1251	23	<pre>self, node, *, list_level: int=0</pre> Render the block.
OpenAllImageGenerationService._generate_openrouter	app/tools/image_io.py:178	23	<pre>self, prompt: str, *, size: str, quality: str, output_format: str, model: str</pre> No summary declared.
backfill	scripts/backfill_knowledge_resources.py:85	23	<pre>*, apply: bool, db=None</pre> Backfill the result.

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cost_overview	app/api/ cost_admin.py:531	22	request: Request, days: int=Query(default=30, ge=1, le=365), user: CurrentUser=Depends(require_admin) Compute the cost overview.
ask_about_node_types	app/api/ node_types_chat.py:153	22	req: AskAboutNodeTypesRequest, request: Request, user: CurrentUser=Depends(require_consultant) Compute the ask about node types.
_build_run_context	app/api/run_chat.py:116	22	run: dict[str, Any] Build the run context.
StructuredDatasetRetrieverAgent.run	app/nodes/ structured_dataset_retriever.py:166	22	self, state: dict[str, Any], resolved_config: dict[str, Any] Run the result.
transform_query	app/retrieval/ query_transform.py:55	22	request: RetrievalQuery, *, llm: Any, metadata_schema: dict[str, Any] Return ``(semantic_query, transformed_queries, generated_filters``).
_validate_nodes	app/runtime/ preflight.py:488	22	spec: WorkflowSpec, report: WorkflowPreflightReport Validate the nodes.
_probe_workflow_model_access	app/runtime/ preflight.py:2860	22	spec: WorkflowSpec, llm: Any, report: WorkflowPreflightReport Confirm provider/project access without making a generation request.
PublicDatabaseLookupService.query_eurostat	app/tools/ database_lookup.py:72	22	self, query: StructuredDatasetQuery, *, timeout_seconds: float=45.0, max_response_bytes: int=20 * 1024 * 1024 Query the eurostat.
retry_failed_run	app/api/runs.py:253	21	source_run_id: str, req: RetryRunRequest, request: Request, user: CurrentUser=Depends(require_consultant) Create a new attempt that reuses every completed node checkpoint.
BoundedDeepResearchAgent.run	app/nodes/ bounded_deep_research_agent.py:141	21	self, state: dict[str, Any], resolved_config: dict[str, Any] Run the result.
run_workflow	app/runtime/ executor.py:83	21	spec: WorkflowSpec, inputs: dict[str, Any], session_id: str None=None, collection_id: str='default', services: dict[str, Any] None=None, run_id: str None=None, reused_node_results: dict[str, dict[str, Any]] None=None, retry_source_run_id: str None=None, hitl_resume_decisions: dict[str, dict[str, Any]] None=None, resume_replay: bool=False Run the workflow.
_node_autofix_scope_valid	app/api/workflows.py:358	20	original_yaml: str, candidate_yaml: str, node_id: str Only the selected node's configuration/model controls may change.
IntegrationService.execute		20	

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	app/integrations/files/service.py:262		<pre>self, *, connection_id: str, operation: IntegrationOperation, folder_id: str list[str] None=None, file_id: str list[str] None=None, query: str='', page_size: int=25, page_token: str None=None</pre> Execute the result.
validate_external_action_url	app/integrations/url_guard.py:42	20	<pre>url: str, *, allow_http: bool=False, resolve: Callable[..., list]=socket.getaddrinfo</pre> Validate an outbound External Action URL and return it unchanged.
_decode_jsonstat	app/nodes/structured_dataset_retriever.py:436	20	<pre>*, query: StructuredDatasetQuery, payload: dict[str, Any], endpoint: str, parameters: dict[str, Any], accessed_at: str, response_sha256: str, snapshot_object_key: str, limit: int, source_version: str</pre> Decode the jsonstat.
_check_subprocess_agents	app/runtime/logic_preflight.py:1642	20	<pre>spec: WorkflowSpec, issue: Callable[..., None], guaranteed_before: Callable[[str, str], bool] None</pre> Everything about a Subprocess step preflight can prove without actually running the child: the referenced workflow exists and parses, every explicit input mapping targets a real declared input of it, every <i>*required*</i> input of it resolves to something, and the whole reference graph this workflow reac
_validate_multiroute_andjoin	app/runtime/preflight.py:1238	20	<pre>spec: WorkflowSpec, report: WorkflowPreflightReport, forward: dict[str, set[str]]</pre> A fan-in target whose declared predecessors depend on DIFFERENT Multi-Route selections can silently never fire, the same failure shape FANIN_UNREACHABLE_ANDJOIN already catches for ordinary (single-select) routers' mutually-exclusive branches — confirmed empirically against real langgraph 1.2.9: a target dispatched to only a SUBSET of a multi-router's own branches is never revisited for the branches that weren't selected, so any node whose AND-join lists all of them hangs forever on the ones that never ran.
probe_model	scripts/openrouter_health_check.py:530	20	<pre>gateway: OpenRouterGateway, entry: CatalogEntry, *, probes: tuple[str, ...], timeout: float, retry_delay: float</pre> Probe the model.
run	scripts/openrouter_health_check.py:1034	20	<pre>args: argparse.Namespace</pre> Run the result.
restart_run	app/api/runs.py:580	19	<pre>source_run_id: str, req: RetryRunRequest, request: Request, user: CurrentUser=Depends(require_consultant)</pre> Create a brand-new run of the same workflow and original inputs.
_field_type_label		19	

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	app/api/ workflow_generation.py:410		<code>schema: dict</code> A short type descriptor for one JSON-schema property, e.g.
<code>_fixture_matches</code>	app/mcp/dynamics/ client.py:716	19	<code>row: dict[str, Any], expression: str</code> Interpret the small OData subset this package generates.
<code>HumanInLoopAgent.run</code>	app/nodes/ human_in_loop.py:211	19	<code>self, state, resolved_config: dict[str, Any]</code> Run the result.
<code>ScientificResearchPlannerAgent.run</code>	app/nodes/ scientific_research_planner.py:157	19	<code>self, state: dict[str, Any], resolved_config: dict[str, Any]</code> Run the result.
<code>workflow_stats</code>	app/workflow/ run_history.py:1246	19	<code>db, session_id: str, workflow_name: str, *, sample_limit: int=200</code> Library "Runs and performance" tab: success rate, median duration, and the most common failure, computed from this session's own run history for one workflow.
<code>_resolve_pending_gate</code>	app/api/runs.py:123	18	<code>db: Any, scope: str, run_id: str, *, parent_run_id: str, visited: set[str] None=None</code> Follow subprocess waits to the deepest actionable human gate.
<code>_node_type_catalog</code>	app/api/ workflow_generation.py:446	18	<code>type_names: list[str] None=None</code> The catalog text embedded in the generation system prompt.
<code>candidate_from_paper</code>	app/evidence/ retrieval.py:202	18	<code>paper: dict[str, Any], *, claim_id: str, query: str, purpose: str, source_hint: str None=None, discovery_lane: str None=None</code> Compute the candidate from paper.
<code>deduplicate_candidates</code>	app/evidence/ retrieval.py:373	18	<code>candidates: list[CandidateSource]</code> Deduplicate the same work for one claim, across queries AND lanes.
<code>DelimitedTextExtractor.extract</code>	app/ingestion/ extractor.py:414	18	<code>self, path: Path</code> Extract the result.
<code>ingest_file</code>	app/ingestion/ pipeline.py:59	18	<code>path: Path, metadata: dict[str, Any] None=None, chunk_config: ChunkConfig None=None, object_store: ObjectStore None=None, embedder: Embedder None=None, weaviate: WeaviateClient None=None, mongo: MongoClient None=None, collection_config: CollectionConfig None=None</code> Ingest one document end-to-end.
<code>MCPToolAgent.required_arguments</code>	app/nodes/ mcp_tool.py:326	18	<code>self, service: Any, cfg: MCPToolConfig</code> The argument names the server itself declares as required.
<code>ScientificSkillCatalog.select</code>	app/research/ skills.py:314	18	<code>self, *, objective: str, requested: Iterable[str]=(), auto_select: bool=True, max_skills: int=3</code> Select the result.

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build_field_index	app/runtime/ logic_preflight.py:743	18	<code>spec: WorkflowSpec</code> Typed paths for every node in the workflow.
_validate_inputs	app/runtime/ preflight.py:2014	18	<code>spec: WorkflowSpec, provided_inputs: dict[str, Any] None, report: WorkflowPreflightReport</code> Validate the inputs.
EntityTokenizerService.detect_spans	app/security/ entity_tokenizer.py:211	18	<code>self, text: str, *, session_id: str, collection_id: str</code> Detect the spans.
upsert_run	app/workflow/ run_history.py:243	18	<code>db, run_id: str, session_id: str, workflow_name: str None=None, status: str None=None, node_types: dict[str, str] None=None, inputs: dict[str, Any] None=None, variables: dict[str, Any] None=None, outputs: dict[str, Any] None=None, workflow_yaml: str None=None, started_at: float None=None, ended_at: float None=None, node_count: int None=None, completed_node_count: int None=None, retry_of_run_id: str None=None, attempt: int None=None, reused_node_count: int None=None, reused_nodes: list[str] None=None, error: str None=None, origin: str None=None, history_visibility: str None=None, workflow_id: str None=None, workflow_version_id: str None=None, conversation_id: str None=None, message_id: str None=None</code> Create or patch one run without erasing fields omitted by the caller.
autofix_node	app/api/workflows.py:501	17	<code>req: AutofixNodeRequest, request: Request, user: CurrentUser=Depends(require_consultant)</code> Repair one selected node while rejecting every unrelated YAML change.
DocxExtractor.extract	app/ingestion/ extractor.py:190	17	<code>self, path: Path</code> Extract the result.
InMemoryEmailAdapter.search	app/integrations/email/ memory.py:57	17	<code>self, connection: EmailConnection, criteria: EmailSearchCriteria</code> Search the result.
ExternalActionService.call	app/integrations/ external_action.py:101	17	<code>self, *, run_id: str, node_id: str, safety_class: str, method: str, url: str, headers: dict[str, str], body: Any, timeout_seconds: float, approval_satisfied: bool</code> Compute the call.
ModelRouter.select	app/llm/ model_router.py:178	17	<code>self, *, method_name: str, kwargs: Mapping[str, Any], input_tokens: int, allowed_models: Sequence[str], is_available: Callable[[str], bool], node_type: str None=None, policy: Mapping[str, Any] None=None</code> Select the result.

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infer_complexity	app/llm/ model_router.py:410	17	<code>method_name: str, kwargs: Mapping[str, Any], *, input_tokens: int, node_type: str None=None</code> Compute the infer complexity.
RegistryLLMGateway_models_for_call	app/llm/registry.py:1141	17	<code>self, intended: str, candidate_models: tuple[str, ...] list[str] None=None</code> Return the permitted, ordered runtime fallback chain.
seed_d365_finance	app/mcp/ business_records/ seed.py:138	17	<code>conn</code> Seed the d365 finance.
find_installed_unit	app/mcp/d365_finance/ handlers.py:446	17	<code>backend: DynamicsBackend, arguments: dict[str, Any]</code> Resolve installed equipment by serial, model, manufacturer or site.
DOCXProposalRender.render_markdown	app/nodes/ docx_renderer.py:279	17	<code>self, doc, md: str, tpl</code> Render the markdown.
ProposalTruthGraphAgent.run	app/nodes/ proposal_truth_graph.py:160	17	<code>self, state: dict[str, Any], resolved_config: dict[str, Any]</code> Run the result.
apply_deterministic_fixes	app/runtime/ autofix.py:664	17	<code>yaml_text: str, report: WorkflowPreflightReport</code> Apply the deterministic fixes.
_check_router	app/runtime/ logic_preflight.py:1059	17	<code>node: NodeSpec, config: dict[str, Any], index: FieldIndex, nodes: dict[str, NodeSpec], issue: Callable[..., None], guaranteed_before: Callable[[str, str], bool] None</code> Check the router.
_check_condition	app/runtime/ logic_preflight.py:1176	17	<code>condition: Condition, *, node: NodeSpec, index: FieldIndex, nodes: dict[str, NodeSpec], path: str, issue: Callable[..., None], guaranteed_before: Callable[[str, str], bool] None, label: str</code> Check the condition.
_resolve	app/runtime/rules.py:195	17	<code>context: dict[str, Any], path: str</code> Resolve the result.
_HorizonDocxBUILDER.render_table	app/tools/ docx_proposal_rendering.py:1550	17	<code>self, table_tag: Tag</code> Render the table.
build_model	scripts/ build_documentation_pdf.py:301	17	No summary declared.
report	scripts/ openrouter_health_check.py:883	17	<code>results: list[ModelResult], excluded: list[tuple[str, str]], args: argparse.Namespace</code> Report the result.
_internal_evidence_from_run	app/api/ candidates.py:224	16	<code>run: dict[str, Any]</code> InternalProjectEvidenceRetrieverAgent facts: no URL exists for these, only a source name and an exact

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			quoted passage — the UI offers a secondary "verify this claim" action for them instead of a link.
ensure_managed_adapter	app/api/ chat_workflows.py:464	16	<pre>request: Request, user: CurrentUser, *, adapter_key: str, display_name: str, yaml_text: str, source: Literal['generated', 'imported', 'existing'], source_workflow_name: str None=None, preferred_slug: str None=None</pre> Reuse one owner-scoped canonical adapter and upgrade it in place.
_managed_adapter_key	app/api/ chat_workspace.py:32	16	<pre>body: PlanWorkspaceRequest, plan</pre> Stable owner-local identity for execution-equivalent Chat adapters.
run_generation_pipeline	app/api/ workflow_generation.py:6 59	16	<pre>*, prompt: str, sample_inputs: dict[str, Any] None, generate_yaml: Callable[[str, str None, str None], Awaitable[str]], static_check: Callable[[str], Awaitable[tuple[WorkflowPreflightReport, dict[str, WorkflowInputSpec]]], execute: Callable[[str, dict[str, Any]], Awaitable[dict[str, Any]]]</pre> The repair-loop control flow, dependency-injected so it's testable without a real LLM/Mongo/executor (see tests/test_workflow_generation.py).
coerce_typed_list_field	app/evidence/ models.py:415	16	<pre>value: Any, model: type[_ModelT], field_name: str</pre> Accept a node's records as a list, JSON text, or empty text.
PdfExtractor.extract	app/ingestion/ extractor.py:81	16	<pre>self, path: Path</pre> Extract the result.
DynamicFigureAgent.run	app/nodes/ dynamic_figure_agent.py: 145	16	<pre>self, state: dict[str, Any], resolved_config: dict[str, Any]</pre> Run the result.
FieldSpec.shape_is_coherent	app/runtime/ field_schema.py:143	16	<pre>self</pre> Compute the shape is coherent.
_static_mcp_required_arguments	app/runtime/ logic_preflight.py:460	16	<pre>server_id: Any, tool_name: Any, arguments: dict[str, Any]</pre> Argument names the tool's own static schema requires but `arguments` never mentions — or `None` when this tool isn't one preflight can see statically (a live/third-party server), in which case there is nothing to report.
_probe_rag_resource_nodes	app/runtime/ preflight.py:2711	16	<pre>spec: WorkflowSpec, repository: Any, report: WorkflowPreflightReport, *, owner_scope_id: str</pre> Confirm every referenced Knowledge resource exists and is usable.
WorkflowSpec.derive_inputs_from_start_node	app/runtime/ schema.py:516	16	<pre>self</pre> A Start node (app/nodes/start.py) is the single place a workflow author declares what this workflow needs — this projects those declarations into the existing `inputs` contract so file-upload validation and the "required input is missing" pre-execution check (app/workflow/

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			file_inputs.py's validate_workflow_inputs) apply to them for free, with no new validator (§37/§38).
WebSearchService._search_kimi	app/tools/web_io.py:349	16	self, query: str, top_k: int Search the kimi.
validate_workflow_inputs	app/workflow/file_inputs.py:224	16	specs: dict[str, WorkflowInputSpec], values: dict[str, Any], *, session_id: str, object_store: Any None Validate and normalize declared workflow file inputs before execution.
collect_insertions	scripts/generate_docstrings.py:524	16	tree: ast.Module, path: Path No summary declared.
_timeline	app/api/builder.py:676	15	spec: WorkflowSpec, state: dict[str, Any], stubbed: set[str] Per-step trace in execution order, with each step's own explanation.
_slice_through	app/api/builder.py:1458	15	spec: WorkflowSpec, node_id: str The smallest valid upstream slice ending at `node_id` (§22).
AnthropicBatchService.results	app/llm/anthropic_batch.py:114	15	self, batch_id: str, *, response_models: dict[str, Type[BaseModel]] None=None Read back results for an ended batch.
MCPIntegrationService.call	app/mcp/service.py:271	15	self, *, server_id: str, tool_name: str, arguments: dict[str, Any], run_id: str='', node_id: str='', session_id: str='', user_role: str None=None, approval_satisfied: bool=False, audit_db: Any=None, timeout_override: float None=None Run one tool through the full gate.
ClaimEvidenceVerifier.run	app/nodes/claim_evidence_verifier.py:110	15	self, state: dict, resolved_config: dict[str, Any] Run the result.
TransformOperation.operation_has_inputs	app/nodes/data_transform.py:67	15	self Compute the operation has its inputs.
IntegrationAgent.run	app/nodes/integration.py:276	15	self, state, resolved_config: dict[str, Any] Run the result.
_fix_template_unknown_structured_field	app/runtime/autofix.py:233	15	raw: dict, issue: PreflightIssue Internal helper for the fix template unknown structured field step.
_resolve_reference	app/runtime/logic_preflight.py:1340	15	reference: str, *, node: NodeSpec, index: FieldIndex, nodes: dict[str, NodeSpec], path: str, issue: Callable[..., None], guaranteed_before: Callable[[str, str], bool] None, label: str Resolve a rule/route reference, reporting what is wrong with it.
_check_mcp_tool	app/runtime/logic_preflight.py:1968	15	node: NodeSpec, issue: Callable[..., None], reviewed_before: Callable[[str],

Logic unit	Location	Branch score	Contract
			<code>bool]</code> Validate an MCP tool step without contacting the server.
<code>prune_absent</code>	<code>app/runtime/templating.py:193</code>	15	<code>config: dict[str, Any], schema: Any</code> Drop config values that resolved to None but cannot be None.
<code>_append_styled_text</code>	<code>app/tools/docx_proposal_rendering.py:241</code>	15	<code>paragraph, text: str, style: '_InlineStyle'</code> Append the styled text.
<code>_class_summary</code>	<code>scripts/generate_docstrings.py:409</code>	15	<code>node: ast.ClassDef</code> No summary declared.
<code>main</code>	<code>scripts/ingest_samples.py:72</code>	15	<code></code> Compute the main.
<code>_structured_dataset_candidates_from_run</code>	<code>app/api/candidates.py:175</code>	14	<code>run: dict[str, Any]</code> StructuredDatasetRetrieverAgent rows, reshaped into the same discovered-candidate table shape as web-search-based discovery.
<code>chat_workflow_labels</code>	<code>app/api/chat_workspace.py:143</code>	14	<code>body: WorkflowLabelsRequest, request: Request, user: CurrentUser=Depends(require_consultant)</code> Return Chat-only business labels without renaming saved workflows.
<code>list_chunks</code>	<code>app/api/inspect.py:192</code>	14	<code>request: Request, collection_id: Optional[str]=Query(None), session_id: Optional[str]=Query(None), limit: int=Query(2000, le=10000), user: CurrentUser=Depends(require_permission('workflow:read'))</code> List ingested chunks grouped by source document.
<code>LocalOpenAICompatibleGateway.chat_with_tools</code>	<code>app/llm/local_openai_gw.py:294</code>	14	<code>self, *, model: str, system: str, messages: list[dict], tools: list[dict], temperature: float=0.0, max_tokens: int=4096</code> Compute the chat with tools.
<code>OpenRouterGateway.chat_with_tools</code>	<code>app/llm/openrouter_gw.py:271</code>	14	<code>self, *, model: str, system: str, messages: list[dict], tools: list[dict], temperature: float=0.0, max_tokens: int=4096</code> Compute the chat with tools.
<code>find_account_ownership</code>	<code>app/mcp/d365_finance/handlers.py:133</code>	14	<code>backend: DynamicsBackend, arguments: dict[str, Any]</code> Resolve account ownership to people who can actually be assigned today.
<code>find_order</code>	<code>app/mcp/dynamics/handlers.py:567</code>	14	<code>backend: DynamicsBackend, arguments: dict[str, Any]</code> Find one sales order by order number or equipment serial number.
<code>field_paths_from_schema</code>	<code>app/mcp/results.py:234</code>	14	<code>schema: dict[str, Any] None, prefix: str='', depth: int=0</code> Flatten a JSON Schema into typed dotted paths.
<code>EmailAgent.run</code>		14	

Logic unit	Location	Branch score	Contract
	app/nodes/ email_integration.py:299		<code>self, state, resolved_config: dict[str, Any]</code> Run the result.
TransformAgent.preflight_output_fields	app/nodes/ transform.py:399	14	<code>cls, config: dict[str, Any]</code> Compute the preflight output fields.
_parse_skill	app/research/ skills.py:424	14	<code>text: str</code> Parse the skill.
compile_workflow	app/runtime/ compiler.py:670	14	<code>spec: WorkflowSpec, checkpointer=None, services=None</code> Compute the compile workflow.
_json_schema_paths	app/runtime/ logic_preflight.py:370	14	<code>schema: dict[str, Any], prefix: str</code> Flatten a JSON-Schema object's `properties` into typed dotted paths.
_kind_from_annotation	app/runtime/ logic_preflight.py:604	14	<code>annotation: Any</code> Best-effort (kind, nullable) for a declared output_schema field.
_check_external_actions	app/runtime/ logic_preflight.py:1533	14	<code>spec: WorkflowSpec, issue: Callable[..., None], guaranteed_before: Callable[[str, str], bool] None</code> An outward-facing action with nothing between it and the model.
_sanitise_fragment	app/tools/ proposal_rendering.py:129	14	<code>value: str, content_format: str</code> Internal helper for the sanitise fragment step.
build_existing_workflow_chat_adapter	app/workflow/ chat_workspace_planner.py:194	14	<code>workflow_name: str, workflow_spec: WorkflowSpec, *, runtime_metadata: dict[str, Any] None=None</code> Wrap any saved workflow with natural-language input/output LLM calls.
main	scripts/ preflight_workflows.py:39	14	 Compute the main.
simulate	app/api/builder.py:507	13	<code>body: SimulateRequest, request: Request, user: CurrentUser=Depends(require_consultant)</code> Run a workflow in memory and return a per-step trace (§23, §24, §47).
list_candidates	app/api/ candidates.py:268	13	<code>run_id: str, request: Request, user: CurrentUser=Depends(require_consultant)</code> Every acquired evidence PDF for a run, plus discovered candidates.
prepare_chat_workspace	app/api/ chat_workspace.py:220	13	<code>body: PlanWorkspaceRequest, request: Request, user: CurrentUser=Depends(require_consultant)</code> No summary declared.
_real_usage_snippet	app/api/ workflow_generation.py:495	13	<code>type_name: str, manifest_entry: dict</code> Step 2's grounding for one node type: its own config, exactly as filled in by a real, already-working workflow on this platform, if one exists.
build_llm_yaml_generator		13	<code>llm: Any, services: dict[str, Any], scope: str, *, mode: str='generate'</code>

Logic unit	Location	Branch score	Contract
	app/api/workflow_generation.py:782		Build a `generate_yaml(base_prompt, prior_yaml, feedback) -> yaml text` closure over a live LLM client.
generate_workflow_endpoint	app/api/workflow_generation.py:940	13	req: GenerateWorkflowRequest, request: Request, user: CurrentUser=Depends(require_consultant) Generate the workflow endpoint.
autofix_workflow	app/api/workflows.py:420	13	req: AutofixWorkflowRequest, request: Request, user: CurrentUser=Depends(require_consultant) Best-effort repair of a workflow's preflight errors: deterministic patches first (free, predictable), then an LLM repair loop for whatever remains.
classify_authority	app/evidence/identifiers.py:331	13	url: str None, work: ResolvedWork “(source_label, authority)” from what is actually known.
PptxExtractor.extract	app/ingestion/extractor.py:256	13	self, path: Path Extract the result.
OcrFallbackDocumentParser.parse	app/ingestion/strategies.py:79	13	self, path: Path, *, config: dict[str, Any] Parse the result.
VisionAugmentedDocumentParser.parse	app/ingestion/strategies.py:159	13	self, path: Path, *, config: dict[str, Any] Parse the result.
AnthropicGateway.chat_with_tools	app/llm/anthropic_gw.py:287	13	self, *, model: str, system: str, messages: list[dict], tools: list[dict], temperature: float=0.0, max_tokens: int=4096 Compute the chat with tools.
OpenAIGateway.complete	app/llm/openai_gw.py:189	13	self, *, model: str, system: str, user: str, temperature: float=0.0, max_tokens: int=1024, reasoning_effort: str None=None, on_token: Callable[[str], Awaitable[None]] None=None Complete the result.
product_search	app/mcp/business_records/handlers.py:185	13	conn: MySQLConnectionAbstract, arguments: dict[str, Any] Compute the product search.
seed_dynamics_crm	app/mcp/business_records/seed.py:278	13	conn Seed the dynamics crm.
evaluate	app/mcp/policy.py:206	13	*, connection: MCPServerConnection, tool_name: str, annotations: Any=None, user_role: str None=None, approval_satisfied: bool=False Run the gate for one tool call.
MCPIntegrationService._describe_tool	app/mcp/service.py:172	13	self, connection: MCPServerConnection, tool: Any Internal helper for the describe tool step.
_candidate_records	app/nodes/bounded_deep_research_agent.py:339	13	dossiers: list[ResearchDossier], *, max_citations_per_brief: int,

Logic unit	Location	Branch score	Contract
			<code>max_candidates_per_claim: int</code> Internal helper for the candidate records step.
<code>check_compatibility</code>	<code>app/nodes/contracts.py:129</code>	13	<code>source: NodeDefinition, target: NodeDefinition, *, available_capabilities: set[str] None=None, available_permissions: set[str] None=None</code> Apply universal policies; never branches on a source/target pair.
<code>_research_questions</code>	<code>app/nodes/internal_project_evidence_retriever.py:417</code>	13	<code>*, graph: Any, research_briefs: Any, limit: int</code> Internal helper for the research questions step.
<code>MCPAgent.run</code>	<code>app/nodes/mcp_agent.py:106</code>	13	<code>self, state, resolved_config: dict[str, Any]</code> Run the result.
<code>scan_snippet_for_warnings</code>	<code>app/nodes/python_snippet.py:269</code>	13	<code>code: str</code> AST-level authoring hints — never a security control.
<code>_metadata_status</code>	<code>app/nodes/research_source_acquirer.py:507</code>	13	<code>candidate: CandidateSource, *, final_url: str, client: httpx.AsyncClient</code> Internal helper for the metadata status step.
<code>deterministic_horizon_blockers</code>	<code>app/proposal_graph/horizon_evaluator.py:142</code>	13	<code>graph: ProposalGraph</code> Compute the deterministic horizon blockers.
<code>RAGService._select_route</code>	<code>app/rag/service.py:89</code>	13	<code>self, config: RetrievalRoutingProfileConfig, query: str, llm: Any</code> Select the route.
<code>_fix_template_unknown_node</code>	<code>app/runtime/autofix.py:297</code>	13	<code>raw: dict, issue: PreflightIssue</code> Internal helper for the fix template unknown node step.
<code>_validate_guided_experience</code>	<code>app/runtime/preflight.py:2136</code>	13	<code>spec: WorkflowSpec, report: WorkflowPreflightReport</code> Validate authored Guided Run copy without penalising legacy workflows.
<code>OpenAllImageGenerationService.generate</code>	<code>app/tools/image_io.py:93</code>	13	<code>self, prompt: str, *, size: str='auto', quality: str='auto', output_format: str='png', model: str None=None, provider: ImageProvider='openai'</code> Generate the result.
<code>preflight_stats</code>	<code>app/workflow/preflight_stats.py:84</code>	13	<code>db: Any, *, since: datetime None=None, sample_limit: int=2000</code> Compute the preflight stats.
<code>_module_doc</code>	<code>scripts/generate_docstrings.py:464</code>	13	<code>path: Path, tree: ast.Module</code> No summary declared.
<code>generate_documents</code>	<code>scripts/generate_reference_workflows.py:32</code>	13	 No summary declared.
<code>fetch_catalog</code>	<code>scripts/openrouter_health_check.py:247</code>	13	 Fetch the catalog.
<code>main</code>		13	

Logic unit	Location	Branch score	Contract
	scripts/ smoke_production.py:26		Compute the main.
get_budgets	app/api/ cost_admin.py:417	12	request: Request, user: CurrentUser=Depends(require_admin) Return the budgets.
oauth_callback	app/api/ email_oauth.py:178	12	provider: Provider, request: Request, code: str None=None, state: str None=None, error: str None=None, error_description: str None=None The provider redirects the user's browser back here after consent.
get_trace_chunk_context	app/api/retrieval.py:276	12	retrieval_request_id: str, chunk_id: str, request: Request, user: CurrentUser=Depends(require_permission('k knowledge:read')) Return cited and adjacent same-document passages from one scoped trace.
resume	app/api/workflows.py:764	12	run_id: str, req: ResumeRequest, request: Request, user: CurrentUser=Depends(require_consultant) Resume doesn't need services for execution — the cached compiled graph in _PAUSED_GRAPHS already has services bound into every node instance.
retrieve_passages	app/evidence/ retrieval.py:599	12	claim: str, pages: list[dict[str, Any]], *, document_id: str, limit: int=8 Page-aware lexical retrieval before semantic stance verification.
chunk_document	app/ingestion/ chunker.py:282	12	doc: ExtractedDocument, config: ChunkConfig None=None, chunk_id_prefix: str None=None Chunk a full ExtractedDocument into Chunk objects with metadata.
infer_task_kind	app/llm/ model_router.py:373	12	method_name: str, kwargs: Mapping[str, Any], *, node_type: str None=None Compute the infer task kind.
_result_from_output_row	app/llm/ openai_batch.py:174	12	row: dict, response_models: dict[str, Type[BaseModel]] Internal helper for the result from output row step.
OpenAIGateway.chat_with_tools	app/llm/openai_gw.py:347	12	self, *, model: str, system: str, messages: list[dict], tools: list[dict], temperature: float=0.0, max_tokens: int=4096 Compute the chat with tools.
RegistryLLMGateway.record_cost	app/llm/registry.py:1078	12	self, intended: str, resolved: str, resp, *, task_type: str='unknown', latency_ms: float None=None, fallback_reason: str None=None, stage: str None=None Write a LedgerEntry if we have context + a ledger.
RegistryLLMGateway.call_resilient	app/llm/registry.py:1240	12	self, method_name: str, *, intended: str, kwargs: dict[str, Any], candidate_models: tuple[str, ...] list[str] None=None Returns (response, resolved_model, latency_ms, fallback_reason).

Logic unit	Location	Branch score	Contract
RegistryLLMGateway_complete_impl	app/llm/registry.py:1399	12	<code>self, *, model: str, stage: str None=None, **kwargs</code> Complete the impl.
call_tool	app/mcp/server.py:137	12	<code>name: str, arguments: dict</code> Compute the call tool.
_review_content	app/nodes/human_in_loop.py:379	12	<code>cfg: HITLConfig, state: dict[str, Any], context: dict[str, Any]</code> Internal helper for the review content step.
_summarise	app/nodes/mcp_tool.py:479	12	<code>data: dict[str, Any]</code> Lift 'count', 'found' and 'first' out of whatever shape the tool returned.
WorkflowFileLoader.run	app/nodes/workflow_file_loader.py:107	12	<code>self, state: dict[str, Any], resolved_config: dict[str, Any]</code> Run the result.
apply_conditional_fields	app/nodes/workflow_input.py:408	12	<code>bindings: list[InputFieldBinding], data: dict[str, Any], missing: list[str]</code> Evaluate visible_when/required_when against this form's OWN already-resolved field values — never workflow graph state, so this reuses app/runtime/rules.py's evaluate_group with zero new evaluation logic, just a context shaped so a bare field name resolves (the same "name matches a node_outputs key" shorthand resolve_path already supports elsewhere).
evaluate_horizon_proposal	app/proposal_graph/horizon_evaluator.py:191	12	<code>llm: Any, *, proposal_text: str, graph: ProposalGraph, generator_model: str None, evaluator_models: list[str], criterion_threshold: float=3.0, total_threshold: float=10.0</code> Compute the evaluate horizon proposal.
_fix_dict_where_string_expected	app/runtime/autofix.py:607	12	<code>raw: dict, issue: PreflightIssue</code> A recurring, purely mechanical generation mistake: giving a field that only accepts one string a dict of several labelled values instead — e.g.
sanitize_preview	app/runtime/node_events.py:15	12	<code>value: Any</code> Short, safe preview for the Cockpit card.
_lookup	app/runtime/templating.py:29	12	<code>path: str, state: dict</code> Walk a dotted path against workflow state.
KimiVisionService.analyze	app/tools/vision_io.py:69	12	<code>self, image_bytes: bytes, *, content_type: str, prompt: str, model: KimiVisionModel='kimi-k3', max_completion_tokens: int=8192</code> Compute the analyze.
_summary_for_function	scripts/generate_docstrings.py:324	12	<code>node, is_property: bool, is_init: bool, class_name: str None</code> No summary declared.
_args_section	scripts/generate_docstrings.py:368	12	<code>node</code> No summary declared.
build_fixtures	scripts/verify_knowledge_studio.py:251	12	<code>root: Path</code> Write one fixture per supported source format.

Logic unit	Location	Branch score	Contract
output_contract	app/api/builder.py:203	11	<code>body: ContractRequest, user: CurrentUser=Depends(require_consultant)</code> What each step guarantees, as typed dotted paths (§40).
_acquired_document_from_run	app/api/candidates.py:53	11	<code>run: dict[str, Any], document_id: str</code> Find an acquired document only in recorded acquisition-node output.
_discovered_candidates_from_run	app/api/candidates.py:115	11	<code>run: dict[str, Any]</code> Every discovery/Deep Research/prior-project candidate this run ever produced.
retrieve_preview	app/api/inspect.py:345	11	<code>request: Request, req: RetrieveRequest, user: CurrentUser=Depends(require_permission('workflow:read'))</code> Run a query through hybrid search + rerank and show the ranked chunks.
oauth_callback	app/api/integration_oauth.py:169	11	<code>provider: Provider, request: Request, code: str None=None, state: str None=None, error: str None=None, error_description: str None=None</code> The provider redirects the user's browser back here after consent.
_page_windows	app/evidence/retrieval.py:568	11	<code>text: str, size: int=1800, overlap: int=250</code> Internal helper for the page windows step.
normalize_openai_base_url	app/llm/local_openai_gw.py:50	11	<code>value: str</code> Validate and normalize a deployment-owned OpenAI-compatible URL.
_provider_error_code	app/llm/registry.py:263	11	<code>exc: BaseException</code> Extract a provider error code without depending on one SDK's shape.
SemanticLLMCache.get	app/llm/semantic_cache.py:71	11	<code>self, *, session_id: str, model: str, system: str, user: str, temperature: float, max_tokens: int</code> Return the result.
order_search	app/mcp/business_records/handlers.py:99	11	<code>conn: MySQLConnectionAbstract, arguments: dict[str, Any]</code> Compute the order search.
EmailNodeConfig.operation_has_what_it_needs	app/nodes/email_integration.py:146	11	<code>self</code> Compute the operation has what it needs.
GraphNormalizer.run	app/nodes/graph_normalizer.py:236	11	<code>self, state: dict, config: dict</code> Run the result.
HorizonDOCXProposalRenderer.run	app/nodes/horizon_docx_renderer.py:135	11	<code>self, state: dict[str, Any], resolved_config: dict[str, Any]</code> Run the result.
HorizonHTMLProposalRenderer.run	app/nodes/html_proposal_renderer.py:121	11	<code>self, state: dict[str, Any], resolved_config: dict[str, Any]</code> Run the result.
_normalise_registry		11	<code>value: Any</code> Internal helper for the normalise registry step.

Logic unit	Location	Branch score	Contract
	app/nodes/ internal_project_evidenc e_retriever.py:466		
KnowledgeRetrieval._retrieve _with_policy	app/nodes/ knowledge_retrieval.py:2 43	11	<code>self, request, scope: str, llm: Any</code> Bound retrieval and retry only recognized transient failures once.
MCPTToolAgent.run	app/nodes/ mcp_tool.py:224	11	<code>self, state, resolved_config: dict[str, Any]</code> Run the result.
ProposalEvidenceFactoryAge nt_gap_for_claim	app/nodes/ proposal_evidence_factor y.py:1013	11	<code>claim_id: str, final_status: str, materiality: str, *, independent_count: int, required_independent: int, contradiction_searched: bool, metadata_ok: bool, retraction_ok: bool, policy: EvidencePolicy</code> Internal helper for the gap for claim step.
_require_public_url	app/nodes/ research_source_acquirer .py:437	11	<code>url: str</code> Internal helper for the require public url step.
RouterAgent._route_by_field_ multi	app/nodes/router.py:430	11	<code>self, cfg: RouterConfig, state: WorkflowState</code> Multi-Route, field mode: 'route_field' resolves to a *list* of values (e.g.
SubprocessAgent._get_or_cr eate_launch	app/nodes/ subprocess.py:199	11	<code>self, cfg: SubprocessConfig, state: WorkflowState</code> Return the or create launch.
CostLedger._group_by	app/observability/ cost_ledger.py:291	11	<code>entries: list[dict], key: str</code> Group the by.
_predicate_to_filter	app/retrieval/ filters.py:202	11	<code>predicate: MetadataFilterPredicate</code> Internal helper for the predicate to filter step.
build_secure_where_filter	app/retrieval/ filters.py:254	11	<code>filters: RetrievalFilters, *, index_id: str None=None, exclude_parent_chunks: bool=False</code> Compile the non-removable security clause, ANDed with user metadata.
_fix_template_unknown_outp ut_field	app/runtime/ autofix.py:186	11	<code>raw: dict, issue: PreflightIssue</code> Internal helper for the fix template unknown output field step.
_fix_router_duplicate_rule	app/runtime/ autofix.py:463	11	<code>raw: dict, issue: PreflightIssue</code> Duplicate rule names in a legacy rule-mode Router — keep the first occurrence of each name.
_iter_model_values	app/runtime/ preflight.py:445	11	<code>config: Any, path: str='config'</code> Internal helper for the iter model values step.
_plain_targets	app/runtime/ preflight.py:1061	11	<code>spec: WorkflowSpec</code> Every target reached by a plain (non-HITL, non-router) edge, mapped to its declared predecessors — mirrors app/runtime/compiler.py's _wire_edges step 3 exactly, which is what actually decides which predecessors get combined into one add_edge(..., target) AND-join.
_matches_declared_field	app/runtime/ preflight.py:1925	11	

Logic unit	Location	Branch score	Contract
			<code>reference: str, declared: set[str]</code> Does a dotted reference match one of a node's declared output paths?
<code>_apply_action</code>	<code>app/runtime/rules.py:827</code>	11	<code>action: Action, values: dict[str, Any], context: dict[str, Any]</code> Write one fact.
<code>check_workflow_inputs</code>	<code>app/security/guardrails.py:164</code>	11	<code>value: Any</code> Block injection and apply the configured PII policy to direct inputs.
<code>OpaClient.evaluate</code>	<code>app/security/llm_policy.py:174</code>	11	<code>self, policy_input: dict</code> Compute the evaluate.
<code>_url_citations</code>	<code>app/tools/web_io.py:497</code>	11	<code>value: Any</code> Internal helper for the url citations step.
<code>package_table</code>	<code>scripts/build_documentation_pdf.py:339</code>	11	<code>model: Model, frontend: bool=False</code> No summary declared.
<code>node_test</code>	<code>app/api/builder.py:358</code>	10	<code>body: NodeTestRequest, request: Request, user: CurrentUser=Depends(require_consultant)</code> Run one node against sample data (§21).
<code>_ancestors</code>	<code>app/api/builder.py:1426</code>	10	<code>spec: WorkflowSpec, node_id: str</code> Internal helper for the ancestors step.
<code>verify_claim_endpoint</code>	<code>app/api/candidates.py:341</code>	10	<code>run_id: str, body: VerifyClaimRequest, request: Request, user: CurrentUser=Depends(require_consultant)</code> Run the gpt-5.6-sol secondary check for one internal-evidence record.
<code>_run_retrieval</code>	<code>app/api/inspect.py:282</code>	10	<code>request: Request, req: RetrieveRequest, user: CurrentUser</code> Run the query through the canonical retrieval path.
<code>download_workflow_input_file</code>	<code>app/api/workflow_files.py:171</code>	10	<code>key: str, request: Request, user: CurrentUser=Depends(require_consultant)</code> Download one uploaded input after enforcing the authenticated scope.
<code>work_identity</code>	<code>app/evidence/identifiers.py:290</code>	10	<code>work: ResolvedWork None, *, title: str None=None, fallback: str None=None</code> The single cross-lane dedup key for one work.
<code>parse_authors</code>	<code>app/evidence/retrieval.py:84</code>	10	<code>value: Any</code> Parse the authors.
<code>_encode</code>	<code>app/integrations/email/gmail.py:247</code>	10	<code>draft: EmailDraft, connection: EmailConnection</code> Build a base64url RFC-2822 message, which is what Gmail accepts.
<code>fetch_profile</code>	<code>app/integrations/email/oauth.py:278</code>	10	<code>provider: Provider, *, access_token: str, client: httpx.AsyncClient None=None</code> Fetch the profile.
<code>fetch_profile</code>	<code>app/integrations/files/oauth.py:276</code>	10	<code>provider: Provider, *, access_token: str, client: httpx.AsyncClient None=None</code> Fetch the profile.

Logic unit	Location	Branch score	Contract
<code>_to_model_info</code>	<code>app/llm/openrouter_catalog.py:94</code>	10	<code>entry: dict[str, Any]</code> Internal helper for the to model info step.
<code>_record_usage</code>	<code>app/llm/registry.py:553</code>	10	<code>intended: str, resolved: str, resp</code> Record token metrics.
<code>RegistryLLMGateway.probe_model_access</code>	<code>app/llm/registry.py:913</code>	10	<code>self, models: list[str] tuple[str, ...] set[str]</code> Probe concrete models without creating a completion.
<code>_product</code>	<code>app/mcp/d365_finance/handlers.py:644</code>	10	<code>row: dict[str, Any]</code> Internal helper for the product step.
<code>normalise_result</code>	<code>app/mcp/results.py:126</code>	10	<code>response: Any, *, max_bytes: int=256000, tool_name: str=''</code> Normalise an MCP `CallToolResult` into a workflow output.
<code>_collect_json_types</code>	<code>app/nodes/contracts.py:108</code>	10	<code>field: dict[str, Any], found: set[DataType]</code> No summary declared.
<code>DOCXProposalRenderer.render_table</code>	<code>app/nodes/docx_renderer.py:335</code>	10	<code>self, doc, lines, start, tpl</code> Render the table.
<code>DOCXProposalRenderer.add_inline</code>	<code>app/nodes/docx_renderer.py:388</code>	10	<code>self, paragraph, text: str, bold: bool=False</code> Add the inline.
<code>_patch_reviewed_content</code>	<code>app/nodes/human_in_loop.py:501</code>	10	<code>state: dict[str, Any], path: str, value: Any</code> Replace only the configured editable path for downstream nodes.
<code>KnowledgeRetrieval.run</code>	<code>app/nodes/knowledge_retrieval.py:113</code>	10	<code>self, state, resolved_config: dict[str, Any]</code> Run the result.
<code>ProposalEvidenceFactoryAgent.verify_document</code>	<code>app/nodes/proposal_evidence_factory.py:747</code>	10	<code>self, *, llm: Any, store: Any, claim_text: str, document: FullTextDocument, candidate: CandidateSource None, model: str, passage_limit: int</code> Verify the document.
<code>ProposalEvidenceFactoryAgent.quantitative_registry</code>	<code>app/nodes/proposal_evidence_factory.py:1182</code>	10	<code>claims: list[VerifiedClaim], links: list[ClaimEvidenceLink]</code> Internal helper for the quantitative registry step.
<code>PythonSnippetAgent.run</code>	<code>app/nodes/python_snippet.py:170</code>	10	<code>self, state, resolved_config: dict[str, Any]</code> Run the result.
<code>_acquire_candidate</code>	<code>app/nodes/research_source_acquirer.py:228</code>	10	<code>candidate: CandidateSource, *, client: httpx.AsyncClient, store: Any, run_id: str, policy: EvidencePolicy, max_redirects: int</code> Acquire the candidate.
<code>StructuredDatasetRetrieverConfig.coerce_queries</code>	<code>app/nodes/structured_dataset_retriever.py:78</code>	10	<code>cls, value: Any</code> Internal helper for the coerce queries step.
<code>SubprocessAgent.finalize</code>		10	

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	app/nodes/ subprocess.py:319		<code>self, cfg: SubprocessConfig, decision: dict[str, Any]</code> Finalize the result.
TransformAgent.preflight_static_output_values	app/nodes/ transform.py:435	10	<code>cls, config: dict[str, Any]</code> Compute the preflight static output values.
TransformAgent.complete_structured_with_retries	app/nodes/ transform.py:713	10	<code>self, llm: Any, *, model: str, system: str, user: str, response_model: Type[BaseModel], temperature: float, max_tokens: int, max_retries: int, fail_on_error: bool=True, reject_empty_fields: list[str] None=None</code> Complete the structured with retries.
ScientificSkillCatalog.refresh	app/research/ skills.py:241	10	<code>self</code> Refresh the result.
_fix_template_unknown_named_root	app/runtime/ autofix.py:365	10	<code>raw: dict, issue: PreflightIssue, *, root: str, noun: str</code> Internal helper for the fix template unknown named root step.
_fix_router_multiple_defaults	app/runtime/ autofix.py:431	10	<code>raw: dict, issue: PreflightIssue</code> A legacy rule-mode Router with more than one `default: true` rule — keep the first and clear the flag on the rest.
_project_output	app/runtime/ executor.py:44	10	<code>spec: WorkflowSpec, state: dict[str, Any], original_inputs: dict[str, Any]</code> Build the caller-facing output declared by the workflow contract.
_field_paths	app/runtime/ field_schema.py:413	10	<code>fields: list[FieldSpec], prefix: str, *, parent_unavailable: bool</code> Internal helper for the field paths step.
_validate_registry	app/runtime/ preflight.py:339	10	<code>report: WorkflowPreflightReport</code> Validate the registry.
_exclusive_branch_groups	app/runtime/ preflight.py:1082	10	<code>spec: WorkflowSpec, forward: dict[str, set[str]]</code> For each SINGLE-selection router (condition+branches) edge with 2+ distinct targets, the node sets exclusively reachable via each individual branch target (i.e.
_multi_route_branch_sets	app/runtime/ preflight.py:1190	10	<code>spec: WorkflowSpec, forward: dict[str, set[str]]</code> Per Multi-Route router edge: branch label -> nodes reachable ONLY via that branch's own target (not via any of the router's OTHER branch targets).
RedisRateLimitMiddleware.dispatch	app/security/ middleware.py:129	10	<code>self, request: Request, call_next: Callable</code> Dispatch the result.
render_horizon_proposal_docx	app/tools/ docx_proposal_rendering.py:1887	10	<code>*, content: str, content_format: Literal['markdown', 'html'], metadata: dict[str, Any], citation_registry: list[dict[str, Any]] None=None, evidence_qa: dict[str, Any] None=None, evidence_blockers: list[str] None=None, include_toc: bool=True, include_bibliography: bool=True, include_evidence_annex: bool=False,</code>

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			<code>page_limit: int None=45,</code> <code>max_embedded_image_bytes: int=10000000,</code> <code>enable_footnotes: bool=False</code> Create an editable Horizon Part B DOCX from sanitised HTML/Markdown.
WebSearchService._search_tavily	app/tools/web_io.py:216	10	<code>self, query: str, top_k: int</code> Search the tavily.
WebSearchService._search_openai	app/tools/web_io.py:283	10	<code>self, query: str, top_k: int</code> Search the openai.
store_upload	app/workflow/file_inputs.py:140	10	<code>upload: UploadFile, *, session_id: str,</code> <code>object_store: Any, remaining_total_bytes: int None=None</code> Stream one multipart upload to disk, enforce limits, then store it.
library_summary	app/workflow/library_metadata.py:92	10	<code>spec: WorkflowSpec</code> The Library card/list-facing metadata dict for one workflow.
_inflate_run_document	app/workflow/run_history.py:113	10	<code>db, doc: dict[str, Any] None</code> Transparently resolve GridFS pointers so callers see the original shape.
main	scripts/generate_docstrings.py:584	10	No summary declared.
assist_rules	app/api/builder.py:1032	9	<code>body: RuleAssistRequest, request: Request, user: CurrentUser=Depends(require_consultant)</code> Turn a described business rule into deterministic configuration (§36).
_previewable	app/api/builder.py:1524	9	<code>value: Any, depth: int=0</code> Trim large values before they cross the wire.
probe_services	app/api/health.py:163	9	<code>services: dict[str, Any]</code> Probe every configured dependency concurrently.
run_proposal_evidence_factory	app/api/proposals.py:569	9	<code>proposal_id: str, body: ProposalEvidenceFactoryRequest, request: Request, user: CurrentUser=Depends(require_consultant)</code> Verify already-fetched full text and return a citable evidence package.
delete_run_endpoint	app/api/runs.py:758	9	<code>run_id: str, request: Request, user: CurrentUser=Depends(require_consultant)</code> Delete the run endpoint.
IngestionJobRunner._embedder_for	app/ingestion/jobs.py:69	9	<code>self, embedding_profile: Any</code> Embedder matching an Embedding Profile, cached per model+dimensions.
load_connections	app/integrations/email/__init__.py:73	9	<code>raw: str None=None</code> Parse configured connections and resolve their credentials.
GmailAdapter._request	app/integrations/email/gmail.py:71	9	<code>self, connection: EmailConnection, method: str, path: str, *, side_effect: bool=False, **kwargs: Any</code> Internal helper for the request step.

Logic unit	Location	Branch score	Contract
<code>_gmail_query</code>	<code>app/integrations/email/gmail.py:225</code>	9	<code>criteria: EmailSearchCriteria</code> Translate neutral criteria into Gmail's `q` grammar.
<code>_extract_bodies</code>	<code>app/integrations/email/gmail.py:305</code>	9	<code>payload: dict[str, Any]</code> Walk the MIME parts tree for text and HTML bodies.
<code>MicrosoftGraphAdapter._request</code>	<code>app/integrations/email/msgraph.py:86</code>	9	<code>self, connection: EmailConnection, method: str, path: str, *, side_effect: bool=False, **kwargs: Any</code> Internal helper for the request step.
<code>_draft_payload</code>	<code>app/integrations/email/msgraph.py:328</code>	9	<code>draft: EmailDraft</code> Draft the payload.
<code>load_connections</code>	<code>app/integrations/files/__init__.py:68</code>	9	<code>raw: str None=None</code> Parse configured connections and resolve their credentials.
<code>_to_meta</code>	<code>app/integrations/files/onedrive.py:233</code>	9	<code>raw: dict[str, Any]</code> Internal helper for the to meta step.
<code>IntegrationService._refreshed</code>	<code>app/integrations/files/service.py:161</code>	9	<code>self, connection: IntegrationConnection</code> Refresh this connection's access token if it's near/past expiry.
<code>_is_model_available</code>	<code>app/llm/registry.py:593</code>	9	<code>model: str</code> Return whether auto-routing may safely select `model`.
<code>build_server_specs</code>	<code>app/mcp/client.py:159</code>	9	<code>app_settings: Settings=settings</code> Build portable MCP launch specs from application configuration.
<code>_quote</code>	<code>app/mcp/d365_finance/handlers.py:211</code>	9	<code>row: dict[str, Any], *, account_id: str, purchase_order_number: str</code> Internal helper for the quote step.
<code>_split_top_level</code>	<code>app/mcp/dynamics/client.py:775</code>	9	<code>expression: str, joiner: str</code> Split on a joiner that is not inside parentheses or a string literal.
<code>find_service_case</code>	<code>app/mcp/dynamics/handlers.py:697</code>	9	<code>backend: DynamicsBackend, arguments: dict[str, Any]</code> Find the service case.
<code>build_query</code>	<code>app/mcp/dynamics/odata.py:142</code>	9	<code>*, select: list[str] None=None, filter_expression: str None=None, order_by: str None=None, top: int None=None, expand: str None=None</code> Assemble OData system query options.
<code>ConsistencyChecker.run</code>	<code>app/nodes/consistency_checker.py:159</code>	9	<code>self, state: dict, config: dict</code> Run the result.
<code>FigureEmbedder.run</code>	<code>app/nodes/figure_embedder.py:121</code>	9	<code>self, state: dict[str, Any], resolved_config: dict[str, Any]</code> Run the result.
<code>_chunk_pages</code>	<code>app/nodes/minio_evidence_ingestion.py:321</code>	9	<code>pages: list[dict[str, Any]], *, chunk_chars: int, overlap: int, max_chunks: int</code> Chunk the pages.
<code>ResearchSourceAcquirer.run</code>		9	

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	app/nodes/ research_source_acquirer.py:123		<code>self, state: dict[str, Any], resolved_config: dict[str, Any]</code> Run the result.
<code>_fetch_public_source</code>	app/nodes/ research_source_acquirer.py:388	9	<code>client: httpx.AsyncClient, url: str, *, max_bytes: int, max_redirects: int</code> Fetch the public source.
<code>SQLQueryAgent.run</code>	app/nodes/ sql_query.py:173	9	<code>self, state, resolved_config: dict[str, Any]</code> Run the result.
<code>StartAgent.run</code>	app/nodes/start.py:212	9	<code>self, state, resolved_config: dict[str, Any]</code> Run the result.
<code>generate_concept_alternatives</code>	app/proposal_graph/ concepts.py:341	9	<code>llm: Any, *, graph: ProposalGraph, model: str='claude-opus-5', judge_model: str None=None, concept_note: str='', skill_catalog: Any None=None</code> Create conservative, balanced, and ambitious options from graph facts.
<code>_sanitize_instructions</code>	app/research/ skills.py:468	9	<code>text: str</code> Keep research guidance while removing execution/install directions.
<code>expand_context</code>	app/retrieval/ context.py:27	9	<code>chunks: list[RetrievedChunk], *, strategy: str, client: Any, collection_name: str, filters: RetrievalFilters, index_id: str None</code> Apply the requested context-expansion strategy to already-kept chunks.
<code>coerce_metadata_filter_group</code>	app/retrieval/ filters.py:174	9	<code>value: dict[str, Any] MetadataFilterGroup None</code> Normalize workflow ``runtime_filters``/API filter payloads to one shape.
<code>rerank</code>	app/retrieval/ reranker.py:81	9	<code>query: str, candidates: list[RetrievedChunk], top_n: int, llm: LLMGateway, model: str='gpt-5'</code> Score all candidates in one LLM call, return top_n sorted by score.
<code>retrieve</code>	app/retrieval/ retriever.py:30	9	<code>q: RetrievalQuery, *, weaviate_client: object, llm: LLMGateway, embedder: Embedder, collection_registry=None</code> Run the four-stage retrieval pipeline.
<code>_fix_unknown_node_config_field</code>	app/runtime/ autofix.py:560	9	<code>raw: dict, issue: PreflightIssue</code> Internal helper for the fix unknown node config field step.
<code>RunEventBus.unsubscribe</code>	app/runtime/ events.py:353	9	<code>self, run_id: str, queue: asyncio.Queue[RunEvent], session_id: str None=None</code> Unsubscribe the result.
<code>_annotation_for</code>	app/runtime/ field_schema.py:258	9	<code>spec: FieldSpec, model_name: str, *, permit_none: bool</code> Internal helper for the annotation for step.
<code>describe_schema</code>		9	

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	app/runtime/ field_schema.py:481		<code>fields: list[FieldSpec]</code> A compact, human-readable contract used in prompts and the About tab.
<code>_find_subprocess_cycle</code>	app/runtime/ logic_preflight.py:1786	9	<code>entry_refs: set[str]</code> Iterative cycle detection over the on-disk subprocess call graph.
<code>_equals</code>	app/runtime/rules.py:594	9	<code>actual: Any, expected: Any</code> Internal helper for the equals step.
<code>analyze</code>	app/security/ presidio_client.py:65	9	<code>text: str, *, language: str='en', score_threshold: float=0.5, client: httpx.AsyncClient None=None</code> Calls POST {presidio_analyzer_url}/analyze.
<code>_set_table_geometry</code>	app/tools/ docx_proposal_rendering. py:401	9	<code>table, widths_dxa: list[int], *, indent_dxa: int=TABLE_INDENT_DXA</code> Apply fixed Word table geometry with matching grid and cell widths.
<code>_HorizonDocxBuilder.table_placements</code>	app/tools/ docx_proposal_rendering. py:1505	9	<code>self, table_tag: Tag</code> Internal helper for the table placements step.
<code>render_horizon_proposal</code>	app/tools/ proposal_rendering.py:248	9	<code>*, content: str, content_format: Literal['markdown', 'html'], metadata: dict[str, Any], citation_registry: list[dict[str, Any]] None=None, evidence_qa: dict[str, Any] None=None, evidence_blockers: list[str] None=None, include_toc: bool=True, include_bibliography: bool=True, include_evidence_annex: bool=False, page_limit: int None=45</code> Render a Horizon Part B proposal and return immutable output metadata.
<code>WebSearchService.search</code>	app/tools/web_io.py:110	9	<code>self, query: str, *, provider: SearchProvider='auto', top_k: int=8, fallback_to_openai: bool=True</code> Search the result.
<code>redact_for_history</code>	app/workflow/ run_history.py:165	9	<code>value: Any</code> Recursively remove prompts and credentials from operator-visible data.
<code>_reconcile_if_stale</code>	app/workflow/ run_history.py:1106	9	<code>db, doc: dict[str, Any] None</code> Flip an orphaned "running" run to "failed" the moment it is read.
<code>source_files</code>	scripts/ build_documentation_pdf. py:182	9	<code></code> No summary declared.
<code>v15</code>	scripts/ build_documentation_pdf. py:578	9	<code>m: Model</code> No summary declared.
<code>validate</code>	scripts/ build_documentation_pdf. py:697	9	<code>directory: Path</code> No summary declared.
<code>_class_doc</code>		9	

Logic unit	Location	Branch score	Contract
	scripts/ generate_docstrings.py:4 30		node: ast.ClassDef No summary declared.
filter_by_provider	scripts/ openrouter_health_check. py:755	9	selected: list[CatalogEntry], excluded: list[tuple[str, str]], *, allowed_slugs: set[str], provider_lookup: dict[str, str], client: httpx.AsyncClient, concurrency: int Keep only models with at least one endpoint on an allowed provider.
main	scripts/ smoke_retrieval.py:33	9	Compute the main.
assist_schema	app/api/builder.py:913	8	body: SchemaAssistRequest, request: Request, user: CurrentUser=Depends(require_consultant) Propose a structured-output schema from a description (\$37).
download_acquired_research _document	app/api/candidates.py:70	8	run_id: str, document_id: str, request: Request, user: CurrentUser=Depends(require_consultant) Authorize against owned run provenance, then issue a short-lived PDF URL.
list_pricing	app/api/cost_admin.py:77	8	request: Request, openrouter_q: str None=Query(default=None, max_length=200), openrouter_limit: int=Query(default=25, ge=1, le=200), user: CurrentUser=Depends(require_admin) Direct/local catalog pricing (editable overrides layered on MODEL_PRICING defaults) plus a live, read-only slice of OpenRouter's own per-model pricing — OpenRouter's ~400-500 models are priced by OpenRouter itself and reported per-call via usage.cost (LedgerEntry.cost_source == "provider_reported"),
cache_summary	app/api/ cost_admin.py:342	8	request: Request, since_days: int=Query(default=30, ge=1, le=365), user: CurrentUser=Depends(require_admin) Compute the cache summary.
draft_prompt	app/api/ node_types_chat.py:383	8	req: DraftPromptRequest, request: Request, user: CurrentUser=Depends(require_consultant) Draft the prompt.
draft_code	app/api/ node_types_chat.py:592	8	req: DraftCodeRequest, request: Request, user: CurrentUser=Depends(require_consultant) Draft the code.
compare	app/api/retrieval.py:191	8	payload: RetrievalCompareRequest, request: Request, user: CurrentUser=Depends(require_permission('r ag:query')) Compare the result.
_reserve_run_id	app/api/workflows.py:70	8	services: dict[str, Any], *, run_id: str, session_id: str Claim a client-supplied run_id for one tenant, idempotently.

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run	app/api/workflows.py:640	8	<code>req: RunRequest, request: Request, user: CurrentUser=Depends(require_consultant)</code> Run the result.
list_chat_workflows	app/api/workflows.py:951	8	<code></code> Return the small published-workflow catalog used by Business Chat.
run_golden_set_with_model	app/evaluation/workflow_golden.py:265	8	<code>cases: list[WorkflowGoldenCase], *, model: str, services: dict[str, Any], run_id_prefix: str</code> Run the golden set with model.
ResolvedWork.is_peer_reviewed	app/evidence/identifiers.py:104	8	<code>self</code> True ONLY when resolved metadata positively supports it.
papers_from_payload	app/evidence/retrieval.py:163	8	<code>raw: Any</code> Compute the papers from payload.
candidate_work_identity	app/evidence/retrieval.py:343	8	<code>candidate: CandidateSource</code> Cross-lane work identity for one candidate.
validate_downloaded_pdf	app/evidence/retrieval.py:519	8	<code>raw_path: str None, *, allowed_root: Path, max_bytes: int</code> Validate the downloaded pdf.
_pack_pieces	app/ingestion/chunker.py:169	8	<code>pieces: list[str], encoding: tiktoken.Encoding, config: ChunkConfig</code> Greedy packer.
MetadataContextEnricher.enrich	app/ingestion/strategies.py:492	8	<code>self, chunks: list[Chunk], *, document: ExtractedDocument, config: dict[str, Any]</code> Compute the enrich.
PdfPageVisionDescriber.describe_pages	app/ingestion/vision.py:120	8	<code>self, path: Path, page_indexes: list[int], *, prompt: str None=None</code> Describe the given pages.
pages_with_visual_content	app/ingestion/vision.py:273	8	<code>path: Path, *, min_images: int=1</code> 0-based indexes of PDF pages that carry figures, images or table rules.
_extract_attachments	app/integrations/email/gmail.py:337	8	<code>payload: dict[str, Any]</code> Extract the attachments.
_to_message	app/integrations/email/msgraph.py:387	8	<code>raw: dict[str, Any]</code> Internal helper for the to message step.
EmailService._refreshed	app/integrations/email/service.py:159	8	<code>self, connection: EmailConnection</code> Refresh this connection's access token if it's near/past expiry.
EmailService.execute	app/integrations/email/service.py:251	8	<code>self, *, connection_id: str, operation: EmailOperation, criteria: EmailSearchCriteria None=None, message_id: str None=None, draft: EmailDraft None=None, idempotency_scope: str None=None</code> Execute the result.
EmailService._perform_write	app/integrations/email/service.py:320	8	<code>self, *, adapter: EmailAdapter, connection: EmailConnection, operation: EmailOperation, draft: EmailDraft,</code>

Logic unit	Location	Branch score	Contract
			<code>idempotency_scope: str None</code> Internal helper for the perform write step.
<code>validate_metadata_schema</code>	<code>app/knowledge/service.py:26</code>	8	<code>schema: dict[str, Any]</code> Validate the metadata schema.
<code>_score_profile</code>	<code>app/llm/model_router.py:549</code>	8	<code>profile: ModelProfile, *, task: TaskKind, complexity: Complexity, priority: AccuracyPriority, estimated_cost: float, max_cost: float, prefer_low_latency: bool, observed_quality: float None</code> Score the profile.
<code>_cosine</code>	<code>app/llm/semantic_cache.py:293</code>	8	<code>left: list[float], right: list[float]</code> Internal helper for the cosine step.
<code>DataverseClient.access_token</code>	<code>app/mcp/dynamics/client.py:237</code>	8	<code>self</code> Client-credentials token for the organisation's `.default` scope.
<code>find_contact</code>	<code>app/mcp/dynamics/handlers.py:399</code>	8	<code>backend: DynamicsBackend, arguments: dict[str, Any]</code> Find the contact.
<code>find_shipment</code>	<code>app/mcp/dynamics/handlers.py:656</code>	8	<code>backend: DynamicsBackend, arguments: dict[str, Any]</code> Find the shipment.
<code>load_servers</code>	<code>app/mcp/registry.py:371</code>	8	<code>raw: str None=None</code> Parse configured servers from JSON.
<code>synthesize_about</code>	<code>app/nodes/about_synthesis.py:192</code>	8	<code>klass: 'Type[NodeType]'</code> Best-effort `about` dict for a node type that didn't author its own.
<code>EndAgent.preflight_output_fields</code>	<code>app/nodes/end.py:167</code>	8	<code>cls, config: dict[str, Any]</code> Compute the preflight output fields.
<code>EndAgent.run</code>	<code>app/nodes/end.py:195</code>	8	<code>self, state, resolved_config: dict[str, Any]</code> Run the result.
<code>MethodologyEngineeringAgent.draft_one_card</code>	<code>app/nodes/methodology_engineering.py:215</code>	8	<code>self, *, llm: Any, catalog: Any, objective_id: str, objective_text: str, model: str, max_skills: int, known_brief_ids: set[str], known_claim_ids: set[str], selected_concept: Any=None</code> Draft the one card.
<code>PaperQAEvidenceSynthesizerAgent.run</code>	<code>app/nodes/paperqa_evidence_synthesizer.py:155</code>	8	<code>self, state: dict[str, Any], resolved_config: dict[str, Any]</code> Run the result.
<code>_messages</code>	<code>app/nodes/proposal_submission_gate.py:78</code>	8	<code>value: str list[Any]</code> Internal helper for the messages step.
<code>_review_queue</code>	<code>app/nodes/proposal_truth_graph.py:325</code>	8	<code>claims: list[VerifiedClaim], gaps: list[EvidenceGap], graph: Any</code> Internal helper for the review queue step.
<code>RouterAgent.parse_literal</code>	<code>app/nodes/router.py:649</code>	8	<code>s: str</code> Parse the literal.

Logic unit	Location	Branch score	Contract
RouterAgent._route_by_llm_multi	app/nodes/router.py:707	8	<pre>self, cfg: RouterConfig</pre> Multi-Route, llm mode: ask the model to choose all routes that apply and parse a comma-separated list back — for genuinely fuzzy "which of these departments does this enquiry touch" judgment.
_is_semantically_empty	app/nodes/transform.py:179	8	<pre>value: Any</pre> No summary declared.
TransformAgent._run_legacy	app/nodes/transform.py:518	8	<pre>self, cfg: TransformConfig, llm: Any</pre> Run the legacy.
InputFieldBinding._default_in_complete_legacy_shape	app/nodes/workflow_input.py:176	8	<pre>cls, data: Any</pre> A binding saved before 'item_type'/'fields'/'enum_values' existed on this model (when 'type' was an unvalidated free-text string) may be missing the piece its type now requires.
_composite_score	app/proposal_graph/concepts.py:196	8	<pre>concept: ConceptAlternative, graph: ProposalGraph</pre> Internal helper for the composite score step.
merge_graph	app/proposal_graph/graph.py:87	8	<pre>left: Any, right: Any</pre> Reducer for the 'proposal_graph' state key.
BoundedToolResearchService._attribute_citations	app/research/deep_research.py:513	8	<pre>self, brief: ResearchBrief, citations: list[ResearchCitation], max_cost_per_call_usd: float</pre> Map each gathered citation to ONE claim, in place.
RetrievalService._record_failed_trace	app/retrieval/service.py:154	8	<pre>self, *, request: RetrievalQuery, request_id: str, owner_scope_id: str, error: Exception, total_ms: float</pre> Record the failed trace.
RunEventBus._pump_redis	app/runtime/events.py:319	8	<pre>self, pubsub: Any, queue: asyncio.Queue[RunEvent], last_seen: int</pre> Internal helper for the pump redis step.
_check_python_snippet_agents	app/runtime/logic_preflight.py:1883	8	<pre>spec: WorkflowSpec, issue: Callable[..., None]</pre> Everything decidable about a Python snippet without running it: a syntax error (would fail every run identically, so it is an error, not a warning), plus authoring hints — an import with no effect inside the sandbox, a dunder-attribute access, or a snippet that never touches 'output' at all.
_has_cycle	app/runtime/preflight.py:793	8	<pre>graph: dict[str, set[str]]</pre> Return whether cycle.
_validate_start_end_edges	app/runtime/preflight.py:1317	8	<pre>spec: WorkflowSpec, report: WorkflowPreflightReport</pre> Start represents the beginning of execution and End its terminus — enforce that structurally rather than treating them as decorative.
_validate_business_logic	app/runtime/preflight.py:2234	8	<pre>spec: WorkflowSpec, report: WorkflowPreflightReport, forward: dict[str, set[str]], reverse: dict[str, set[str]]</pre> Check visually authored rules, routes and schemas (zero tokens).

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<code>_contains</code>	<code>app/runtime/rules.py:548</code>	8	<code>haystack: Any, needle: Any</code> Internal helper for the contains step.
<code>evaluate_rules</code>	<code>app/runtime/rules.py:776</code>	8	<code>rules: list[Rule], context: dict[str, Any], *, initial: dict[str, Any] None=None</code> Evaluate every rule in order and accumulate the facts they set.
<code>_HorizonDocxBuilder.add_cover</code>	<code>app/tools/docx_proposal_rendering.py:1090</code>	8	<code>self, evidence_blockers: list[str]</code> Add the cover.
<code>WorkflowBuilderStore.list_versions</code>	<code>app/workflow/builder_store.py:270</code>	8	<code>self, name: str</code> List the versions.
<code>validate_workflow_file_reference</code>	<code>app/workflow/file_inputs.py:303</code>	8	<code>raw_ref: WorkflowFileRef dict[str, Any], *, session_id: str, object_store: Any None, require_parseable_text: bool=False</code> Validate one file reference before a download, extraction, or HITL edit.
<code>BackgroundRunManager.run_owned</code>	<code>app/workflow/orchestration.py:301</code>	8	<code>self, coro: Awaitable[dict[str, Any]], *, db: Any, run_id: str, session: str, record_rejection_reason: bool, services: dict[str, Any] None=None</code> Run the owned.
<code>deliver_by_token</code>	<code>app/workflow/subprocess_callback.py:76</code>	8	<code>db: Any, services: dict[str, Any], token: str, *, status: str, output: Any, node_outputs: dict[str, Any], error: str None</code> Deliver the by token.
<code>class_value</code>	<code>scripts/build_documentation_pdf.py:225</code>	8	<code>node: ast.ClassDef, name: str</code> No summary declared.
<code>v23</code>	<code>scripts/build_documentation_pdf.py:664</code>	8	<code>m: Model</code> No summary declared.
<code>build</code>	<code>scripts/build_eu_proposal_copy_bundle.py:69</code>	8	<code>repo: Path, output: Path</code> Build the result.
<code>build</code>	<code>scripts/build_workflow_file_inputs_bundle.py:48</code>	8	<code>repo: Path, output: Path</code> Build the result.
<code>_returns_section</code>	<code>scripts/generate_docstrings.py:351</code>	8	<code>node, is_property: bool</code> No summary declared.
<code>main_async</code>	<code>scripts/load_test.py:20</code>	8	<code>args: argparse.Namespace</code> Compute the main async.
<code>sweep</code>	<code>scripts/openrouter_health_check.py:823</code>	8	<code>entries: list[CatalogEntry], args: argparse.Namespace</code> Sweep the result.
<code>main</code>		8	<code></code> Compute the main.

Logic unit	Location	Branch score	Contract
	scripts/ production_preflight.py: 52		
main	scripts/ verder_process_emails.py :44	8	 Compute the main.
main	scripts/ verder_setup_knowledge.p y:63	8	 Compute the main.
login	app/api/auth.py:76	7	<code>request: Request, response: Response,</code> <code>form: OAuth2PasswordRequestForm=Depends()</code> Compute the login.
mcp_test_tool	app/api/builder.py:1207	7	<code>body: MCPToolTestRequest, request:</code> <code>Request, user:</code> <code>CurrentUser=Depends(require_consultant)</code> Run one MCP tool with literal arguments (§21).
_structured_dataset_query_ur l	app/api/ candidates.py:154	7	<code>record: dict[str, Any]</code> Reconstruct the exact bounded Eurostat query that produced this row.
_repair_workspace_existing_ adapter	app/api/ chat_workflows.py:366	7	<code>record: ChatWorkflowRecord, store:</code> <code>ChatWorkflowStore</code> Refresh planner-created saved-workflow wrappers after adapter fixes.
score_output	app/api/eval.py:71	7	<code>req: ScoreOutputRequest, request:</code> <code>Request, user:</code> <code>CurrentUser=Depends(require_permission('e</code> <code>val:run'))</code> Score the output.
_health_payload	app/api/health.py:207	7	<code>request: Request</code> Internal helper for the health payload step.
_build_node_type_catalog	app/api/ node_types_chat.py:63	7	 Always reads NodeRegistry.manifest() fresh — the same live source GET /api/node-types serves — so a node type added to the codebase shows up here automatically with no other change needed.
_describe_fields	app/api/ node_types_chat.py:484	7	<code>fields: list[DraftInstructionsFieldSpec]</code> Internal helper for the describe fields step.
_graph_from_checkpoint	app/api/proposals.py:83	7	<code>checkpoint: dict[str, Any]</code> Internal helper for the graph from checkpoint step.
_node_configs_from_workflo w_yaml	app/api/run_chat.py:85	7	<code>workflow_yaml: str None</code> Internal helper for the node configs from workflow yaml step.
ask_about_run	app/api/run_chat.py:248	7	<code>run_id: str, req: AskAboutRunRequest,</code> <code>request: Request, user:</code> <code>CurrentUser=Depends(require_consultant)</code> Compute the ask about run.
_hitl_config_fallback	app/api/runs.py:204	7	<code>checkpoint: dict[str, Any], node_id: str</code> <code> None</code> Best-effort HITL gate fields straight from the workflow's own node config, for a checkpoint whose captured interrupt payload is missing or unusable.

Logic unit	Location	Branch score	Contract
resume_paused_run	app/api/runs.py:500	7	<code>run_id: str, request: Request, user: CurrentUser=Depends(require_consultant)</code> Resume a run paused by the run-history "pause" action.
upload_workflow_input_files	app/api/workflow_files.py:72	7	<code>files: Annotated[list[UploadFile], File(...)], request: Request, user: CurrentUser=Depends(require_consultant)</code> Store files and return stable references suitable for workflow state.
_synthesize_sample_inputs	app/api/workflow_generation.py:597	7	<code>specs: dict[str, WorkflowInputSpec], provided: dict[str, Any] None</code> Fill in placeholder values for a workflow's declared inputs so it can actually be run.
validate_workflow	app/api/workflows.py:231	7	<code>req: ValidateWorkflowRequest, request: Request, user: CurrentUser=Depends(require_consultant)</code> Validate a workflow without running a node or consuming LLM tokens.
get_file	app/api/workflows.py:1345	7	<code>key: str, request: Request, download: bool=False</code> Stream a file from object storage by key (used by the Output Viewer).
run_eval	app/evaluation/runner.py:60	7	<code>*, workflow_name: str, examples: list[GoldenExample], judge: LLMJudge, produce_answer: ProduceAnswer</code> Run the whole golden set through produce_answer + judge, aggregate.
run_workflow_golden_case	app/evaluation/workflow_golden.py:205	7	<code>case: WorkflowGoldenCase, *, model: str, services: dict[str, Any], run_id: str</code> Run the workflow golden case.
formatted_citation	app/evidence/retrieval.py:476	7	<code>candidate: CandidateSource</code> Compute the formatted citation.
main	app/ingestion/chunker.py:368	7	<code></code> Quick CLI: `python -m app.ingestion.chunker <path>`.
select_embedding_model	app/ingestion/embedding_catalog.py:141	7	<code>*, doc_types: list[str] None=None, document_count: int=0, total_bytes: int=0, sample_text: str=''</code> Deterministically pick an embedding model for a corpus.
XlsxExtractor.extract	app/ingestion/extractor.py:148	7	<code>self, path: Path</code> Extract the result.
MarkdownExtractor.extract	app/ingestion/extractor.py:357	7	<code>self, path: Path</code> Extract the result.
main	app/ingestion/extractor.py:591	7	<code></code> Quick CLI: `python -m app.ingestion.extractor path/to/file.<ext>`.
SemanticChunkingStrategy.chunk	app/ingestion/strategies.py:450	7	<code>self, document: ExtractedDocument, *, config: dict[str, Any], chunk_id_prefix: str</code> Chunk the result.
_to_message	app/integrations/email/gmail.py:270	7	<code>raw: dict[str, Any]</code> Internal helper for the to message step.

Logic unit	Location	Branch score	Contract
MicrosoftGraphAdapter.read	app/integrations/email/msgraph.py:176	7	<pre>self, connection: EmailConnection, message_id: str</pre> Read the result.
exchange_code	app/integrations/email/oauth.py:149	7	<pre>provider: Provider, *, code: str, code_verifier: str None=None, client: httpx.AsyncClient None=None</pre> Compute the exchange code.
refresh_access_token	app/integrations/email/oauth.py:200	7	<pre>provider: Provider, *, refresh_token: str, client: httpx.AsyncClient None=None</pre> Refresh the access token.
GoogleDriveProvider._request	app/integrations/files/google_drive.py:81	7	<pre>self, connection: IntegrationConnection, method: str, url: str, **kwargs: Any</pre> Internal helper for the request step.
_to_meta	app/integrations/files/google_drive.py:261	7	<pre>raw: dict[str, Any], *, parent_id: str None=None</pre> Internal helper for the to meta step.
exchange_code	app/integrations/files/oauth.py:147	7	<pre>provider: Provider, *, code: str, code_verifier: str None=None, client: httpx.AsyncClient None=None</pre> Compute the exchange code.
refresh_access_token	app/integrations/files/oauth.py:198	7	<pre>provider: Provider, *, refresh_token: str, client: httpx.AsyncClient None=None</pre> Refresh the access token.
OneDriveProvider._request	app/integrations/files/onedrive.py:66	7	<pre>self, connection: IntegrationConnection, method: str, url: str, *, follow_redirects: bool=False, **kwargs: Any</pre> Internal helper for the request step.
AnthropicGateway.complete_structured	app/llm/anthropic_gw.py:204	7	<pre>self, *, model: str, system: str, user: str, response_model: Type[T], temperature: float=0.0, max_tokens: int=4096, reasoning_effort: str None=None</pre> Complete the structured.
OpenAIGateway.complete_structured	app/llm/openai_gw.py:274	7	<pre>self, *, model: str, system: str, user: str, response_model: Type[T], temperature: float=0.0, max_tokens: int=4096, reasoning_effort: str None=None</pre> Complete the structured.
_usage_fields	app/llm/openrouter_gw.py:102	7	<pre>usage: dict[str, Any] None</pre> Internal helper for the usage fields step.
_retry_after_seconds	app/llm/registry.py:320	7	<pre>exc: BaseException</pre> Retry the after seconds.
update_order	app/mcp/business_records/handlers.py:371	7	<pre>conn: MySQLConnectionAbstract, arguments: dict[str, Any]</pre> Update the order.
update_case		7	

Logic unit	Location	Branch score	Contract
	app/mcp/business_records/handlers.py:420		conn: MySQLConnectionAbstract, arguments: dict[str, Any] Update the case.
_finance_scm_spec	app/mcp/client.py:102	7	app_settings: Settings Launch spec for the Dynamics 365 Finance & Operations MCP server.
MCPClient.start	app/mcp/client.py:268	7	self, servers: list[str] None=None Launch the requested servers (default: all declared).
build_registry	app/mcp/connections.py:252	7	app_settings: Settings=settings Assemble every configured connection.
_sales_order	app/mcp/d365_finance/handlers.py:303	7	row: dict[str, Any] Internal helper for the sales order step.
_shipment	app/mcp/d365_finance/handlers.py:499	7	row: dict[str, Any] Internal helper for the shipment step.
FixtureBackend.query	app/mcp/dynamics/client.py:603	7	self, entity_set: str, *, select: list[str], filter_expression: str None=None, order_by: str None=None, top: int None=None, expand: str None=None Query the result.
classify_by_name	app/mcp/policy.py:103	7	tool_name: str Best-effort classification from the tool's name alone.
parse_json_text	app/mcp/results.py:95	7	text: str, *, max_bytes: int Parse a text payload that looks like JSON, safely.
human_approved	app/nodes/approval.py:13	7	state: dict[str, Any], approved_by: str None=None Did a human review approve something earlier on this run's path?
_search_audit	app/nodes/bounded_deep_research_agent.py:452	7	briefs: list[ResearchBrief], dossiers: list[ResearchDossier], failures: list[dict[str, str]] Search the audit.
ClaimEvidenceVerifier.source_text	app/nodes/claim_evidence_verifier.py:85	7	self, source, cap: int Internal helper for the source text step.
NodeDefinition.validate_contract	app/nodes/contracts.py:65	7	self Reject metadata gaps and contradictory declarations early.
GraphNormalizer.parse_json	app/nodes/graph_normalizer.py:321	7	raw: str Parse the LLM's JSON and fail loudly when extraction is malformed.
IntegrationNodeConfig.operation_has_what_it_needs	app/nodes/integration.py:124	7	self Compute the operation has what it needs.
KnowledgeRetrieval.validate_resources	app/nodes/knowledge_retrieval.py:203	7	self, cfg: KnowledgeRetrievalConfig, scope: str Revalidate saved references immediately before every execution.
MethodologyEngineeringAgent.run		7	

Logic unit	Location	Branch score	Contract
	app/nodes/ methodology_engineering. py:154		<code>self, state: dict[str, Any], resolved_config: dict[str, Any]</code> Run the result.
OpenAllImageGenerationAgent.run	app/nodes/ openai_image_generation_ agent.py:114	7	<code>self, state: dict[str, Any], resolved_config: dict[str, Any]</code> Run the result.
ProposalEvidenceFactoryAgent.citation_registry	app/nodes/ proposal_evidence_factor y.py:1080	7	<code>self, retained: dict[str, FullTextDocument], candidates: dict[str, CandidateSource], claims: list[VerifiedClaim]</code> Internal helper for the citation registry step.
_string_list	app/nodes/ proposal_truth_graph.py: 383	7	<code>value: Any</code> Internal helper for the string list step.
_with_model_catalog	app/nodes/ registry.py:173	7	<code>schema: dict</code> Expose the model catalog as a Builder dropdown.
RouterConfig.mode_has_what_it_needs	app/nodes/router.py:155	7	<code>self</code> Compute the mode has what it needs.
RouterAgent.run	app/nodes/router.py:330	7	<code>self, state, resolved_config: dict[str, Any]</code> Run the result.
RouterAgent.route_by_conditions_multi	app/nodes/router.py:541	7	<code>self, cfg: RouterConfig, state: WorkflowState</code> Multi-Route, conditions mode: every case is evaluated (never stop at the first match) and every matching route is collected — business policy here is additive, the same reasoning DecisionAgent's rule engine already applies (app/runtime/rules.py).
_dimension_code	app/nodes/ structured_dataset_retriever.py:578	7	<code>dimension: dict[str, Any], position: int</code> Internal helper for the dimension code step.
_resolve_child_inputs	app/nodes/ subprocess.py:385	7	<code>cfg: SubprocessConfig, child_spec: Any, state: WorkflowState</code> Explicit mapping -> same-named parent workflow input -> same-named parent node's whole output -> None.
_coerce_child_input	app/nodes/ subprocess.py:414	7	<code>value: Any, input_spec: Any</code> No summary declared.
resolve_field_bindings	app/nodes/ workflow_input.py:377	7	<code>bindings: list[InputFieldBinding], sample: dict[str, Any], state: dict[str, Any]</code> Resolve each declared field's source path against workflow state.
reject_values_outside_declared_shape	app/nodes/ workflow_input.py:519	7	<code>specs: list[FieldSpec], data: dict[str, Any]</code> Validate what actually arrived against the declared shape before any downstream node can read it — e.g.
score_concept	app/proposal_graph/ concepts.py:142	7	<code>concept: ConceptAlternative, graph: ProposalGraph</code> Evidence-weighted score based only on IDs that exist in the graph.
validate_independent_models		7	

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	app/proposal_graph/horizon_evaluator.py:120		evaluator_models: list[str], generator_model: str None Validate the independent models.
ProposalWorkspaceStore.register_source_version	app/proposal_graph/workspace_store.py:175	7	self, *, session_id: str, proposal_id: str, source_id: str, content: str, title: str, created_by: str, identifier: str None=None, authority: str='unverified', metadata: dict[str, Any] None=None Register the source version.
_check_value	app/retrieval/filters.py:111	7	field: str, value: Any, schema: dict[str, Any] Reject a value whose type contradicts the collection's declared schema.
_build_where_filter	app/retrieval/hybrid_search.py:22	7	f: RetrievalFilters Translate filters into a Weaviate Filter expression.
_fix_unknown_node_type	app/runtime/autofix.py:496	7	raw: dict, issue: PreflightIssue Internal helper for the fix unknown node type step.
_fix_model_not_in_catalog	app/runtime/autofix.py:522	7	raw: dict, issue: PreflightIssue Internal helper for the fix model not in catalog step.
_workflow_identity	app/runtime/autofix.py:736	7	yaml_text: str (name, node ids) for a parseable workflow dict, or (None, set()) if 'yaml_text' isn't one — used to sanity-check an LLM repair actually revised the given workflow rather than returning an unrelated one.
RunEventBus.publish	app/runtime/events.py:132	7	self, evt: RunEvent Publish the result.
RunEventBus.subscribe	app/runtime/events.py:202	7	self, run_id: str, session_id: str None=None, *, after_event_id: int None=None Subscribe the result.
validate_fields	app/runtime/field_schema.py:192	7	fields: list[FieldSpec], *, where: str='output' Reject duplicate names per level and runaway nesting depth.
_build_model	app/runtime/field_schema.py:297	7	name: str, fields: list[FieldSpec] Build the model.
_infer_kind	app/runtime/logic_preflight.py:318	7	value: Any Internal helper for the infer kind step.
_shallow_child_node_paths	app/runtime/logic_preflight.py:520	7	child_spec: WorkflowSpec, node_id: str One node's typed output paths, one level deep, deliberately NOT a recursive call into build_field_index(child_spec): a SubprocessAgent output node's own further-nested child is left untyped rather than resolved, so a long — or cyclic — subprocess reference chain can never blow the Python call stack just to compute field paths.
validate_business_logic	app/runtime/logic_preflight.py:840	7	spec: WorkflowSpec, *, issue: Callable[..., None], guaranteed_before: Callable[[str, str], bool] None=None Run every logic check.
_parse_spec		7	

Logic unit	Location	Branch score	Contract
	app/runtime/ preflight.py:261		yaml_text: str, report: WorkflowPreflightReport Parse the spec.
_topological_order	app/runtime/ preflight.py:726	7	reachable: set[str], forward: dict[str, set[str]], reverse: dict[str, set[str]] Kahn's-algorithm topological sort restricted to 'reachable'.
_validate_fanin_reachability	app/runtime/ preflight.py:1142	7	spec: WorkflowSpec, report: WorkflowPreflightReport, forward: dict[str, set[str]] A plain-edge fan-in target (2+ declared predecessors combined into one add_edge(..., target) AND-join — see app/runtime/compiler.py's _wire_edges) can only ever fire if ALL of its predecessors actually run in the same invocation.
_previewable	app/runtime/rules.py:708	7	value: Any Keep traces small — they are stored in run history and sent over SSE.
resolve	app/runtime/ templating.py:110	7	value: Any, state: dict Recursively walk a config value and replace template strings.
EntityTokenizerService._detokenize_text	app/security/ entity_tokenizer.py:350	7	self, text: str, *, session_id: str, collection_id: str Detokenize the text.
_set_run_font	app/tools/ docx_proposal_rendering.py:164	7	run, *, name: str='Arial', size: float None=None, color: str None=None, bold: bool None=None, italic: bool None=None Set the run font.
_toc_with_pdf_pages	app/tools/ docx_proposal_rendering.py:788	7	entries: list[dict[str, Any]], pdf: bytes Attach deterministic page numbers from the matching PDF layout.
_HorizonDocxBuilder.add_toc	app/tools/ docx_proposal_rendering.py:1187	7	self, entries: list[dict[str, Any]] Add the toc.
_HorizonDocxBuilder.decode_data_image	app/tools/ docx_proposal_rendering.py:1684	7	self, source: str Decode the data image.
_HorizonDocxBuilder.add_bibliography	app/tools/ docx_proposal_rendering.py:1771	7	self, citation_registry: list[dict[str, Any]] Add the bibliography.
_model_unavailable	app/tools/ image_io.py:286	7	exc: BaseException Internal helper for the model unavailable step.
select_candidate_node_types	app/workflow/ capability_selection.py:93	7	prompt: str, manifest: list[dict[str, Any]], *, max_candidates: int=12 Return the type names the generator's system prompt should describe: every core node type, Literal/Echo, and up to 'max_candidates' specialized types ranked by keyword overlap with 'prompt'.
select_generation_model		7	prompt: str, manifest: list[dict[str, Any]]

Logic unit	Location	Branch score	Contract
	app/workflow/capability_selection.py:163		Deterministic complexity → one of two model tiers, no LLM call.
_externalize_if_large	app/workflow/run_history.py:57	7	db, *, run_id: str, node_id: str, field: str, value: Any, force: bool=False Return `value` unchanged, or a small GridFS pointer if it's too big.
get_retry_checkpoint	app/workflow/run_history.py:1021	7	db, session_id: str, run_id: str Return a session-scoped private checkpoint for the retry endpoint.
cleanup_stale_runs	app/workflow/run_history.py:1391	7	db Periodic sweep (see the background loop started in app/main.py) that hard-deletes runs stuck in "running" or "paused" for at least ``settings.run_auto_cleanup_after_seconds``.
inspect_frontend	scripts/build_documentation_pdf.py:287	7	path: Path, model: Model No summary declared.
build	scripts/build_documentation_pdf.py:735	7	only: int None=None, output_dir: Path None=None No summary declared.
build	scripts/build_hitl_editor_bundle.py:40	7	repo: Path, output: Path Build the result.
category	scripts/build_workflow_qa_matrix.py:49	7	path: Path No summary declared.
process_file	scripts/generate_docstrings.py:559	7	path: Path, write: bool No summary declared.
main	scripts/generate_production_env.py:45	7	 Compute the main.
main	scripts/inspect_weaviate.py:13	7	 Compute the main.
fetch_provider_slugs	scripts/openrouter_health_check.py:284	7	client: httpx.AsyncClient Live provider registry as {lowercased name or slug: slug}, so an allowlist can be given in either form and a typo fails loudly instead of silently matching nothing.
_http_detail	scripts/openrouter_health_check.py:423	7	error: httpx.HTTPStatusError OpenRouter puts the useful part in `error.message`; the raw body is noisy.
_friendly_validation_message	app/api/builder.py:88	6	error: ValidationError One line per Pydantic error, without the "N validation errors for ModelName" header or the errors.pydantic.dev URL — those name an internal class the Builder never shows the author, and would otherwise leak straight into the Test tab's error banner verbatim.
_waiting_steps	app/api/builder.py:639	6	spec: WorkflowSpec, interrupt: Any One step per gate the run is parked at.

Logic unit	Location	Branch score	Contract
<code>_executed_path</code>	<code>app/api/builder.py:722</code>	6	<code>state: dict[str, Any]</code> Internal helper for the executed path step.
<code>run</code>	<code>app/api/eval.py:138</code>	6	<code>req: RunEvalRequest, request: Request, user: CurrentUser=Depends(require_permission('eval:run'))</code> Run the result.
<code>verify_claims</code>	<code>app/api/proposals.py:261</code>	6	<code>proposal_id: str, body: VerifyClaimsRequest, request: Request, user: CurrentUser=Depends(require_consultant)</code> Verify the claims.
<code>query_agent</code>	<code>app/api/rag_agents.py:121</code>	6	<code>rag_agent_id: str, payload: RAGQueryRequest, request: Request, user: CurrentUser=Depends(require_permission('rag:query'))</code> Query the agent.
<code>my_run_detail</code>	<code>app/api/runs.py:67</code>	6	<code>run_id: str, request: Request, user: CurrentUser=Depends(require_consultant)</code> Compute the my run detail.
<code>_hitl_display_name</code>	<code>app/api/runs.py:232</code>	6	<code>checkpoint: dict[str, Any], node_id: str None</code> No summary declared.
<code>_real_usage_examples</code>	<code>app/api/workflow_generation.py:531</code>	6	<code>type_names: list[str], manifest: list[dict]</code> Step 2's text block: one real config snippet per candidate type that has one (see <code>`_real_usage_snippet`</code>).
<code>_reference_workflow_examples</code>	<code>app/api/workflow_generation.py:555</code>	6	<code>type_names: list[str], *, limit: int=3</code> Build a bounded, type-ranked context slice from the hidden corpus.
<code>stream_run_events</code>	<code>app/api/workflows.py:1275</code>	6	<code>run_id: str, request: Request, last_event_id: int None=Header(default=None, alias='Last-Event-ID'), user: CurrentUser=Depends(require_consultant)</code> Authenticated Server-Sent Events stream for one workflow run.
<code>_acquire_lock</code>	<code>app/db/migrations.py:166</code>	6	<code>collection: Any, *, owner: str, wait_timeout_seconds: float</code> Acquire the lock.
<code>run_migrations</code>	<code>app/db/migrations.py:233</code>	6	<code>db: Any, *, owner: str None=None, wait_timeout_seconds: float=60.0</code> Run every pending migration and return the applied migration ids.
<code>StructuredDatasetQuery_safe_filters</code>	<code>app/evidence/database_models.py:79</code>	6	<code>cls, value: dict[str, list[str]]</code> Internal helper for the safe filters step.
<code>download_path_from_payload</code>	<code>app/evidence/retrieval.py:499</code>	6	<code>raw: Any</code> Download the path from payload.
<code>_split_sections</code>	<code>app/ingestion/chunker.py:121</code>	6	<code>text: str</code> Split on top-level section headers.

Logic unit	Location	Branch score	Contract
PdfPageVisionDescriber.unavailable_reason	app/ingestion/vision.py:104	6	<code>self</code> Compute the unavailable reason.
_graph_filters	app/integrations/email/msgraph.py:284	6	<code>criteria: EmailSearchCriteria</code> Internal helper for the graph filters step.
authorize_url	app/integrations/email/oauth.py:100	6	<code>provider: Provider, *, state: str, code_challenge: str None=None</code> Authorize the url.
authorize_url	app/integrations/files/oauth.py:98	6	<code>provider: Provider, *, state: str, code_challenge: str None=None</code> Authorize the url.
_load_kek	app/integrations/files/token_vault.py:35	6	<code></code> Derive the vault's Key Encryption Key from settings, failing closed.
KnowledgeService.activate_index	app/knowledge/service.py:259	6	<code>self, *, owner_scope_id: str, collection_id: str, index_id: str</code> Compute the activate index.
OpenAIBatchService.results	app/llm/openai_batch.py:126	6	<code>self, batch_id: str, *, response_models: dict[str, Type[BaseModel]] None=None</code> Read back results for an ended batch.
_fetch_raw_catalog	app/llm/openrouter_catalog.py:125	6	<code>client: httpx.AsyncClient None</code> Fetch the raw catalog.
OpenRouterGateway.complete	app/llm/openrouter_gw.py:164	6	<code>self, *, model: str, system: str, user: str, temperature: float=0.0, max_tokens: int=1024, reasoning_effort: str None=None</code> Complete the result.
OpenRouterGateway.complete_structured	app/llm/openrouter_gw.py:210	6	<code>self, *, model: str, system: str, user: str, response_model: Type[T], temperature: float=0.0, max_tokens: int=1024, reasoning_effort: str None=None</code> Complete the structured.
_promote_tier_peers	app/llm/registry.py:130	6	<code>requested: list[str]</code> Move each model's documented cross-provider peer to directly follow it.
_is_retryable_error	app/llm/registry.py:234	6	<code>exc: BaseException</code> Return whether another generation may succeed.
_gateway_class_for	app/llm/registry.py:385	6	<code>model_name: str</code> Internal helper for the gateway class for step.
_construct	app/llm/registry.py:410	6	<code>gw_cls: type[LLMGateway]</code> Internal helper for the construct step.
_estimate_input_tokens	app/llm/registry.py:573	6	<code>kwargs: dict[str, Any]</code> Cheap pre-call token estimate for deterministic routing.
RegistryLLMGateway.ensure_routed	app/llm/registry.py:743	6	<code>self, *, node_type: str None=None</code> Return a gateway guaranteed to resolve AUTO_MODEL via ModelRouter.
RegistryLLMGateway.tokenize_value	app/llm/registry.py:812	6	<code>self, value: Any</code> Tokenize the value.

Logic unit	Location	Branch score	Contract
RegistryLLMGateway.tokenize_messages	app/llm/registry.py:832	6	<code>self, messages: list[dict]</code> Tokenize the messages.
customer_search	app/mcp/business_records/handlers.py:55	6	<code>conn: MySQLConnectionAbstract, arguments: dict[str, Any]</code> Compute the customer search.
call_tool	app/mcp/business_records/server.py:91	6	<code>name: str, arguments: dict[str, Any] None</code> Compute the call tool.
run_readonly_query	app/mcp/business_records/sql_guard.py:80	6	<code>sql: str, params: dict[str, Any], *, connect: Any, max_rows: int, timeout_seconds: float</code> Run one read-only query with every layer above applied.
_dynamics_env	app/mcp/client.py:79	6	<code>app_settings: Settings</code> Environment for the Dynamics 365 MCP subprocess.
finance_scm_connection	app/mcp/connections.py:134	6	<code>app_settings: Settings=settings</code> The Dynamics 365 Finance & Operations connector.
_resolve_pump_models	app/mcp/d365_finance/handlers.py:385	6	<code>backend: DynamicsBackend, arguments: dict[str, Any]</code> product_name/product_family describe a catalogue product, not an inventory record directly — resolve to the pump model(s) inventory is actually keyed by before querying availability.
find_inventory_availability	app/mcp/d365_finance/handlers.py:406	6	<code>backend: DynamicsBackend, arguments: dict[str, Any]</code> Find the inventory availability.
_invoice	app/mcp/d365_finance/handlers.py:549	6	<code>row: dict[str, Any]</code> Internal helper for the invoice step.
_contract	app/mcp/d365_finance/handlers.py:598	6	<code>row: dict[str, Any]</code> Internal helper for the contract step.
DataverseClient.request	app/mcp/dynamics/client.py:331	6	<code>self, method: str, path: str, *, params: dict[str, str] None=None, json_body: dict[str, Any] None=None, write: bool=False</code> Internal helper for the request step.
DataverseClient.create	app/mcp/dynamics/client.py:479	6	<code>self, entity_set: str, payload: dict[str, Any]</code> Create the result.
_order	app/mcp/dynamics/handlers.py:167	6	<code>row: dict[str, Any]</code> Internal helper for the order step.
call_tool	app/mcp/dynamics/server.py:164	6	<code>name: str, arguments: dict[str, Any] None</code> Execute one tool.
_depth	app/mcp/results.py:58	6	<code>value: Any, current: int=0</code> Internal helper for the depth step.
_json_type	app/mcp/results.py:291	6	<code>schema: dict[str, Any]</code> Internal helper for the json type step.
_tool_error_payload	app/mcp/service.py:594	6	<code>result: MCPToolResult</code> Recognise a structured error a server returned in place of a result.

Logic unit	Location	Branch score	Contract
discover_nodes	app/nodes/___init___py:23	6	<code>*, force: bool=False</code> Import all built-in node modules and return module import failures.
_edge_targets	app/nodes/about_synthesis.py:55	6	<code>edge: dict[str, Any]</code> Internal helper for the edge targets step.
_important_config_fields	app/nodes/about_synthesis.py:172	6	<code>config_schema: Type[Any] None</code> Internal helper for the important config fields step.
LanguagePolicy.directive	app/nodes/ai_task.py:207	6	<code>self</code> Compute the directive.
AITaskAgent._user_prompt	app/nodes/ai_task.py:579	6	<code>self, cfg: AITaskConfig, fields: list[FieldSpec]</code> Internal helper for the user prompt step.
DataTransformAgent.run	app/nodes/data_transform.py:254	6	<code>self, state, resolved_config: dict[str, Any]</code> Run the result.
_as_text	app/nodes/data_transform.py:379	6	<code>value: Any</code> Internal helper for the as text step.
_as_number	app/nodes/data_transform.py:399	6	<code>value: Any</code> Internal helper for the as number step.
_summarise	app/nodes/email_integration.py:400	6	<code>message: Any</code> Internal helper for the summarise step.
_ScholarlyCandidateDiscover y_search_plan	app/nodes/evidence_agent.py:405	6	<code>self, llm: Any, claim: Claim, model: str None, *, skill_guidance: str=''</code> Search the plan.
ExternalActionAgent.run	app/nodes/external_action.py:193	6	<code>self, state, resolved_config: dict[str, Any]</code> Run the result.
IntegrationAgent.prelight_output_fields	app/nodes/integration.py:255	6	<code>cls, config: dict[str, Any]</code> Compute the preflight output fields.
KimiVisionAgent.run	app/nodes/kimi_vision_agent.py:93	6	<code>self, state: dict[str, Any], resolved_config: dict[str, Any]</code> Run the result.
PaperQAEvidenceSynthesizer Agent._synthesize_claim	app/nodes/paperqa_evidence_synthesizer.py:221	6	<code>self, *, claim_id: str, question: str, documents: list[FullTextDocument], store: Any, settings: Any, paperqa: Any</code> Internal helper for the synthesize claim step.
PriorProjectRetrieverConfig._coerce_briefs	app/nodes/prior_project_retriever.py:87	6	<code>cls, value: Any</code> Internal helper for the coerce briefs step.
_evidence_score	app/nodes/proposal_evidence_factor.py:230	6	<code>*, document: FullTextDocument, stance: str, confidence: float, exact_locator: bool, corroborated: bool</code> Internal helper for the evidence score step.
ProposalEvidenceFactoryAgent._cited_markdown	app/nodes/proposal_evidence_factor.py:1130	6	<code>claims: list[VerifiedClaim]</code> Internal helper for the cited markdown step.
_bounded_candidates		6	

Logic unit	Location	Branch score	Contract
	app/nodes/ research_source_acquirer.py:344		<code>candidates: list[CandidateSource], *, per_claim: int, total: int</code> Internal helper for the bounded candidates step.
RouterConfig.route_names	app/nodes/router.py:191	6	<code>self</code> Every route this router can emit — read by preflight to compare against the edge's declared branches, and by the Builder to draw them.
RouterAgent._route_by_field	app/nodes/router.py:364	6	<code>self, cfg: RouterConfig, state: WorkflowState</code> Internal helper for the route by field step.
RouterAgent._route_by_rule	app/nodes/router.py:592	6	<code>self, cfg, state</code> Internal helper for the route by rule step.
RouterAgent._eval_condition	app/nodes/router.py:618	6	<code>self, condition: str None, state: dict</code> Safe expression eval — supports only `path OP literal`.
RouterAgent._route_by_llm	app/nodes/router.py:670	6	<code>self, cfg</code> Internal helper for the route by llm step.
StartAgent.preflight_static_output_values	app/nodes/start.py:197	6	<code>cls, config: dict[str, Any]</code> Compute the preflight static output values.
TransformConfig.deterministic_mode_has_operations	app/nodes/ transform.py:129	6	<code>self</code> Compute the deterministic mode has operations.
TransformAgent._run_deterministic	app/nodes/ transform.py:484	6	<code>self, cfg: TransformConfig, state: Any</code> Run the deterministic.
TransformAgent._new_style_system_prompt	app/nodes/ transform.py:650	6	<code>self, cfg: TransformConfig</code> Internal helper for the new style system prompt step.
CostLedger.query	app/observability/ cost_ledger.py:388	6	<code>self, *, since: datetime None=None, until: datetime None=None, collection_id: str None=None</code> Raw entries across the whole ledger (not scoped to one run/session), for account-wide reporting (app/api/cost_admin.py).
ProposalWorkspaceStore.request_approval	app/proposal_graph/ workspace_store.py:387	6	<code>self, *, session_id: str, proposal_id: str, graph: ProposalGraph, stage: str, requested_by: str, selected_concept_id: str None=None</code> Compute the request approval.
_build_sources	app/rag/service.py:27	6	<code>citations: list[RAGCitation], chunk_by_id: dict[str, Any]</code> Group citations into one entry per source document (§37 dedup shape).
_field_type	app/retrieval/ filters.py:63	6	<code>field: str, schema: dict[str, Any]</code> Internal helper for the field type step.
_group_to_filter	app/retrieval/ filters.py:235	6	<code>group: MetadataFilterGroup</code> Group the to filter.
deduplicate	app/retrieval/ fusion.py:36	6	<code>candidates: list[RetrievedChunk]</code> Drop repeat `chunk_id`s, keeping the highest-scoring occurrence.
upsert_objects_on	app/retrieval/ weaviate_client.py:100	6	<code>client: weaviate.WeaviateClient, collection_name: str, objects: list[dict[str, Any]], vectors:</code>

Logic unit	Location	Branch score	Contract
			<code>list[list[float]]</code> Batch-insert ``objects``/``vectors`` into ``collection_name``.
WeaviateClient.upsert_chunks	app/retrieval/weaviate_client.py:251	6	<code>self, chunks: list[dict[str, Any]], vectors: list[list[float]]</code> Insert or replace chunks.
_validate_saved_decision	app/runtime/hitl.py:70	6	<code>checkpoint: dict[str, Any], decision: dict[str, Any]</code> Validate the saved decision.
_declared_output_paths	app/runtime/logic_preflight.py:640	6	<code>klass: Any</code> Typed paths for a node's own output_schema fields.
_check_threshold	app/runtime/logic_preflight.py:1283	6	<code>condition: Condition, described: FieldPath, value: float, node: NodeSpec, path: str, issue: Callable[..., None], label: str</code> Catch a threshold that can never do anything.
_check_output_field_enums	app/runtime/logic_preflight.py:1495	6	<code>node: NodeSpec, fields: list[FieldSpec], issue: Callable[..., None]</code> Check the output field enums.
_load_subprocess_refs	app/runtime/logic_preflight.py:1764	6	<code>name: str</code> Workflow names a *saved* workflow's own SubprocessAgent nodes reference.
_check_sql_query_agents	app/runtime/logic_preflight.py:1840	6	<code>spec: WorkflowSpec, issue: Callable[..., None]</code> A literal ``{...}`` inside the ``sql`` field text would be substituted by the compiler's generic templating pass before run() ever sees it — landing a mapped value directly in SQL text instead of going through ``params`` as a bound ``%(name)s`` placeholder.
_validate_typed_router	app/runtime/preflight.py:967	6	<code>source: NodeSpec, config: dict[str, Any], mode: str, edge: EdgeSpec, edge_path: str, report: WorkflowPreflightReport</code> Branch/route agreement for the Builder's visual router modes.
_mutually_exclusive	app/runtime/preflight.py:1120	6	<code>a: str, b: str, exclusive_groups: list[list[set[str]]]</code> True if ``a`` and ``b`` are exclusively reachable via two DIFFERENT branches of the SAME router — i.e.
_iter_strings	app/runtime/preflight.py:1599	6	<code>value: Any, path: str</code> Internal helper for the iter strings step.
_validate_templates	app/runtime/preflight.py:1965	6	<code>spec: WorkflowSpec, report: WorkflowPreflightReport, forward: dict[str, set[str]], reverse: dict[str, set[str]]</code> Validate the templates.
Condition.value_present_when_needed	app/runtime/rules.py:267	6	<code>self</code> Compute the value present when needed.
Rule.rule_is_decidable	app/runtime/rules.py:378	6	<code>self</code> Compute the rule is decidable.
RuleEvaluation.explanation_lines	app/runtime/rules.py:469	6	<code>self</code> Flat, human-readable "matched rules" list for the run trace (§24).

Logic unit	Location	Branch score	Contract
<code>_summarise</code>	<code>app/runtime/rules.py:719</code>	6	<code>condition: Condition, actual: Any, missing: bool, matched: bool</code> Internal helper for the summarise step.
<code>evaluate_group</code>	<code>app/runtime/rules.py:747</code>	6	<code>group: ConditionGroup, context: dict[str, Any]</code> Compute the evaluate group.
<code>SnippetRunnerClient._call</code>	<code>app/runtime/snippet_client.py:80</code>	6	<code>self, request: dict[str, Any], timeout_seconds: float</code> Internal helper for the call step.
<code>EntityTokenizerService.tokenize</code>	<code>app/security/entity_tokenizer.py:157</code>	6	<code>self, text: str, *, session_id: str, collection_id: str='default', mode: ProcessingMode=ProcessingMode.PSEUDONYMISED</code> Tokenize the result.
<code>EntityTokenizerService.detokenize</code>	<code>app/security/entity_tokenizer.py:311</code>	6	<code>self, value: Any, *, session_id: str, collection_id: str='default'</code> Recursively walks str/dict/list, restoring placeholders in every leaf string.
<code>EntityVault.get_or_create_placeholder</code>	<code>app/security/entity_vault.py:225</code>	6	<code>self, *, session_id: str, collection_id: str, entity_type: str, real_value: str, source: str</code> Stable placeholder for one entity within a scope.
<code>_walk</code>	<code>app/security/guardrails.py:119</code>	6	<code>value: Any, *, path: str, transform</code> Internal helper for the walk step.
<code>get_provider_compliance_metadata</code>	<code>app/security/llm_policy.py:113</code>	6	<code>provider: str None</code> Operator-configured ZDR/data-collection facts — a business/legal fact about Eurskem's actual vendor agreements, not derivable from code.
<code>ObjectStore.list_objects</code>	<code>app/storage/minio_client.py:215</code>	6	<code>self, prefix: str</code> List objects under a key prefix.
<code>_HorizonDocxBUILDER.configure_styles</code>	<code>app/tools/docx_proposal_rendering.py:920</code>	6	<code>self</code> Configure the styles.
<code>_HorizonDocxBUILDER.append_text_with_footnotes</code>	<code>app/tools/docx_proposal_rendering.py:1377</code>	6	<code>self, paragraph, text: str, style: InlineStyle</code> Append the text with footnotes.
<code>_HorizonDocxBUILDER.render_image</code>	<code>app/tools/docx_proposal_rendering.py:1712</code>	6	<code>self, image: Tag, caption: Tag None=None</code> Render the image.
<code>_HorizonDocxBUILDER.render_figure</code>	<code>app/tools/docx_proposal_rendering.py:1748</code>	6	<code>self, figure: Tag</code> Render the figure.
<code>_HorizonDocxBUILDER.add_evidence_annex</code>	<code>app/tools/docx_proposal_rendering.py:1816</code>	6	<code>self, evidence_qa: dict[str, Any], evidence_blockers: list[str]</code> Add the evidence annex.
<code>read_tables_from_xlsx</code>	<code>app/tools/excel_io.py:9</code>	6	<code>file_bytes: bytes</code> Return a dict of {sheet_name: rows} where rows is a list of lists.
<code>_safe_url_fetcher</code>		6	

Logic unit	Location	Branch score	Contract
	app/tools/ proposal_rendering.py:232		<code>url: str, *args: Any, **kwargs: Any</code> Allow embedded images and renderer-owned local assets only.
readiness_summary	app/workflow/ library_metadata.py:153	6	<code>report: WorkflowPreflightReport</code> Maps a structural preflight report to the Library's three-state readiness model (ready / ready_with_warnings / blocked), reusing the report's own plain-language issue text (message/suggestion) instead of inventing new copy or surfacing only a technical code.
finalize_run_result	app/workflow/ orchestration.py:126	6	<code>result: dict[str, Any], *, db: Any, run_id: str, session: str, record_rejection_reason: bool=False, services: dict[str, Any] None=None</code> Persist a terminal or re-paused outcome.
matching	scripts/ build_documentation_pdf.py:335	6	<code>model: Model, prefixes: Sequence[str], terms: Sequence[str]=()</code> No summary declared.
v05	scripts/ build_documentation_pdf.py:461	6	<code>m: Model</code> No summary declared.
node_summary	scripts/ build_workflow_qa_matrix.py:94	6	<code>document: dict[str, Any]</code> No summary declared.
main	scripts/ build_workflow_qa_matrix.py:231	6	 No summary declared.
CatalogEntry.worst_case_cost_usd	scripts/ openrouter_health_check.py:214	6	<code>self, probes: Iterable[str]</code> Compute the worst case cost usd.
fetch_model_providers	scripts/ openrouter_health_check.py:319	6	<code>client: httpx.AsyncClient, model_id: str, semaphore: asyncio.Semaphore</code> Which providers actually serve this model, via GET / models/{id}/endpoints.
_run_probe	scripts/ openrouter_health_check.py:492	6	<code>call: Any, *, timeout: float, retry_delay: float</code> Run one gateway call under a wall-clock cap, with a single retry on transient failures (429 / 5xx / dropped connection).
main	scripts/ verder_reingest_production.py:73	6	 Compute the main.
_write_pdf	scripts/ verify_knowledge_studio.py:348	6	<code>path: Path, lines: list[str] None=None</code> Write the pdf.
ingest	scripts/ verify_knowledge_studio.py:412	6	<code>api: Api, collection_id: str, paths: list[Path], *, parser: dict[str, Any] None=None, chunking: dict[str, Any] None=None, embedding: dict[str, Any] None=None, metadata: dict[str, Any] None=None, wait: bool=True</code> Ingest the result.

Logic unit	Location	Branch score	Contract
get_session_audit	app/api/audit.py:15	5	<pre>session_id: str, request: Request, run_id: str None=None, user: CurrentUser=Depends(require_consultant)</pre> Return the session audit.
operators	app/api/builder.py:112	5	<pre>user: CurrentUser=Depends(require_consultant)</pre> Which operators apply to which field type, and how to render each one.
_interrupt_values	app/api/builder.py:620	5	<pre>interrupt: Any</pre> Pull the real payloads out of LangGraph's Interrupt objects.
list_private_chat_workflows	app/api/chat_workflows.py:547	5	<pre>request: Request, user: CurrentUser=Depends(require_consultant), include_managed: bool=False</pre> No summary declared.
_humanize_workflow_name	app/api/chat_workspace.py:99	5	<pre>value: str</pre> No summary declared.
_weak_workflow_label	app/api/chat_workspace.py:105	5	<pre>item: WorkflowLabelInput</pre> No summary declared.
list_infra_allocations	app/api/cost_admin.py:225	5	<pre>request: Request, user: CurrentUser=Depends(require_admin)</pre> List the infra allocations.
set_infra_allocation	app/api/cost_admin.py:256	5	<pre>model: str, body: InfraAllocationRequest, request: Request, user: CurrentUser=Depends(require_admin)</pre> Set the infra allocation.
_allocated_infra_cost	app/api/cost_admin.py:300	5	<pre>entries: list[dict], allocations: dict[str, dict]</pre> Internal helper for the allocated infra cost step.
_cache_savings_usd	app/api/cost_admin.py:327	5	<pre>entry: dict</pre> Mirrors CostLedger.calculate()'s own cache multipliers — what THIS entry's cached tokens would have cost at full input price, minus what they actually cost.
_html_result	app/api/email_oauth.py:124	5	<pre>*, ok: bool, message: str</pre> A minimal, self-contained page for the OAuth popup window — no external assets (this is a plain HTTP response, not a Builder route), just enough to tell the user what happened and let the opener refresh its connection list before the window closes.
workflow_compare	app/api/eval.py:255	5	<pre>req: WorkflowCompareRequest, request: Request, user: CurrentUser=Depends(require_permission('eval:run'))</pre> Run the same golden set of real customer messages through the flagship triage workflow once per candidate model, and compare which one actually reaches the right business outcome — not a judged quality score, since there's a deterministic right answer (intent/complexity/route/review) to check against
_html_result	app/api/integration_oauth.py:118	5	<pre>*, ok: bool, message: str</pre> A minimal, self-contained page for the OAuth popup window.

Logic unit	Location	Branch score	Contract
browse_path	app/api/integration_oauth.py:327	5	<pre>connection_id: str, file_id: str, request: Request, user: CurrentUser=Depends(require_consultant)</pre> Breadcrumb chain for the file browser: walk `parent_id` up from `file_id` via repeated lookups, bounded so a cyclic/misbehaving provider response can never hang this request.
_configured	app/api/llm_providers.py:22	5	<pre>model: str, provider: str</pre> Internal helper for the configured step.
list_models	app/api/llm_providers.py:44	5	<pre>user: CurrentUser=Depends(require_consultant)</pre> Return public capability metadata without URLs or credentials.
_compact_catalog	app/api/node_types_chat.py:134	5	<pre>type_names: list[str]</pre> Same one-line-per-type shape as <code>_build_node_type_catalog()</code>, but for an explicit shortlist rather than the whole registry — what a node- or field-scoped question actually needs.
draft_instructions	app/api/node_types_chat.py:507	5	<pre>req: DraftInstructionsRequest, request: Request, user: CurrentUser=Depends(require_consultant)</pre> Draft the instructions.
duplicate_prompt_template	app/api/prompt_templates.py:60	5	<pre>template_id: str, request: Request, user: CurrentUser=Depends(require_consultant)</pre> No summary declared.
pause_run	app/api/runs.py:454	5	<pre>run_id: str, request: Request, user: CurrentUser=Depends(require_consultant)</pre> Ask a running workflow to pause at its next node boundary.
list_workflows	app/api/workflows.py:901	5	<pre></pre> Library view in the UI calls this to render the workflow collection.
load_golden_set	app/evaluation/golden_set.py:33	5	<pre>path: str Path</pre> Load a JSONL golden set.
recommend_model	app/evaluation/workflow_golden.py:303	5	<pre>comparisons: list[ModelComparisonResult]</pre> Highest pass rate wins; ties broken by lower cost, then lower latency.
verify_claim	app/evidence/claim_verification.py:59	5	<pre>content: str, *, source_name: str, llm: Any, web_search: Any</pre> Verify the claim.
parse_mcp_payload	app/evidence/retrieval.py:141	5	<pre>raw: Any</pre> Parse the mcp payload.
_apply_overlap	app/ingestion/chunker.py:237	5	<pre>chunks: list[str], encoding: tiktoken.Encoding, overlap_tokens: int</pre> Prepend the last `overlap_tokens` of chunk N-1 onto the start of chunk N.
ParentChildChunkingStrategy.chunk	app/ingestion/strategies.py:318	5	<pre>self, document: ExtractedDocument, *, config: dict[str, Any], chunk_id_prefix: str</pre> Chunk the result.

Logic unit	Location	Branch score	Contract
ContextualChunkingStrategy.chunk	app/ingestion/strategies.py:367	5	<pre>self, document: ExtractedDocument, *, config: dict[str, Any], chunk_id_prefix: str</pre> Chunk the result.
PdfPageVisionDescriber.describe	app/ingestion/vision.py:176	5	<pre>self, index: int, png: bytes, prompt: str</pre> Internal helper for the describe step.
_render_pages	app/ingestion/vision.py:252	5	<pre>path: Path, indexes: list[int], scale: float</pre> Render the given 0-based page indexes to PNG bytes.
_resolve_credentials	app/integrations/email/__init__.py:115	5	<pre>raw: dict[str, Any]</pre> Turn `*_env` indirections into their values.
load_dynamic_connections	app/integrations/email/connections_store.py:74	5	<pre>db: Any</pre> Every OAuth-established connection, with its tokens attached.
MicrosoftGraphAdapter.search	app/integrations/email/msgraph.py:141	5	<pre>self, connection: EmailConnection, criteria: EmailSearchCriteria</pre> Search the result.
_address	app/integrations/email/msgraph.py:356	5	<pre>raw: dict[str, Any] None</pre> Internal helper for the address step.
idempotency_key	app/integrations/email/service.py:425	5	<pre>*, connection_id: str, operation: str, draft: EmailDraft, scope: str None=None</pre> Stable fingerprint of "this exact email side effect".
_load_kek	app/integrations/email/token_vault.py:35	5	<pre></pre> Derive the vault's Key Encryption Key from settings, failing closed.
_resolve_credentials	app/integrations/files/__init__.py:106	5	<pre>raw: dict[str, Any]</pre> Turn `*_env` indirections into their values.
load_dynamic_connections	app/integrations/files/connections_store.py:77	5	<pre>db: Any</pre> Every OAuth-established connection, with its tokens attached.
_as_id_list	app/integrations/files/service.py:57	5	<pre>value: str list[str] None</pre> Normalize a config field that may hold one id, several, or none.
RetrievalRoutingProfileConfig.validate_routes	app/knowledge/models.py:318	5	<pre>self</pre> Validate the routes.
KnowledgeRepository.get_profile	app/knowledge/repository.py:335	5	<pre>self, owner_scope_id: str, profile_id: str, version: int None=None, expected_type: ProfileType None=None</pre> Return the profile.
KnowledgeRepository.list_rag_agents	app/knowledge/repository.py:600	5	<pre>self, owner_scope_id: str, search: str None=None</pre> List the rag agents.
KnowledgeService.create_rag_agent	app/knowledge/service.py:289	5	<pre>self, *, owner_scope_id: str, name: str, collection_id: str, retrieval_profile_id: str, generation_profile_id: str, routing_profile_id: str None=None, description: str=''</pre> Create the rag agent.

Logic unit	Location	Branch score	Contract
KnowledgeService.ensure_default_profiles	app/knowledge/service.py:347	5	<code>self, owner_scope_id: str</code> Ensure the default profiles.
_build_body	app/llm/openai_batch.py:35	5	<code>item: BatchRequestItem</code> Build one /v1/chat/completions request body for a batch JSONL row.
OpenRouterCatalogCache.get_models	app/llm/openrouter_catalog.py:172	5	<code>self, *, client: httpx.AsyncClient None=None, force_refresh: bool=False</code> Return the models.
RegistryLLMGateway._complete_structured_impl	app/llm/registry.py:1551	5	<code>self, *, model: str, stage: str None=None, **kwargs</code> Complete the structured impl.
RegistryLLMGateway._chat_with_tools_impl	app/llm/registry.py:1654	5	<code>self, *, model: str, **kwargs</code> Internal helper for the chat with tools impl step.
apply_schema	app/mcp/business_records/seed.py:112	5	<code>conn</code> Apply the schema.
_customer	app/mcp/d365_finance/handlers.py:90	5	<code>row: dict[str, Any]</code> Internal helper for the customer step.
find_quote	app/mcp/d365_finance/handlers.py:246	5	<code>backend: DynamicsBackend, arguments: dict[str, Any]</code> Find a quotation by quotation number or purchase-order number.
find_order_fulfilment_status	app/mcp/d365_finance/handlers.py:344	5	<code>backend: DynamicsBackend, arguments: dict[str, Any]</code> Ask why an order is where it is — the customer's wording is identical for a material shortage and a quality hold; this is what tells them apart.
find_products	app/mcp/d365_finance/handlers.py:666	5	<code>backend: DynamicsBackend, arguments: dict[str, Any]</code> Find the products.
call_tool	app/mcp/d365_finance/server.py:99	5	<code>name: str, arguments: dict[str, Any] None</code> Compute the call tool.
DataverseClient.whoami	app/mcp/dynamics/client.py:398	5	<code>self</code> Compute the whoami.
_activity	app/mcp/dynamics/handlers.py:230	5	<code>row: dict[str, Any]</code> Internal helper for the activity step.
find_quotation	app/mcp/dynamics/handlers.py:511	5	<code>backend: DynamicsBackend, arguments: dict[str, Any]</code> Find one quotation by its customer-facing number.
update_account_contact_details	app/mcp/dynamics/handlers.py:901	5	<code>backend: DynamicsBackend, arguments: dict[str, Any]</code> Update the account contact details.
escape_literal	app/mcp/dynamics/odata.py:62	5	<code>value: Any, *, field: str='value'</code> Escape a string for use inside an OData string literal.
any_of	app/mcp/dynamics/odata.py:101	5	<code>*filters: str</code> OR the given filters, parenthesised so precedence cannot surprise.

Logic unit	Location	Branch score	Contract
<code>all_of</code>	<code>app/mcp/dynamics/odata.py:111</code>	5	<code>*filters: str</code> Compute the all of.
<code>_order_by</code>	<code>app/mcp/dynamics/odata.py:190</code>	5	<code>clause: str</code> Internal helper for the order by step.
<code>classify_by_annotations</code>	<code>app/mcp/policy.py:123</code>	5	<code>annotations: Any</code> Read the server's own hints, if it declared any.
<code>MCPIntegrationService.discover_tools</code>	<code>app/mcp/service.py:131</code>	5	<code>self, server_id: str, *, refresh: bool=False</code> Ask the server what it can do.
<code>_schema_fields_summary</code>	<code>app/nodes/about_synthesis.py:153</code>	5	<code>schema: Type[Any] None</code> Internal helper for the schema fields summary step.
<code>AITaskAgent.run_structured</code>	<code>app/nodes/ai_task.py:672</code>	5	<code>self, cfg: AITaskConfig, llm: Any, system: str, user: str, fields: list[FieldSpec]</code> Run the structured.
<code>BoundedDeepResearchAgent.record_costs</code>	<code>app/nodes/bounded_deep_research_agent.py:301</code>	5	<code>self, state: dict[str, Any], dossiers: list[ResearchDossier]</code> Record the costs.
<code>DataTransformAgent.preflight_output_fields</code>	<code>app/nodes/data_transform.py:224</code>	5	<code>cls, config: dict[str, Any]</code> Compute the preflight output fields.
<code>_as_boolean</code>	<code>app/nodes/data_transform.py:424</code>	5	<code>value: Any</code> Internal helper for the as boolean step.
<code>DecisionAgent.preflight_static_output_values</code>	<code>app/nodes/decision.py:238</code>	5	<code>cls, config: dict[str, Any]</code> Compute the preflight static output values.
<code>_ScholarlyCandidateDiscovery.query_planning_guidance</code>	<code>app/nodes/evidence_agent.py:377</code>	5	<code>self, cfg: ScholarlyCandidateDiscoveryConfig</code> Search methodology text from the configured Scientific Agent Skill.
<code>_set_nested</code>	<code>app/nodes/human_in_loop.py:535</code>	5	<code>container: Any, parts: list[str], value: Any</code> Set the nested.
<code>_source_blocks</code>	<code>app/nodes/internal_project_evidence_retriever.py:505</code>	5	<code>text: str</code> Internal helper for the source blocks step.
<code>_empty_arguments</code>	<code>app/nodes/mcp_tool.py:463</code>	5	<code>arguments: dict[str, Any]</code> Argument names that were supplied but resolved to nothing.
<code>_allowed_official_url</code>	<code>app/nodes/prior_project_retriever.py:346</code>	5	<code>url: str, hosts: tuple[str, ...]</code> Internal helper for the allowed official url step.
<code>_strength</code>	<code>app/nodes/proposal_evidence_factor.py:210</code>	5	<code>score: int</code> Internal helper for the strength step.
<code>ProposalEvidenceFactoryAgent.bibtex</code>	<code>app/nodes/proposal_evidence_factor.py:1158</code>	5	<code>entry: CitationRegistryEntry</code> Internal helper for the bibtex step.
<code>assigns_output</code>	<code>app/nodes/python_snippet.py:298</code>	5	

Logic unit	Location	Branch score	Contract
			<code>code: str</code> Best-effort: does this snippet's top level ever touch `output` at all?
<code>_manifest_entry</code>	<code>app/nodes/registry.py:97</code>	5	<code>klass: Type[NodeType]</code> Internal helper for the manifest entry step.
<code>RouterAgent._route_by_conditions</code>	<code>app/nodes/router.py:501</code>	5	<code>self, cfg: RouterConfig, state: WorkflowState</code> Internal helper for the route by conditions step.
<code>_matched_summaries</code>	<code>app/nodes/router.py:766</code>	5	<code>trace: GroupTrace</code> Internal helper for the matched summaries step.
<code>StartAgent.preflight_output_fields</code>	<code>app/nodes/start.py:172</code>	5	<code>cls, config: dict[str, Any]</code> Compute the preflight output fields.
<code>_coordinates</code>	<code>app/nodes/structured_dataset_retriever.py:558</code>	5	<code>flat_index: int, sizes: list[int]</code> Internal helper for the coordinates step.
<code>TransformConfig.has_a_prompt_source</code>	<code>app/nodes/transform.py:113</code>	5	<code>self</code> Return whether a prompt source.
<code>_refresh_pricing_overrides_if_stale</code>	<code>app/observability/cost_ledger.py:142</code>	5	 Refresh the pricing overrides if stale.
<code>CostLedger.run_summary</code>	<code>app/observability/cost_ledger.py:339</code>	5	<code>self, run_id: str, session_id: str None=None</code> Run the summary.
<code>verify_claim_against_text</code>	<code>app/proposal_graph/evidence_verification.py:54</code>	5	<code>llm: Any, *, claim: str, source_text: str, model: str='claude-sonnet-4-5'</code> Return a typed verdict and fail closed when the quote is not traceable.
<code>ProposalWorkspaceStore.save_snapshot</code>	<code>app/proposal_graph/workspace_store.py:338</code>	5	<code>self, *, session_id: str, proposal_id: str, graph: ProposalGraph, created_by: str, reason: str</code> Save the snapshot.
<code>ScientificSkillCatalog.prompt_bundle</code>	<code>app/research/skills.py:389</code>	5	<code>self, selection: SkillSelection</code> Build a bounded prompt that represents every selected skill.
<code>reciprocal_rank_fusion</code>	<code>app/retrieval/fusion.py:9</code>	5	<code>result_sets: list[list[RetrievedChunk]], *, limit: int</code> Fuse independently-ranked result sets (dense+sparse, or per-query multi-query results) into one ranking via Reciprocal Rank Fusion.
<code>WeaviateClient.connect</code>	<code>app/retrieval/weaviate_client.py:157</code>	5	<code>self</code> Open the gRPC + REST connection.
<code>_replace_template_reference</code>	<code>app/runtime/autofix.py:108</code>	5	<code>text: str, node_id: str, old_remainder: str, new_remainder: str</code> Rewrite the field portion of a `{(node_id.old_remainder)}` (or `{(outputs.node_id.old_remainder)}`) reference to `new_remainder`, leaving everything else in the string untouched.
<code>_replace_template_node_id</code>	<code>app/runtime/autofix.py:142</code>	5	<code>text: str, old_node_id: str, new_node_id: str</code> Rewrite the node-id portion of a template reference, keeping whatever field path followed it.

Logic unit	Location	Branch score	Contract
<code>_node_by_id</code>	<code>app/runtime/autofix.py:170</code>	5	<code>raw: dict, node_id: str</code> Internal helper for the node by id step.
<code>_preserves_identity</code>	<code>app/runtime/autofix.py:755</code>	5	<code>original_yaml: str, candidate_yaml: str</code> Guards against a repaired workflow that is well-formed but is not actually a fix of the input — the observed failure mode was an LLM repair call returning a different, generic workflow (one whose own job was "repair workflow YAML") instead of the user's real workflow with its validation errors fixed.
<code>repair_with_llm</code>	<code>app/runtime/autofix.py:801</code>	5	<code>yaml_text: str, report: WorkflowPreflightReport, *, static_check: Callable[[str], Awaitable[WorkflowPreflightReport]], generate_yaml: Callable[[str, str None, str None], Awaitable[str]]</code> Feed remaining preflight errors back to the LLM up to MAX_LLM_REPAIR_ATTEMPTS times.
<code>_merge_mapping</code>	<code>app/runtime/domain_state.py:16</code>	5	<code>left: Any, right: Any</code> Safe default for simple dict-shaped domain state.
<code>DomainStateRegistry.register</code>	<code>app/runtime/domain_state.py:41</code>	5	<code>cls, namespace: str, reducer: DomainReducer</code> Register the result.
<code>merge_domain_state</code>	<code>app/runtime/domain_state.py:70</code>	5	<code>left: dict[str, Any] None, right: dict[str, Any] None</code> LangGraph reducer for state written by independent use-case packs.
<code>RunEventBus._subscribe_rediss</code>	<code>app/runtime/events.py:242</code>	5	<code>self, run_id: str, session_id: str None, *, after_event_id: int None</code> Subscribe the redis.
<code>RunEventBus.close</code>	<code>app/runtime/events.py:393</code>	5	<code>self</code> Stop Redis subscription pumps before the shared client is closed.
<code>_find_rejection</code>	<code>app/runtime/executor.py:29</code>	5	<code>state: dict[str, Any]</code> Find the rejection.
<code>FieldSpec.name_is_an_identifier</code>	<code>app/runtime/field_schema.py:122</code>	5	<code>cls, value: str</code> Compute the name is an identifier.
<code>_safe_model_name</code>	<code>app/runtime/field_schema.py:336</code>	5	<code>name: str</code> Internal helper for the safe model name step.
<code>parse_fields</code>	<code>app/runtime/field_schema.py:469</code>	5	<code>raw: Any</code> Coerce YAML/JSON rows into FieldSpec, with a clear error on bad shape.
<code>_find_rejection</code>	<code>app/runtime/hitl.py:34</code>	5	<code>state: dict</code> Any node output carrying decision == 'reject' means a HITL gate was rejected.
<code>_declared_value</code>	<code>app/runtime/logic_preflight.py:299</code>	5	<code>parsed: Any, name: str</code> Internal helper for the declared value step.
<code>_static_subprocess_child_spec</code>	<code>app/runtime/logic_preflight.py:496</code>	5	<code>config: dict[str, Any]</code> Load the referenced workflow's spec purely to read its shape (declared output contract) — None for anything

Logic unit	Location	Branch score	Contract
			not statically resolvable (missing name, bad charset, file not found, fails to parse).
FieldIndex.node_of	app/runtime/logic_preflight.py:708	5	<code>self, reference: str</code> Compute the node of.
_check_transform_agent	app/runtime/logic_preflight.py:1471	5	<code>node: NodeSpec, config: dict[str, Any], issue: Callable[..., None]</code> Same contract sanity as `_check_ai_task`, for a new-style TransformAgent node's `output_fields`.
_adjacency	app/runtime/preflight.py:685	5	<code>spec: WorkflowSpec</code> Internal helper for the adjacency step.
_multi_router_ids	app/runtime/preflight.py:1045	5	<code>spec: WorkflowSpec</code> RouterAgent nodes configured for Multi-Route (selection="multi") — their branches are OPTIONAL (any subset may fire), not EXCLUSIVE (exactly one fires), so they must never enter the mutual-exclusivity analysis below (that would either miss a real deadlock or misreport its cause).
_probe_integration_nodes	app/runtime/preflight.py:2572	5	<code>spec: WorkflowSpec, service: Any, report: WorkflowPreflightReport</code> Zero-token: checks the referenced connection exists on the service — no provider HTTP call, just a dict lookup.
_probe_image_generation_node	app/runtime/preflight.py:2645	5	<code>spec: WorkflowSpec, service: Any, report: WorkflowPreflightReport</code> Zero-token: OpenAllImageGenerationService.available() checks settings only — see app/tools/image_io.py.
_as_number	app/runtime/rules.py:509	5	<code>value: Any</code> Internal helper for the as number step.
_boolish	app/runtime/rules.py:576	5	<code>value: Any</code> Read a value as a boolean, or None if it isn't one.
WorkflowInputSpec.validate_file_options	app/runtime/schema.py:75	5	<code>self</code> Validate the file options.
_handle	app/runtime/snippet_daemon.py:172	5	<code>reader: asyncio.StreamReader, writer: asyncio.StreamWriter</code> Handle the result.
_accepts_none	app/runtime/templating.py:169	5	<code>annotation: Any</code> Whether a value of None satisfies this annotation.
extract_entities	app/security/entity_ner.py:63	5	<code>text: str</code> PERSON/ORG spans via spaCy.
EntityRegistry.sync_from_partners	app/security/entity_registry.py:121	5	<code>self, *, session_id: str, collection_id: str, graph: Any</code> Register every partner on a finalized ProposalGraph.
EntityVault.resolve_placeholders	app/security/entity_vault.py:331	5	<code>self, *, session_id: str, collection_id: str, placeholders: set[str]</code> Batch placeholder -> real value lookup, for detokenizing a response.
check_generated_output	app/security/guardrails.py:219	5	<code>value: Any</code> Remove credentials/private keys from model and tool output.
enforce_policy		5	<code>*, workspace_id: str, data_class: DataClass, gateway_class_name: str,</code>

Logic unit	Location	Branch score	Contract
	app/security/ llm_policy.py:229		<pre>model: str, capabilities: list[str] None=None, opa_client: OpaClient None=None</pre> Evaluates the routing policy for one candidate (provider, model) pair.
authenticate_local_user	app/security/users.py:75	5	<pre>db: Any, *, username: str, password: str</pre> Authenticate the local user.
_set_paragraph_borders	app/tools/ docx_proposal_rendering.py:276	5	<pre>paragraph, *, left: tuple[str, int, int] None=None, bottom: tuple[str, int, int] None=None</pre> Set the paragraph borders.
_set_cell_margins	app/tools/ docx_proposal_rendering.py:318	5	<pre>cell, margins: dict[str, int] None=None</pre> Set the cell margins.
_column_widths	app/tools/ docx_proposal_rendering.py:450	5	<pre>rows: list[list[str]], column_count: int, *, total_width: int=TABLE_WIDTH_DXA</pre> Internal helper for the column widths step.
_image_request_kwargs	app/tools/ image_io.py:240	5	<pre>*, model: str, prompt: str, size: str, quality: str, output_format: str</pre> Internal helper for the image request kwargs step.
WebSearchService.available_providers	app/tools/web_io.py:88	5	<pre>self</pre> Compute the available providers.
WebSearchService._select_provider	app/tools/web_io.py:193	5	<pre>self, requested: SearchProvider</pre> Select the provider.
WorkflowBuilderStore.save_draft	app/workflow/ builder_store.py:133	5	<pre>self, name: str, yaml_text: str, *, canvas: dict[str, Any] None=None</pre> Save the draft.
_entry_text	app/workflow/ capability_selection.py:74	5	<pre>entry: dict[str, Any]</pre> Internal helper for the entry text step.
_terminal_ids	app/workflow/ chat_output_compatibility.py:31	5	<pre>spec: WorkflowSpec</pre> No summary declared.
extract_workflow_file_text	app/workflow/ file_inputs.py:339	5	<pre>raw_ref: WorkflowFileRef dict[str, Any], *, session_id: str, object_store: Any None, max_chars: int=1000000</pre> Extract one scoped upload into plain text for the HITL editor.
_terminal_node_types	app/workflow/ library_metadata.py:48	5	<pre>spec: WorkflowSpec</pre> Internal helper for the terminal node types step.
infer_output_types	app/workflow/ library_metadata.py:62	5	<pre>spec: WorkflowSpec</pre> Compute the infer output types.
_reconcile_subprocess_callback	app/workflow/ orchestration.py:409	5	<pre>db: Any, run_id: str, session: str, services: dict[str, Any] None=*, status: str, output: Any, node_outputs: dict[str, Any], error: str None</pre> Best-effort: if this run_id is a Subprocess node's child, deliver its result to the waiting parent (app.workflow.subprocess_callback).
record_node_completed		5	

Logic unit	Location	Branch score	Contract
	app/workflow/run_history.py:464		<code>db, *, run_id: str, session_id: str, node_id: str, output: dict[str, Any], model_selections: list[dict[str, Any]] None=None, ended_at: float, duration_s: float</code> Persist a full node output as soon as that node completes.
list_runs	app/workflow/run_history.py:1205	5	<code>db, session_id: str, limit: int=50, *, include_conversation_only: bool=False</code> Return light run summaries, newest first, including active runs.
_select_result	app/workflow/subprocess_callback.py:52	5	<code>launch: dict[str, Any], *, output: Any, node_outputs: dict[str, Any]</code> Project the child's result the way the SubprocessConfig author asked for (result_from, recorded on the launch doc at reserve time — see app/nodes/subprocess.py's SubprocessConfig): workflow_output the child's own declared 'output:' contract (what every other caller of that workflow would get back) n
_group_manifests_by_owner_and_collection	scripts/backfill_knowledge_resources.py:67	5	<code>db</code> Group the manifests by owner and collection.
table	scripts/build_documentation_pdf.py:171	5	<code>headers: Sequence[str], rows: Iterable[Sequence[Any]], widths: Sequence[str]=()</code> No summary declared.
v22	scripts/build_documentation_pdf.py:659	5	<code>m: Model</code> No summary declared.
main	scripts/build_documentation_pdf.py:771	5	No summary declared.
input_summary	scripts/build_workflow_qa_matrix.py:76	5	<code>document: dict[str, Any]</code> No summary declared.
_function_doc	scripts/generate_docstrings.py:448	5	<code>node, is_property: bool, class_name: str None</code> No summary declared.
_render_docstring	scripts/generate_docstrings.py:496	5	<code>lines: list[str], indent: str</code> No summary declared.
_classify	scripts/openrouter_health_check.py:441	5	<code>error: BaseException</code> Classify the result.
parse_env	scripts/production_preflight.py:31	5	<code>path: Path</code> Parse the env.
_execution_kind	app/api/builder.py:1384	4	<code>type_name: str</code> Internal helper for the execution kind step.
_discovery_candidate_url	app/api/candidates.py:97	4	<code>candidate: dict[str, Any]</code> Internal helper for the discovery candidate url step.

Logic unit	Location	Branch score	Contract
build_deep_research_chat_workflow	app/api/chat_workflows.py:267	4	<pre>rag_agent_id: str None=None</pre> Add a saved RAG Agent lane without exposing collection/index filters.
_private_yaml	app/api/chat_workflows.py:333	4	<pre>yaml_text: str</pre> Force private lifecycle metadata without changing executable semantics.
_save	app/api/chat_workflows.py:421	4	<pre>request: Request, user: CurrentUser, *, slug: str, display_name: str, yaml_text: str, source: Literal['generated', 'imported', 'existing'], source_workflow_name: str None=None, managed: bool=False, adapter_key: str None=None, adapter_fingerprint: str None=None</pre> No summary declared.
ensure_deep_research_chat_workflow	app/api/chat_workflows.py:562	4	<pre>request: Request, user: CurrentUser=Depends(require_consultant), rag_agent_id: str None=None</pre> Return the owner's vetted Deep Research workflow for one Knowledge binding.
get_builder_chat_execution_adapter	app/api/chat_workflows.py:602	4	<pre>workflow_name: str, user: CurrentUser=Depends(require_consultant)</pre> Build a universal, execution-only Chat wrapper around a saved workflow.
generate_private_chat_workflow	app/api/chat_workflows.py:688	4	<pre>body: GenerateChatWorkflowRequest, request: Request, user: CurrentUser=Depends(require_consultant)</pre> No summary declared.
_clean_workflow_label	app/api/chat_workspace.py:121	4	<pre>value: str, fallback: str</pre> No summary declared.
session_cost	app/api/cost.py:34	4	<pre>session_id: str, request: Request, user: CurrentUser=Depends(get_current_user)</pre> Compute the session cost.
update_connection	app/api/email_oauth.py:274	4	<pre>connection_id: str, body: AllowSendUpdate, request: Request, user: CurrentUser=Depends(require_consultant)</pre> The only lever to actually let a connection send — every connection (OAuth-established or static) starts refused; this is the explicit second step, same reasoning as every other write-capable connection in this platform (MCP, F&O).
_probe	app/api/health.py:35	4	<pre>name: str, probe: Probe None</pre> Probe the result.
browse_files	app/api/integration_oauth.py:295	4	<pre>connection_id: str, request: Request, folder_id: str None=None, query: str='', page_size: int=25, page_token: str None=None, user: CurrentUser=Depends(require_consultant)</pre> Live folder/search browsing for the node config panel's file picker — not workflow execution, just a thin proxy through the service for whichever operation the presence of `query` selects.
download_file		4	

Logic unit	Location	Branch score	Contract
	app/api/integration_oauth.py:359		connection_id: str, file_id: str, request: Request, user: CurrentUser=Depends(require_consultant) Streams one file's real bytes straight to the browser — a plain navigation (an ` <a href="">` click, not a fetch), authenticated by the same HttpOnly cookie the OAuth popup flow relies on.
probe_model	app/api/llm_providers.py:112	4	model: str, user: CurrentUser=Depends(require_admin) Probe an enabled local endpoint; never return its private address.
_store	app/api/proposals.py:60	4	request: Request, *, require_objects: bool=False Internal helper for the store step.
review_run	app/api/proposals.py:129	4	run_id: str, request: Request, user: CurrentUser=Depends(require_consultant) Compute the review run.
_llm	app/api/retrieval.py:52	4	request: Request, scope: str, node_id: str, collection_id: str None=None Internal helper for the llm step.
_summarize_value	app/api/run_chat.py:59	4	value: Any, limit: int=800 Summarize the value.
_strip_code_fence	app/api/workflow_generation.py:576	4	text: str Internal helper for the strip code fence step.
_scope	app/api/workflows.py:127	4	user: CurrentUser, body_session: str None The value a run is scoped to for storage + retrieval.
source_errors_from_payload	app/evidence/retrieval.py:185	4	raw: Any Per-source failures from a multi-source search payload (e.g.
IngestionCoordinator.run_safely	app/ingestion/coordinator.py:93	4	self, job: IngestionJob Run the safely.
FixedTokenChunkingStrategy.chunk	app/ingestion/strategies.py:250	4	self, document: ExtractedDocument, *, config: dict[str, Any], chunk_id_prefix: str Chunk the result.
SentenceWindowChunkingStrategy.chunk	app/ingestion/strategies.py:391	4	self, document: ExtractedDocument, *, config: dict[str, Any], chunk_id_prefix: str Chunk the result.
PdfPageVisionDescriber.available	app/ingestion/vision.py:90	4	self Compute the available.
_parse_address	app/integrations/email/gmail.py:390	4	value: str None Parse the address.
_parse_addresses	app/integrations/email/gmail.py:407	4	value: str None Parse the addresses.
MicrosoftGraphAdapter.reply	app/integrations/email/msggraph.py:257	4	self, connection: EmailConnection, draft: EmailDraft

Logic unit	Location	Branch score	Contract
			Use Graph's own reply endpoint so it quotes and threads correctly.
<code>_addresses</code>	<code>app/integrations/email/msgraph.py:374</code>	4	<code>raw: list[dict[str, Any]] None</code> Internal helper for the addresses step.
<code>OneDriveProvider.list_folder</code>	<code>app/integrations/files/onedrive.py:120</code>	4	<code>self, connection: IntegrationConnection, *, folder_id: str None, page_size: int, page_token: str None</code> List the folder.
<code>OneDriveProvider.search_files</code>	<code>app/integrations/files/onedrive.py:152</code>	4	<code>self, connection: IntegrationConnection, *, query: str, folder_id: str None, page_size: int, page_token: str None</code> Search the files.
<code>ExternalOperationLedger.reserve</code>	<code>app/integrations/operations.py:85</code>	4	<code>self, key: str, record: dict[str, Any]</code> Claim the key.
<code>KnowledgeRepository.index_document_stats</code>	<code>app/knowledge/repository.py:488</code>	4	<code>self, owner_scope_id: str, index_id: str</code> Return "(document_count, chunk_count)" from authoritative mappings.
<code>_build_params</code>	<code>app/llm/anthropic_batch.py:40</code>	4	<code>item: BatchRequestItem</code> Build one MessageCreateParamsNonStreaming-shaped dict for a batch row.
<code>_mark_last_block_cacheable</code>	<code>app/llm/anthropic_gw.py:78</code>	4	<code>messages: list[dict]</code> Add a cache breakpoint to the last content block of the last message.
<code>_reasoning_content</code>	<code>app/llm/local_openai_gw.py:77</code>	4	<code>message: Any</code> Internal helper for the reasoning content step.
<code>LocalOpenAICompatibleGateway.complete_structured</code>	<code>app/llm/local_openai_gw.py:229</code>	4	<code>self, *, model: str, system: str, user: str, response_model: Type[T], temperature: float=0.0, max_tokens: int=4096, reasoning_effort: str None=None</code> Complete the structured.
<code>LocalOpenAICompatibleGateway.probe_details</code>	<code>app/llm/local_openai_gw.py:405</code>	4	<code>self</code> Probe the details.
<code>_request_text</code>	<code>app/llm/model_router.py:462</code>	4	<code>kwargs: Mapping[str, Any], *, node_type: str None</code> Internal helper for the request text step.
<code>_selection_reason</code>	<code>app/llm/model_router.py:610</code>	4	<code>profile: ModelProfile, *, task: TaskKind, complexity: Complexity, priority: AccuracyPriority, source: Literal['policy', 'evaluation'], estimated_cost: float, ceiling: float None</code> Internal helper for the selection reason step.
<code>_cached_tokens_from_usage</code>	<code>app/llm/openai_gw.py:74</code>	4	<code>usage: Any</code> Read OpenAI's automatic-prompt-cache hit count off a usage payload.
<code>OpenRouterCatalogCache.search</code>	<code>app/llm/openrouter_catalog.py:191</code>	4	<code>self, query: str None=None, *, limit: int=50, client: httpx.AsyncClient None=None</code> Search the result.

Logic unit	Location	Branch score	Contract
build_auto_router_plugin	app/llm/openrouter_ranking.py:56	4	<pre>*, accuracy_priority: str='balanced', max_estimated_cost_usd: float None=None, quality_scores: Mapping[str, float] None=None, faithfulness_scores: Mapping[str, float] None=None</pre> Builds one `plugins[]` entry for OpenRouter's auto-router from Eurskem's ModelRoutingPolicy fields.
_anthropic_error_type	app/llm/registry.py:218	4	<pre>exc: BaseException</pre> Anthropic streams some errors (e.g.
_is_model_unavailable_error	app/llm/registry.py:303	4	<pre>exc: BaseException</pre> Return whether this candidate cannot serve the requested model.
configured_local_model_probes	app/llm/registry.py:523	4	<pre></pre> Return health probes for enabled local providers without exposing URLs.
RegistryLLMGateway.tokenize_text	app/llm/registry.py:793	4	<pre>self, text: str None</pre> Tokenize the text.
RegistryLLMGateway.auto_candidates	app/llm/registry.py:1206	4	<pre>self</pre> Concrete models "auto" may resolve to, best first.
RegistryLLMGateway.delay_for	app/llm/registry.py:1212	4	<pre>self, attempt: int, exc: BaseException</pre> Internal helper for the delay for step.
RegistryLLMGateway.chat_with_tools	app/llm/registry.py:1625	4	<pre>self, *, model: str, **kwargs</pre> Public entry point — tokenizes system + every message's content/ tool_calls[].arguments, delegates, then detokenizes text/ reasoning_content/ tool_calls[].arguments on the response.
query_readonly	app/mcp/business_records/handlers.py:244	4	<pre>conn: MySQLConnectionAbstract, arguments: dict[str, Any]</pre> Deliberately ignores `conn` — the shared module-level connection (server.py's backend()) authenticates as the full-privilege app user.
classify_sql_verb	app/mcp/business_records/sql_guard.py:63	4	<pre>sql: str</pre> "read" or "write", from the statement's first keyword alone.
MCPClient.stop	app/mcp/client.py:309	4	<pre>self</pre> Stop the result.
find_customer	app/mcp/d365_finance/handlers.py:108	4	<pre>backend: DynamicsBackend, arguments: dict[str, Any]</pre> Search customers by name — a customer writing 'Meridian' should find 'Meridian Process Systems', same contains-or-exact convention as the CRM connector's find_account.
_order_filter	app/mcp/d365_finance/handlers.py:281	4	<pre>arguments: dict[str, Any]</pre> Customers quote their own purchase-order number at least as often as our sales-order number, and a workflow may only have one ambiguous reference field to give us — search whichever value(s) were supplied against *both* columns, since which kind of reference it is often is not knowable in advance.
DataverseClient.get	app/mcp/dynamics/client.py:455	4	

Logic unit	Location	Branch score	Contract
			<pre>self, entity_set: str, record_id: str, *, select: list[str]</pre> Return the result.
<code>_safe_error_text</code>	<code>app/mcp/dynamics/client.py:526</code>	4	<pre>response: httpx.Response</pre> Extract Dataverse's error message without echoing an entire payload.
<code>_project</code>	<code>app/mcp/dynamics/client.py:709</code>	4	<pre>row: dict[str, Any], select: list[str]</pre> Mimic <code>`\$select`</code>: a fixture row returns only the requested columns.
<code>_account</code>	<code>app/mcp/dynamics/handlers.py:106</code>	4	<pre>row: dict[str, Any]</pre> Internal helper for the account step.
<code>_text</code>	<code>app/mcp/dynamics/handlers.py:794</code>	4	<pre>arguments: dict[str, Any], key: str, limit: int</pre> Read an optional write field, bounded.
<code>create_lead</code>	<code>app/mcp/dynamics/handlers.py:817</code>	4	<pre>backend: DynamicsBackend, arguments: dict[str, Any]</pre> Create the lead.
<code>create_followup_activity</code>	<code>app/mcp/dynamics/handlers.py:853</code>	4	<pre>backend: DynamicsBackend, arguments: dict[str, Any]</pre> Create the followup activity.
<code>build_backend</code>	<code>app/mcp/dynamics/server.py:72</code>	4	<pre></pre> Choose a backend from the environment.
<code>_normalize_openaire_legacy_json</code>	<code>app/mcp/paper_search_server.py:71</code>	4	<pre>payload: Any</pre> Guard against <code>"response": null</code> / <code>"results": null</code> in the OpenAIRE legacy search payload, which otherwise crashes the caller's <code>`data.get('response', {}).get('results', {}).get('result', [])`</code> chain (a present <code>`None`</code> value bypasses <code>`dict.get`</code>'s default).
<code>patch_openaire_query_param</code>	<code>app/mcp/paper_search_server.py:123</code>	4	<pre>module_name: str='paper_search_mcp.academic_platforms.openaire'</pre> Rename the legacy fallback's <code>`query`</code> param to the supported <code>`keywords`</code>, and normalize its response so a null <code>`response`/`results`</code> doesn't raise in the caller's <code>`get()`</code> chain.
<code>classify_tool</code>	<code>app/mcp/policy.py:151</code>	4	<pre>tool_name: str, *, connection: MCPServerConnection None=None, annotations: Any=None</pre> The platform's classification of one tool.
<code>PolicyDecision.raise_if_denied</code>	<code>app/mcp/policy.py:197</code>	4	<pre>self</pre> Raise the if denied.
<code>MCPServerConnection.permits_tool</code>	<code>app/mcp/registry.py:215</code>	4	<pre>self, tool_name: str</pre> Compute the permits tool.
<code>_collect_text</code>	<code>app/mcp/results.py:202</code>	4	<pre>content: Any</pre> Internal helper for the collect text step.
<code>MCPIntegrationService._audit</code>	<code>app/mcp/service.py:551</code>	4	<pre>self, audit_db: Any, *, run_id: str, node_id: str, session_id: str, server_id: str, tool_name: str, operation: str, status: str, started: float, detail: dict[str, Any]</pre> Record the invocation (§27).

Logic unit	Location	Branch score	Contract
<code>_confidence_fields</code>	<code>app/nodes/ai_task.py:368</code>	4	<code>fields: list[FieldSpec]</code> Internal helper for the confidence fields step.
<code>effective_fields</code>	<code>app/nodes/ai_task.py:409</code>	4	<code>config: dict[str, Any]</code> The schema the model is actually held to, generated fields included.
<code>AITaskAgent._examples_block</code>	<code>app/nodes/ai_task.py:606</code>	4	<code>cfg: AITaskConfig</code> Internal helper for the examples block step.
<code>_detected_language</code>	<code>app/nodes/ai_task.py:828</code>	4	<code>parsed: dict[str, Any]</code> Surface a language field from the author's schema as a runtime field.
<code>NodeType.definition</code>	<code>app/nodes/base.py:73</code>	4	<code>cls, config: dict[str, Any] None=None</code> Return the complete standardized contract for this node type.
<code>BoundedDeepResearchConfig.coerce_briefs</code>	<code>app/nodes/bounded_deep_research_agent.py:72</code>	4	<code>cls, value: Any</code> Internal helper for the coerce briefs step.
<code>ConceptFreezeAgent.run</code>	<code>app/nodes/concept_freeze.py:74</code>	4	<code>self, state: dict[str, Any], resolved_config: dict[str, Any]</code> Run the result.
<code>DataTransformConfig.targets_are_unique</code>	<code>app/nodes/data_transform.py:134</code>	4	<code>self</code> Compute the targets are unique.
<code>_is_blank</code>	<code>app/nodes/data_transform.py:281</code>	4	<code>value: Any</code> Return whether blank.
<code>DOCXProposalRenderer.render_docx_bytes</code>	<code>app/nodes/docx_renderer.py:187</code>	4	<code>self, sections: dict[str, str], template_name: str, client: str, title: str</code> Render the docx bytes.
<code>_draft</code>	<code>app/nodes/email_integration.py:379</code>	4	<code>cfg: EmailNodeConfig</code> Draft the result.
<code>_resolve_path</code>	<code>app/nodes/human_in_loop.py:329</code>	4	<code>path: str, state: dict</code> Resolve the path.
<code>_resolve_source_name</code>	<code>app/nodes/internal_project_evidence_retriever.py:561</code>	4	<code>requested: str, blocks: dict[str, str]</code> Resolve the source name.
<code>MCPToolAgent._failure</code>	<code>app/nodes/mcp_tool.py:408</code>	4	<code>self, cfg: MCPToolConfig, error: Exception</code> Internal helper for the failure step.
<code>RAGConfig.reject_ambiguous_saved_agent_config</code>	<code>app/nodes/rag.py:71</code>	4	<code>self</code> Compute the reject ambiguous saved agent config.
<code>_html_pages</code>	<code>app/nodes/research_source_acquirer.py:478</code>	4	<code>raw: bytes, *, chars_per_page: int=9000</code> Internal helper for the html pages step.
<code>_branch_key</code>	<code>app/nodes/router.py:752</code>	4	<code>value: Any</code> Normalise a routed value into a branch key.
<code>_indexed_value</code>	<code>app/nodes/structured_dataset_retriever.py:599</code>	4	<code>values: Any, index: int</code> Internal helper for the indexed value step.

Logic unit	Location	Branch score	Contract
TransformAgent.new_style_user_prompt	app/nodes/transform.py:681	4	<code>self, cfg: TransformConfig</code> Internal helper for the new style user prompt step.
WebSearchAgent.run	app/nodes/web_search_agent.py:108	4	<code>self, state: dict[str, Any], resolved_config: dict[str, Any]</code> Run the result.
record_llm_usage	app/observability/metrics.py:230	4	<code>model: str, prompt_tokens: int, completion_tokens: int, ok: bool</code> Record token usage and call outcome.
configure_tracing	app/observability/tracing.py:19	4	<code>app</code> Configure the tracing.
ConceptAlternativeSet.one_of_each_posture	app/proposal_graph/concepts.py:69	4	<code>self</code> Compute the one of each posture.
proposal_graph_from_state	app/proposal_graph/state.py:10	4	<code>state: dict[str, Any]</code> Compute the proposal graph from state.
ProposalWorkspaceStore.source_text	app/proposal_graph/workspace_store.py:295	4	<code>self, *, session_id: str, proposal_id: str, source_id: str, version_id: str None=None</code> Compute the source text.
_build_relevant_context	app/rag/service.py:57	4	<code>chunks: list[Any]</code> One entry per retrieved chunk actually used to ground the answer.
_compress_one	app/retrieval/compressor.py:47	4	<code>query: str, chunk: RetrievedChunk, llm: LLMGateway, model: str</code> Compress a single chunk.
_validate_group	app/retrieval/filters.py:142	4	<code>group: MetadataFilterGroup, schema: dict[str, Any]</code> Validate the group.
hybrid_search	app/retrieval/hybrid_search.py:44	4	<code>client: Any, embedder: Embedder, collection_name: str, query: str, filters: RetrievalFilters, top_k: int, alpha: float</code> Hybrid BM25 + vector search with metadata pre-filter.
fetch_chunks_by_id	app/retrieval/strategies.py:197	4	<code>*, client: Any, collection_name: str, chunk_ids: list[str], filters: RetrievalFilters, index_id: str None=None</code> Fetch the chunks by id.
_accepts_plain_string	app/runtime/autofix.py:597	4	<code>annotation: Any</code> Whether a pydantic field annotation permits an ordinary `str` — true for `str` itself and for `str   None`/`Optional[str]`.
RedisLease.keep_alive	app/runtime/coordination.py:94	4	<code>self</code> Return only when the lease is lost; cancellation stops the loop.
RunEventBus.decode_redis_event	app/runtime/events.py:288	4	<code>fields: dict[Any, Any]</code> Decode the redis event.
_reaching_branches	app/runtime/logic_preflight.py:969	4	<code>edge: Any, node_id: str, forward: dict[str, set[str]]</code> Which of this router's branches can reach `node_id`.

Logic unit	Location	Branch score	Contract
<code>_reachable_from</code>	<code>app/runtime/logic_preflight.py:980</code>	4	<code>start: str, forward: dict[str, set[str]]</code> Internal helper for the reachable from step.
<code>_edge_targets</code>	<code>app/runtime/logic_preflight.py:1001</code>	4	<code>edge: Any</code> Internal helper for the edge targets step.
<code>_check_decision</code>	<code>app/runtime/logic_preflight.py:1019</code>	4	<code>node: NodeSpec, config: dict[str, Any], index: FieldIndex, nodes: dict[str, NodeSpec], issue: Callable[..., None], guaranteed_before: Callable[[str, str], bool] None</code> Check the decision.
<code>_check_external_action</code>	<code>app/runtime/logic_preflight.py:1606</code>	4	<code>node: NodeSpec, issue: Callable[..., None], reviewed_before: Callable[[str], bool]</code> A write or external_action call with nothing between it and the model.
<code>unreachable_branch_targets</code>	<code>app/runtime/logic_preflight.py:2100</code>	4	<code>spec: WorkflowSpec, reachable: Iterable[str]</code> Router branch targets no run can reach.
<code>interrupt_payload</code>	<code>app/runtime/node_events.py:48</code>	4	<code>exc: BaseException</code> Recover the actual value passed to <code>“langgraph.types.interrupt(…)”</code> .
<code>_add_check</code>	<code>app/runtime/preflight.py:221</code>	4	<code>report: WorkflowPreflightReport, name: str, before: int, detail: str</code> Add the check.
<code>_required_services_for_node</code>	<code>app/runtime/preflight.py:470</code>	4	<code>node: NodeSpec</code> Delegates to the node type's own <code>required_services()</code> override (see <code>app/nodes/base.py</code>) — every node-type-specific service dependency lives beside the node that declares it, not in a central if/elif chain here.
<code>_edge_targets</code>	<code>app/runtime/preflight.py:667</code>	4	<code>edge: EdgeSpec</code> Internal helper for the edge targets step.
<code>_annotation_permits_none</code>	<code>app/runtime/preflight.py:1622</code>	4	<code>annotation: Any</code> True if a declared field type allows None (<code>“X None”</code> / <code>Optional[X]</code>).
<code>_compile_dry_run</code>	<code>app/runtime/preflight.py:2095</code>	4	<code>spec: WorkflowSpec, services: dict[str, Any] None, report: WorkflowPreflightReport</code> Internal helper for the compile dry run step.
<code>_probe_web_search_nodes</code>	<code>app/runtime/preflight.py:2609</code>	4	<code>spec: WorkflowSpec, service: Any, report: WorkflowPreflightReport</code> Zero-token: <code>WebSearchService.resolve_provider</code> does a settings check only, no HTTP request — see <code>app/tools/web_io.py</code> .
<code>_probe_kimi_vision_nodes</code>	<code>app/runtime/preflight.py:2682</code>	4	<code>spec: WorkflowSpec, service: Any, report: WorkflowPreflightReport</code> Zero-token: <code>KimiVisionService.available()</code> checks settings only — see <code>app/tools/vision_io.py</code> .
<code>ConditionGroup.group_is_usable</code>	<code>app/runtime/rules.py:300</code>	4	<code>self</code> Group the is usable.
<code>_leaf_traces</code>	<code>app/runtime/rules.py:488</code>	4	

Logic unit	Location	Branch score	Contract
			<code>node: GroupTrace ConditionTrace None</code> Internal helper for the leaf traces step.
<code>_is_empty</code>	<code>app/runtime/rules.py:530</code>	4	<code>value: Any</code> Return whether empty.
<code>collect_condition_fields</code>	<code>app/runtime/rules.py:863</code>	4	<code>group: ConditionGroup None</code> Every field path a group reads — used by preflight to check references.
<code>collect_conditions</code>	<code>app/runtime/rules.py:876</code>	4	<code>group: ConditionGroup None</code> Every leaf condition in a group, for type checking in preflight.
<code>NodeSpec.data_protection_mode_must_be_valid</code>	<code>app/runtime/schema.py:308</code>	4	<code>self</code> Compute the data protection mode must be valid.
<code>acquire</code>	<code>app/runtime/sleep_guard.py:46</code>	4	<code></code> Block idle sleep for the duration of one workflow run.
<code>release</code>	<code>app/runtime/sleep_guard.py:66</code>	4	<code></code> Release this run's hold; sleep is allowed again once none remain.
<code>_clamped</code>	<code>app/runtime/snippet_daemon.py:56</code>	4	<code>request: dict[str, Any]</code> Never trust the client's own limits past the sidecar's own ceiling — a compromised or buggy app process must not be able to ask this container to run something for an hour.
<code>_execute</code>	<code>app/runtime/snippet_daemon.py:102</code>	4	<code>request: SnippetRequest</code> Execute the result.
<code>read_message</code>	<code>app/runtime/snippet_protocol.py:21</code>	4	<code>reader: asyncio.StreamReader, limit: int=MAX_MESSAGE_BYTES</code> Read one EOF-terminated message, accumulating across as many socket reads as it takes.
<code>_element_annotation</code>	<code>app/runtime/templating.py:178</code>	4	<code>annotation: Any, container: type, position: int</code> The declared type of a dict value / list item, or <code>_UNKNOWN</code> .
<code>summarize_payload</code>	<code>app/security/audit.py:59</code>	4	<code>data: dict[str, Any] None</code> Record SHAPE, not CONTENT.
<code>_load_pipeline</code>	<code>app/security/entity_ner.py:42</code>	4	<code></code> Load the pipeline.
<code>_load_kek</code>	<code>app/security/entity_vault.py:47</code>	4	<code></code> Derive the vault's Key Encryption Key from settings, failing closed.
<code>EntityVault.list_scope_entities</code>	<code>app/security/entity_vault.py:296</code>	4	<code>self, *, session_id: str, collection_id: str</code> Decrypted known entities for a scope — feeds the registry-first literal-match detection pass and the manual-registration listing API.
<code>RequestContextMiddleware.dispatch</code>	<code>app/security/middleware.py:27</code>	4	<code>self, request: Request, call_next: Callable</code> Dispatch the result.
<code>RequestSizeLimitMiddleware.dispatch</code>	<code>app/security/middleware.py:89</code>	4	

Logic unit	Location	Branch score	Contract
			<pre>self, request: Request, call_next: Callable</pre> Dispatch the result.
<code>_rate_limit_subject</code>	<code>app/security/middleware.py:209</code>	4	<pre>request: Request</pre> Internal helper for the rate limit subject step.
<code>ObjectStore.ensure_bucket</code>	<code>app/storage/minio_client.py:116</code>	4	<pre>self, bucket: str None=None</pre> Create the bucket if it doesn't exist.
<code>ObjectStore.put_file</code>	<code>app/storage/minio_client.py:134</code>	4	<pre>self, local_path: Path, key: str None=None, content_type: str None=None, extra_metadata: dict[str, str] None=None</pre> Upload a local file.
<code>_pairs_for_audit</code>	<code>app/tools/database_lookup.py:191</code>	4	<pre>pairs: list[tuple[str, str]]</pre> Internal helper for the pairs for audit step.
<code>_set_table_borders</code>	<code>app/tools/docx_proposal_rendering.py:370</code>	4	<pre>table, *, style: str='single', color: str=BORDER, size: int=4</pre> Set the table borders.
<code>_HorizonDocxBuilder.configure_header_footer</code>	<code>app/tools/docx_proposal_rendering.py:1027</code>	4	<pre>self, section</pre> Configure the header footer.
<code>_HorizonDocxBuilder.render_list</code>	<code>app/tools/docx_proposal_rendering.py:1485</code>	4	<pre>self, list_tag: Tag, *, list_level: int</pre> Render the list.
<code>_HorizonDocxBuilder.placeholder_text</code>	<code>app/tools/docx_proposal_rendering.py:1631</code>	4	<pre>self, figure: Tag</pre> Internal helper for the placeholder text step.
<code>build_proposal_pptx</code>	<code>app/tools/powerpoint_io.py:9</code>	4	<pre>sections: dict[str, str], title: str, client_name: str</pre> Title slide + one content slide per section.
<code>WorkflowBuilderStore.read_draft</code>	<code>app/workflow/builder_store.py:172</code>	4	<pre>self, name: str</pre> Read the draft.
<code>ChatConversationStore.get_or_create</code>	<code>app/workflow/chat_conversation_store.py:87</code>	4	<pre>self, owner_scope_id: str, *, workflow_source: ChatWorkflowSource, workflow_id: str</pre> No summary declared.
<code>ChatConversationStore.append_message</code>	<code>app/workflow/chat_conversation_store.py:139</code>	4	<pre>self, owner_scope_id: str, conversation_id: str, *, message_id: str, role: ChatMessageRole, content: dict[str, Any], run_id: str None</pre> No summary declared.
<code>ChatWorkflowStore.create</code>	<code>app/workflow/chat_workflow_store.py:95</code>	4	<pre>self, *, owner_scope_id: str, slug: str, display_name: str, description: str, yaml_text: str, source: Literal['generated', 'imported', 'existing'], source_workflow_name: str None=None, output_compatibility: dict[str, Any] None=None, managed: bool=False, adapter_key: str None=None, adapter_fingerprint: str None=None</pre> No summary declared.

Logic unit	Location	Branch score	Contract
build_artifact_adapter	app/workflow/ chat_workspace_planner.py:608	4	<pre>title: str, output: Literal['pdf', 'docx', 'pptx'], *, with_files: bool, previous_result: Any=None</pre> No summary declared.
start_new_run_record	app/workflow/ orchestration.py:28	4	<pre>db: Any, *, run_id: str, session: str, spec: WorkflowSpec, workflow_yaml: str, inputs: dict[str, Any], collection_id: str, retry_of_run_id: str None=None, attempt: int=1, origin: str='direct', history_visibility: str='global', workflow_id: str None=None, workflow_version_id: str None=None, conversation_id: str None=None, message_id: str None=None</pre> Create the durable "running" record and checkpoint before execution.
record_attempt	app/workflow/ preflight_stats.py:39	4	<pre>db: Any, *, source: AttemptSource, workflow_name: str None, success: bool, error_codes: list[str], initial_error_codes: list[str] None=None, deterministic_fixes_applied: int=0, llm_attempts: int=0, total_attempts: int=1</pre> Record the attempt.
_inflate_value	app/workflow/ run_history.py:94	4	<pre>db, value: Any</pre> Reverse `_externalize_if_large`: fetch and decode a GridFS pointer.
build_node_input	app/workflow/ run_history.py:193	4	<pre>state: dict[str, Any], resolved_config: dict[str, Any]</pre> Build the inspectable input for one node without duplicating all outputs.
record_node_started	app/workflow/ run_history.py:390	4	<pre>db, *, run_id: str, session_id: str, node_id: str, type_name: str, node_input: dict[str, Any], started_at: float</pre> Mark one node active and persist its effective, redacted input.
record_node_reused	app/workflow/ run_history.py:545	4	<pre>db, *, run_id: str, session_id: str, node_id: str, type_name: str, output: dict[str, Any], source_run_id: str, ended_at: float, duration_s: float</pre> Persist a node result replayed from a previous failed attempt.
record_checkpoint_node_completed	app/workflow/ run_history.py:834	4	<pre>db, *, run_id: str, session_id: str, node_id: str, output: dict[str, Any], extra_state: dict[str, Any]</pre> Save the exact replayable result immediately after node completion.
get_resume_checkpoint	app/workflow/ run_history.py:1049	4	<pre>db, session_id: str, run_id: str</pre> Return a restart-safe checkpoint only when it is currently paused.
_delete_run_blobs	app/workflow/ run_history.py:1345	4	<pre>db, *, run_id: str</pre> Remove every GridFS blob externalized for this run (see `_externalize_if_large`).

Logic unit	Location	Branch score	Contract
deliver_by_child_run_id	app/workflow/ subprocess_callback.py:178	4	<code>db: Any, services: dict[str, Any], child_run_id: str, *, status: str, output: Any, node_outputs: dict[str, Any], error: str None</code> Deliver completion when the run belongs to a Subprocess launch.
main	scripts/ backfill_knowledge_resources.py:217	4	 Compute the main.
sentence	scripts/ build_documentation_pdf.py:156	4	<code>value: str, fallback: str='No summary declared.'</code> No summary declared.
literal	scripts/ build_documentation_pdf.py:205	4	<code>node: ast.AST None, default: str=''</code> No summary declared.
dependencies	scripts/ build_documentation_pdf.py:427	4	 No summary declared.
v17	scripts/ build_documentation_pdf.py:601	4	<code>m: Model</code> No summary declared.
v19	scripts/ build_documentation_pdf.py:616	4	<code>m: Model</code> No summary declared.
write_csv	scripts/ build_workflow_qa_matrix.py:219	4	<code>path: Path, rows: list[dict[str, Any]]</code> No summary declared.
registered_routes	scripts/ check_api_routes.py:43	4	 Return every HTTP method/path pair mounted on the running app object.
main	scripts/ check_api_routes.py:69	4	 Compute the main.
main	scripts/ debug_compressor.py:20	4	 Compute the main.
_default_text	scripts/ generate_docstrings.py:308	4	<code>default: ast.expr None</code> No summary declared.
_enclosing_class	scripts/ generate_docstrings.py:517	4	<code>tree: ast.Module, target</code> No summary declared.
run	scripts/ manage_user.py:44	4	<code>args: argparse.Namespace</code> Run the result.
resolve_providers	scripts/ openrouter_health_check.py:301	4	<code>requested: Iterable[str], lookup: dict[str, str]</code> Map user-facing provider names onto slugs.
_is_retryable	scripts/ openrouter_health_check.py:476	4	<code>error: BaseException</code> Return whether retryable.

Logic unit	Location	Branch score	Contract
<code>_paths</code>	<code>scripts/ preflight_workflows.py:13</code>	4	<code>values: list[str]</code> Internal helper for the paths step.
<code>main</code>	<code>scripts/ smoke_events.py:58</code>	4	<code></code> Compute the main.
<code>Report.check</code>	<code>scripts/ verify_knowledge_studio. py:82</code>	4	<code>self, area: str, name: str, fn: Callable[[], Any]</code> Run fn; PASS when it returns without raising.
<code>Api.send</code>	<code>scripts/ verify_knowledge_studio. py:144</code>	4	<code>self, method: str, path: str, **kw: Any</code> Issue a request, waiting out the API's per-minute rate limit.
<code>_catalog</code>	<code>scripts/ verify_knowledge_studio. py:1579</code>	4	<code>api: Api</code> Internal helper for the catalog step.
<code>schema_preview</code>	<code>app/api/builder.py:296</code>	3	<code>body: SchemaPreviewRequest, user: CurrentUser=Depends(require_consultant)</code> Compile visual schema rows and show what the model will be held to.
<code>_contract_sort_key</code>	<code>app/api/builder.py:1329</code>	3	<code>field: Any</code> Order a step's contract so the author's own fields come first.
<code>_label_of</code>	<code>app/api/builder.py:1377</code>	3	<code>node: Any</code> The business label if the author set one, else the technical id (§17).
<code>MessageBody.content_is_bounded</code>	<code>app/api/ chat_conversations.py:50</code>	3	<code>cls, value: dict[str, Any]</code> No summary declared.
<code>get_conversation</code>	<code>app/api/ chat_conversations.py:90</code>	3	<code>conversation_id: str, request: Request, user: CurrentUser=Depends(require_consultant)</code> No summary declared.
<code>copy_existing_chat_workflow</code>	<code>app/api/ chat_workflows.py:667</code>	3	<code>body: CopyChatWorkflowRequest, request: Request, user: CurrentUser=Depends(require_consultant)</code> No summary declared.
<code>run_cost</code>	<code>app/api/cost.py:14</code>	3	<code>run_id: str, request: Request, user: CurrentUser=Depends(get_current_user)</code> Run the cost.
<code>_cost_ledger_entries</code>	<code>app/api/cost_admin.py:42</code>	3	<code>db, *, since: datetime None=None</code> Internal helper for the cost ledger entries step.
<code>set_pricing_override</code>	<code>app/api/ cost_admin.py:125</code>	3	<code>model: str, body: PricingOverrideRequest, request: Request, user: CurrentUser=Depends(require_admin)</code> Set the pricing override.
<code>_effective_per_call_cost</code>	<code>app/api/ cost_admin.py:207</code>	3	<code>allocation: dict[str, Any]</code> Internal helper for the effective per call cost step.
<code>connect</code>	<code>app/api/ email_oauth.py:146</code>	3	<code>provider: Provider, request: Request, user: CurrentUser=Depends(require_consultant)</code> Starts the authorization-code flow: stash a one-time

Logic unit	Location	Branch score	Contract
			state (+ PKCE verifier for Gmail), redirect the user's browser to the provider.
list_golden_set	app/api/eval.py:116	3	<pre>name: str='document_qa', _user: CurrentUser=Depends(require_permission('eval:run'))</pre> List the golden set.
list_workflow_golden_set	app/api/eval.py:234	3	<pre>name: str='verder_customer_triage', _user: CurrentUser=Depends(require_permission('eval:run'))</pre> List the workflow golden set.
connect	app/api/integration_oauth.py:137	3	<pre>provider: Provider, request: Request, user: CurrentUser=Depends(require_consultant)</pre> Starts the authorization-code flow: stash a one-time state (+ PKCE verifier for Google), redirect the user's browser to the provider.
list_openrouter_models	app/api/llm_providers.py:90	3	<pre>q: str None=Query(default=None, max_length=200), limit: int=Query(default=50, ge=1, le=200), user: CurrentUser=Depends(require_consultant)</pre> Live OpenRouter model catalog (via OpenRouter's own `GET /v1/models`), cached with a short TTL — see app/llm/openrouter_catalog.py.
_schema_field_names	app/api/node_types_chat.py:49	3	<pre>schema: dict[str, Any] None</pre> Internal helper for the schema field names step.
_manifest_entry	app/api/node_types_chat.py:81	3	<pre>type_name: str</pre> Live lookup — same freshness guarantee as _build_node_type_catalog.
update_prompt_template	app/api/prompt_templates.py:50	3	<pre>template_id: str, body: TemplateBody, request: Request, user: CurrentUser=Depends(require_consultant)</pre> No summary declared.
favorite_prompt_template	app/api/prompt_templates.py:75	3	<pre>template_id: str, body: FavoriteBody, request: Request, user: CurrentUser=Depends(require_consultant)</pre> No summary declared.
delete_prompt_template	app/api/prompt_templates.py:85	3	<pre>template_id: str, request: Request, user: CurrentUser=Depends(require_consultant)</pre> No summary declared.
horizon_evaluation	app/api/proposals.py:507	3	<pre>proposal_id: str, body: HorizonEvaluationRequest, request: Request, user: CurrentUser=Depends(require_consultant)</pre> Compute the horizon evaluation.
render_proposal	app/api/proposals.py:659	3	<pre>proposal_id: str, body: RenderProposalRequest, request: Request, user: CurrentUser=Depends(require_consultant)</pre> Render a proposal and persist both sanitised HTML and final PDF.

Logic unit	Location	Branch score	Contract
render_proposal_docx	app/api/proposals.py:696	3	<pre>proposal_id: str, body: RenderDOCXProposalRequest, request: Request, user: CurrentUser=Depends(require_consultant)</pre> Render a proposal as editable DOCX and persist its sanitised source.
my_runs	app/api/runs.py:50	3	<pre>request: Request, limit: Annotated[int, Query(ge=1, le=200)]=50, user: CurrentUser=Depends(require_consultant)</pre> Return a bounded, newest-first list of the caller's global workflow runs.
_run_preflight	app/api/workflows.py:398	3	<pre>yaml_text: str, req: AutofixWorkflowRequest, services: dict[str, Any]</pre> Same branching `validate_workflow` uses, so Auto-fix and Preflight agree on what "valid" means for identical input.
subprocess_callback	app/api/workflows.py:858	3	<pre>token: str, req: SubprocessCallbackRequest, request: Request</pre> Deliver a Subprocess child's terminal result to its waiting parent.
get_workflow_detail	app/api/workflows.py:997	3	<pre>name: str</pre> Workflow Library details panel calls this to refresh `library` and `readiness` for one workflow on demand (e.g.
save_workflow	app/api/workflows.py:1061	3	<pre>req: SaveWorkflowRequest</pre> Builder calls this on Save.
delete_workflow	app/api/workflows.py:1082	3	<pre>name: str, user: CurrentUser=Depends(require_consultant)</pre> Permanently removes a workflow's YAML together with its autosave draft and full version history — there is no undo.
_version_compatibility	app/api/workflows.py:1140	3	<pre>yaml_text: str</pre> Return cached structural compatibility for immutable version content.
restore_workflow_version	app/api/workflows.py:1224	3	<pre>name: str, version_id: str</pre> Restoring an old version is Save-equivalent: it must pass the same zero-token preflight gate before it can become the live workflow again.
_valid_http_url	app/config.py:630	3	<pre>value: str</pre> Internal helper for the valid http url step.
_backfill_run_documents_v1	app/db/migrations.py:61	3	<pre>db: Any</pre> Give legacy run and checkpoint documents the explicit v1 shape.
_remove_business_and_pipeline_persistence_v3	app/db/migrations.py:131	3	<pre>db: Any</pre> Remove persistence owned exclusively by the retired product concepts.
MongoClient.upsert_manifest	app/db/mongo.py:127	3	<pre>self, manifest: dict[str, Any]</pre> Insert or replace a manifest by _id.
MongoClient.list_scorecards	app/db/mongo.py:196	3	

Logic unit	Location	Branch score	Contract
			<code>self, workflow_name: str None=None, limit: int=20</code> List scorecards newest-first, optionally for one workflow.
ExampleResult.mean	app/evaluation/models.py:58	3	<code>self</code> The mean.
load_workflow_golden_set	app/evaluation/workflow_golden.py:50	3	<code>path: str Path</code> Load the workflow golden set.
_actual_value	app/evaluation/workflow_golden.py:186	3	<code>result: dict[str, Any], field: str</code> Internal helper for the actual value step.
_format_results	app/evidence/claim_verification.py:42	3	<code>results: list[dict[str, Any]]</code> Format the results.
StructuredDatasetQuery.safe_period	app/evidence/database_models.py:59	3	<code>cls, value: str None</code> Internal helper for the safe period step.
ResolvedWork.as_dict	app/evidence/identifiers.py:76	3	<code>self</code> Only the populated identifier fields, for storage on a candidate.
normalise_doi_value	app/evidence/identifiers.py:139	3	<code>value: Any</code> Lowercased bare DOI from anything that might contain one.
_arxiv_from	app/evidence/identifiers.py:155	3	<code>text: str</code> Internal helper for the arxiv from step.
stable_id	app/evidence/retrieval.py:52	3	<code>prefix: str, *parts: Any, length: int=16</code> Compute the stable id.
normalise_doi	app/evidence/retrieval.py:68	3	<code>value: Any</code> Compute the normalise doi.
parse_year	app/evidence/retrieval.py:111	3	<code>value: Any</code> Parse the year.
first_value	app/evidence/retrieval.py:124	3	<code>data: dict[str, Any], *keys: str</code> Compute the first value.
_tokens	app/evidence/retrieval.py:552	3	<code>text: str</code> Internal helper for the tokens step.
_split_oversized	app/ingestion/chunker.py:220	3	<code>text: str, encoding: tiktoken.Encoding, config: ChunkConfig</code> Recursive descent: paragraphs → sentences → hard cut.
chunk_text	app/ingestion/chunker.py:261	3	<code>text: str, config: ChunkConfig None=None</code> Chunk a raw text string.
_parse_metadata	app/ingestion/cli.py:18	3	<code>raw: list[str]</code> Parse repeated --meta key=value flags into a dict.
_amain	app/ingestion/cli.py:29	3	<code>args: argparse.Namespace</code> Internal helper for the amain step.
CollectionRegistry.get	app/ingestion/collections.py:67	3	<code>self, collection_id: str</code> Return the result.
Embedder.embed	app/ingestion/embedder.py:86	3	<code>self, texts: list[str]</code> Embed a list of texts.

Logic unit	Location	Branch score	Contract
Embedder.embed_batch_with_retry	app/ingestion/embedder.py:116	3	<code>self, batch: list[str]</code> Single batch with tenacity-managed retry.
_looks_multilingual	app/ingestion/embedding_catalog.py:126	3	<code>sample: str</code> Internal helper for the looks multilingual step.
PlainTextExtractor.extract	app/ingestion/extractor.py:314	3	<code>self, path: Path</code> Extract the result.
IngestionJobRunner._raise_if_cancelled	app/ingestion/jobs.py:101	3	<code>self, job: IngestionJob</code> Raise the if cancelled.
StructureAwareChunkingStrategy.chunk	app/ingestion/strategies.py:297	3	<code>self, document: ExtractedDocument, *, config: dict[str, Any], chunk_id_prefix: str</code> Chunk the result.
PdfPageVisionDescriber.model	app/ingestion/vision.py:81	3	<code>self</code> The model.
PdfPageVisionDescriber._build_client	app/ingestion/vision.py:212	3	<code>self</code> Build the client.
build_email_service	app/integrations/email/__init__.py:133	3	<code>*, db: Any=None, connections: dict[str, EmailConnection] None=None, include_memory_adapter: bool=True</code> Assemble the service with every adapter registered.
EmailAdapter.reply	app/integrations/email/base.py:235	3	<code>self, connection: EmailConnection, draft: EmailDraft</code> Reply defaults to a threaded send.
GmailAdapter.search	app/integrations/email/gmail.py:130	3	<code>self, connection: EmailConnection, criteria: EmailSearchCriteria</code> Search the result.
GmailAdapter.create_draft	app/integrations/email/gmail.py:178	3	<code>self, connection: EmailConnection, draft: EmailDraft</code> Create the draft.
GmailAdapter.send	app/integrations/email/gmail.py:202	3	<code>self, connection: EmailConnection, draft: EmailDraft</code> Send the result.
_parse_internal_date	app/integrations/email/gmail.py:422	3	<code>value: Any</code> Parse the internal date.
InMemoryEmailAdapter.send	app/integrations/email/memory.py:141	3	<code>self, connection: EmailConnection, draft: EmailDraft</code> Send the result.
MicrosoftGraphAdapter._mailbox	app/integrations/email/msggraph.py:76	3	<code>self, connection: EmailConnection</code> Graph addresses either the signed-in user or a specific mailbox.
_parse_timestamp	app/integrations/email/msggraph.py:418	3	<code>value: Any</code> Parse the timestamp.
_parse_token_response	app/integrations/email/oauth.py:252	3	<code>provider: Provider, response: httpx.Response</code> Parse the token response.
EmailService.connection	app/integrations/email/service.py:128	3	<code>self, connection_id: str</code> Compute the connection.

Logic unit	Location	Branch score	Contract
EmailService.describe_connections	app/integrations/email/service.py:235	3	<code>self</code> Connection list for the Builder's picker.
TokenVault.load	app/integrations/email/token_vault.py:159	3	<code>self, connection_id: str</code> Returns `{access_token, refresh_token, expires_at}` (expires_at a real datetime), or None if this connection has no stored tokens.
ExternalActionService._replay_or_refuse	app/integrations/external_action.py:248	3	<code>self, existing: dict[str, Any], key: str</code> Internal helper for the replay or refuse step.
_parse_body	app/integrations/external_action.py:277	3	<code>response: httpx.Response</code> Parse the body.
GoogleDriveProvider.download_file	app/integrations/files/google_drive.py:225	3	<code>self, connection: IntegrationConnection, *, file_id: str</code> Download the file.
_parse_timestamp	app/integrations/files/google_drive.py:300	3	<code>value: Any</code> Parse the timestamp.
_parse_token_response	app/integrations/files/oauth.py:250	3	<code>provider: Provider, response: httpx.Response</code> Parse the token response.
_parse_timestamp	app/integrations/files/onedrive.py:271	3	<code>value: Any</code> Parse the timestamp.
IntegrationService.connection	app/integrations/files/service.py:130	3	<code>self, connection_id: str</code> Compute the connection.
IntegrationService.describe_connections	app/integrations/files/service.py:246	3	<code>self</code> Connection list for the Builder's picker.
TokenVault.load	app/integrations/files/token_vault.py:165	3	<code>self, connection_id: str</code> Returns `{access_token, refresh_token, expires_at}` (expires_at a real datetime), or None if this connection has no stored tokens.
ChunkingProfileConfig.validate_token_windows	app/knowledge/models.py:218	3	<code>self</code> Validate the token windows.
KnowledgeRepository.list_profiles	app/knowledge/repository.py:367	3	<code>self, owner_scope_id: str, profile_type: ProfileType None=None</code> List the profiles.
KnowledgeRepository.list_ingestion_jobs	app/knowledge/repository.py:537	3	<code>self, owner_scope_id: str, collection_id: str None=None</code> List the ingestion jobs.
KnowledgeService.create_collection	app/knowledge/service.py:93	3	<code>self, *, owner_scope_id: str, name: str, description: str='', metadata_schema: dict[str, Any] None=None, doc_types: list[str] None=None</code> Create the collection.
_cache_usage_fields	app/llm/anthropic_gw.py:102	3	<code>usage</code> Extract cache_creation/cache_read counts off a Message.usage object.
AnthropicGateway._create	app/llm/anthropic_gw.py:138	3	<code>self, *, on_token: Callable[[str], Awaitable[None]] None=None, **kwargs</code> Issue a request via streaming and return the assembled Message.

Logic unit	Location	Branch score	Contract
AnthropicGateway.complete	app/llm/anthropic_gw.py:156	3	<code>self, *, model: str, system: str, user: str, temperature: float=0.0, max_tokens: int=1024, reasoning_effort: str None=None, on_token: Callable[[str], Awaitable[None]] None=None</code> Complete the result.
ModelDefinition.permits_data_class	app/llm/catalog.py:41	3	<code>self, data_class: str None</code> Compute the permits data class.
_usage	app/llm/local_openai_gw.py:94	3	<code>response: Any</code> Internal helper for the usage step.
LocalOpenAICompatibleGateway.complete	app/llm/local_openai_gw.py:178	3	<code>self, *, model: str, system: str, user: str, temperature: float=0.0, max_tokens: int=1024, reasoning_effort: str None=None, on_token: object None=None</code> Complete the result.
_hosted_bonus	app/llm/model_router.py:514	3	<code>profile: ModelProfile, priority: AccuracyPriority</code> Break same-score ties toward hosted providers under max accuracy.
_target_tier	app/llm/model_router.py:528	3	<code>complexity: Complexity, priority: AccuracyPriority</code> Internal helper for the target tier step.
_system_content	app/llm/openai_gw.py:120	3	<code>text: str, model: str None</code> Render system-prompt text as a bare string, or (gpt-5.6+ only) as a text content-part carrying an explicit prompt_cache_breakpoint.
_price_per_million	app/llm/openrouter_catalog.py:84	3	<code>raw: Any</code> OpenRouter reports pricing as a per-token USD string (e.g.
_parse_structured	app/llm/openrouter_gw.py:83	3	<code>content: str, response_model: Type[T]</code> Parse a structured response, retrying once on the unfenced body.
OpenRouterGateway.probe_model_access	app/llm/openrouter_gw.py:140	3	<code>self, model: str</code> Cheap existence check via the catalog — OpenRouter has no per-model equivalent of OpenAI's GET /models/{id}.
_status_code	app/llm/registry.py:201	3	<code>exc: BaseException</code> Internal helper for the status code step.
resolve_model	app/llm/registry.py:351	3	<code>intended: str</code> Map an intended model name to the model name we'll actually call.
local_model_enabled	app/llm/registry.py:507	3	<code>model: str</code> Compute the local model enabled.
probe_local_model	app/llm/registry.py:536	3	<code>model: str</code> Probe the local model.
RegistryLLMGateway.detonize_value	app/llm/registry.py:861	3	<code>self, value: Any</code> Detokenize the value.
RegistryLLMGateway.cached_model_access	app/llm/registry.py:879	3	<code>self, model: str</code> Internal helper for the cached model access step.

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RegistryLLMGateway._available_for_auto	app/llm/registry.py:905	3	<code>self, model: str</code> Combine configured-provider checks with verified project access.
RegistryLLMGateway._select_model	app/llm/registry.py:1034	3	<code>self, *, method_name: str, requested: str, kwargs: dict[str, Any]</code> Deterministic, zero-token model choice for one call.
RegistryLLMGateway.complete	app/llm/registry.py:1381	3	<code>self, *, model: str, stage: str None=None, **kwargs</code> Public entry point — tokenizes system/user, delegates to the unchanged implementation, then detokenizes the response.
RegistryLLMGateway.complete_structured	app/llm/registry.py:1536	3	<code>self, *, model: str, stage: str None=None, **kwargs</code> Public entry point — tokenizes system/user, delegates, then detokenizes every string field of the parsed structured result by walking its model_dump() and re-validating.
SemanticLLMCache.put	app/llm/semantic_cache.py:158	3	<code>self, *, session_id: str, model: str, system: str, user: str, temperature: float, max_tokens: int, response: dict[str, Any], query_embedding: list[float] None</code> Store the result.
SemanticLLMCache._prune	app/llm/semantic_cache.py:216	3	<code>self, scope: str</code> Prune the result.
inventory_check	app/mcp/business_records/handlers.py:155	3	<code>conn: MySQLConnectionAbstract, arguments: dict[str, Any]</code> Compute the inventory check.
backend	app/mcp/business_records/server.py:45	3	<code></code> Compute the backend.
_kill_query	app/mcp/business_records/sql_guard.py:146	3	<code>connect: Any, connection_id: int</code> A second, short-lived connection to cancel the first's running query — verified this actually stops a running SELECT SLEEP() and leaves the original connection usable (which is then closed anyway by run_readonly_query's own finally, since its result is now meaningless).
MCPClient.call_tool	app/mcp/client.py:379	3	<code>self, name: str, arguments: dict, server: str=DEFAULT_SERVER</code> Call a tool on `server` (default: primary).
MCPClient.call_tool_raw	app/mcp/client.py:390	3	<code>self, name: str, arguments: dict, *, server: str=DEFAULT_SERVER, timeout_seconds: float None=None</code> Call a tool and return the full CallToolResult.
_any_of_present	app/mcp/d365_finance/handlers.py:53	3	<code>*filters: str None</code> Internal helper for the any of present step.
_priority_filter	app/mcp/d365_finance/handlers.py:66	3	<code>*fields: tuple[str, str None]</code> Search by the first-given reference only.
find_credit_status	app/mcp/d365_finance/handlers.py:186	3	

Logic unit	Location	Branch score	Contract
			<pre>backend: DynamicsBackend, arguments: dict[str, Any]</pre> Find the credit status.
find_sales_order	app/mcp/d365_finance/handlers.py:323	3	<pre>backend: DynamicsBackend, arguments: dict[str, Any]</pre> Find the sales order.
_availability	app/mcp/d365_finance/handlers.py:369	3	<pre>row: dict[str, Any]</pre> Internal helper for the availability step.
find_shipment	app/mcp/d365_finance/handlers.py:518	3	<pre>backend: DynamicsBackend, arguments: dict[str, Any]</pre> Find the shipment.
find_invoice	app/mcp/d365_finance/handlers.py:568	3	<pre>backend: DynamicsBackend, arguments: dict[str, Any]</pre> Find the invoice.
find_contract	app/mcp/d365_finance/handlers.py:616	3	<pre>backend: DynamicsBackend, arguments: dict[str, Any]</pre> Find the contract.
DataverseClient.close	app/mcp/dynamics/client.py:229	3	<pre>self</pre> Close the result.
FixtureBackend.get	app/mcp/dynamics/client.py:640	3	<pre>self, entity_set: str, record_id: str, *, select: list[str]</pre> Return the result.
FixtureBackend.update	app/mcp/dynamics/client.py:678	3	<pre>self, entity_set: str, record_id: str, payload: dict[str, Any]</pre> Update the result.
_contact	app/mcp/dynamics/handlers.py:128	3	<pre>row: dict[str, Any]</pre> Internal helper for the contact step.
_opportunity	app/mcp/dynamics/handlers.py:147	3	<pre>row: dict[str, Any]</pre> Internal helper for the opportunity step.
_quote	app/mcp/dynamics/handlers.py:196	3	<pre>row: dict[str, Any]</pre> Map a Dataverse quotation row onto business field names.
_product	app/mcp/dynamics/handlers.py:213	3	<pre>row: dict[str, Any]</pre> Internal helper for the product step.
_shipment	app/mcp/dynamics/handlers.py:248	3	<pre>row: dict[str, Any]</pre> Internal helper for the shipment step.
_service_case	app/mcp/dynamics/handlers.py:268	3	<pre>row: dict[str, Any]</pre> Internal helper for the service case step.
guid	app/mcp/dynamics/odata.py:47	3	<pre>value: Any, *, field: str='id'</pre> Validate an identifier as a GUID and return its canonical form.
entity_path	app/mcp/dynamics/odata.py:128	3	<pre>entity_set: str, record_id: str None=None</pre> Build an entity path, validating the key predicate.
list_tools	app/mcp/dynamics/server.py:136	3	<pre></pre> List the tools.
_prepare_source_checkout		3	

Logic unit	Location	Branch score	Contract
	app/mcp/paper_search_server.py:49		Internal helper for the prepare source checkout step.
inject_openalex_api_key	app/mcp/paper_search_server.py:59	3	<code>searcher: Any, api_key: str</code> Inject the key into a searcher session without logging the secret.
patch_semantic_scholar_empty_results	app/mcp/paper_search_server.py:214	3	<code>module_name: str='paper_search_mcp.academic_platforms.semantic', min_interval_seconds: float=1.1</code> Make zero-result Semantic Scholar responses parse instead of KeyError, and pace requests to stay under the 1 req/s cumulative keyed rate limit.
main	app/mcp/paper_search_server.py:246	3	Compute the main.
MCPServerConnection.stdio_needs_a_command	app/mcp/registry.py:138	3	<code>self</code> Compute the stdio needs a command.
MCPServerConnection.secrets_are_references	app/mcp/registry.py:149	3	<code>self</code> Reject a literal secret pasted where a reference belongs.
MCPServerConnection.resolve_environment	app/mcp/registry.py:170	3	<code>self</code> Build the subprocess environment, reading secrets at launch time.
MCPServerRegistry.require	app/mcp/registry.py:297	3	<code>self, server_id: str</code> Compute the require.
_as_object	app/mcp/results.py:80	3	<code>value: Any</code> Normalise a parsed payload into an object, or reject it.
MCPIntegrationService.describe_servers	app/mcp/service.py:122	3	<code>self</code> Connections for the Builder's server picker.
MCPIntegrationService.find_tool	app/mcp/service.py:227	3	<code>self, server_id: str, tool_name: str</code> Find the tool.
MCPIntegrationService.require_connection	app/mcp/service.py:480	3	<code>self, server_id: str, tool_name: str</code> Internal helper for the require connection step.
MCPIntegrationService.replay_or_refuse	app/mcp/service.py:508	3	<code>self, existing: dict[str, Any], key: str, connection: MCPServerConnection, tool_name: str, decision: PolicyDecision</code> Internal helper for the replay or refuse step.
AITaskConfig.structured_tasks_declare_a_schema	app/nodes/ai_task.py:330	3	<code>self</code> Compute the structured tasks declare a schema.
AITaskAgent.preflight_output_fields	app/nodes/ai_task.py:481	3	<code>cls, config: dict[str, Any]</code> Authorise `result.<path>` references for the author's own schema.
AITaskAgent.preflight_static_output_values	app/nodes/ai_task.py:504	3	<code>cls, config: dict[str, Any]</code> Compute the preflight static output values.
AITaskAgent._system_prompt	app/nodes/ai_task.py:549	3	<code>self, cfg: AITaskConfig, fields: list[FieldSpec]</code> Internal helper for the system prompt step.
AITaskAgent._success	app/nodes/ai_task.py:761	3	

Logic unit	Location	Branch score	Contract
			<code>self, cfg: AITaskConfig, parsed: dict[str, Any], attempt: int</code> Internal helper for the success step.
<code>execution_kind_for</code>	<code>app/nodes/categories.py:201</code>	3	<code>type_name: str, *, uses_llm: bool</code> Compute the execution kind for.
<code>CitationRegistryBuilderConfig.coerce_documents</code>	<code>app/nodes/citation_registry_builder.py:69</code>	3	<code>cls, value: Any</code> Internal helper for the coerce documents step.
<code>ConceptAlternativesAgent.run</code>	<code>app/nodes/concept_alternatives.py:76</code>	3	<code>self, state: dict, resolved_config: dict</code> Run the result.
<code>DataTransformAgent.preflight_static_output_values</code>	<code>app/nodes/data_transform.py:241</code>	3	<code>cls, config: dict[str, Any]</code> Compute the preflight static output values.
<code>DecisionConfig.output_field_names</code>	<code>app/nodes/decision.py:150</code>	3	<code>self</code> Compute the output field names.
<code>DecisionAgent.preflight_output_fields</code>	<code>app/nodes/decision.py:228</code>	3	<code>cls, config: dict[str, Any]</code> Authorise `decisions.<field>` for every field the rules can set.
<code>DecisionAgent.run</code>	<code>app/nodes/decision.py:251</code>	3	<code>self, state, resolved_config: dict[str, Any]</code> Run the result.
<code>DOCXProposalRenderer.style_heading</code>	<code>app/nodes/docx_renderer.py:257</code>	3	<code>self, paragraph, tpl, top_level: bool</code> Internal helper for the style heading step.
<code>EmailAgent.preflight_output_fields</code>	<code>app/nodes/email_integration.py:284</code>	3	<code>cls, config: dict[str, Any]</code> Authorise the message shape as dotted paths.
<code>ExternalActionAgent.failure</code>	<code>app/nodes/external_action.py:245</code>	3	<code>self, cfg: ExternalActionConfig, error: Exception</code> Internal helper for the failure step.
<code>_review_panels</code>	<code>app/nodes/human_in_loop.py:350</code>	3	<code>cfg: HITLConfig, state: dict[str, Any]</code> Resolve each configured panel against state.
<code>_content_as_text</code>	<code>app/nodes/human_in_loop.py:471</code>	3	<code>value: Any</code> Internal helper for the content as text step.
<code>_edited_value</code>	<code>app/nodes/human_in_loop.py:487</code>	3	<code>content: HITLReviewContent</code> Restore structured editor content before patching workflow state.
<code>_RichTextSanitizer.handle_endtag</code>	<code>app/nodes/human_in_loop.py:601</code>	3	<code>self, tag: str</code> Handle the endtag.
<code>_explicitly_approved</code>	<code>app/nodes/internal_project_evidence_retriever.py:543</code>	3	<code>metadata: dict[str, Any]</code> Internal helper for the explicitly approved step.
<code>MinIOEvidenceIngestionConfig.reject_unresolved</code>	<code>app/nodes/minio_evidence_ingestion.py:74</code>	3	<code>cls, value: Any</code> Internal helper for the reject unresolved step.
<code>_load_pages</code>	<code>app/nodes/minio_evidence_ingestion.py:291</code>	3	<code>raw: bytes</code> Load the pages.
<code>_materiality</code>		3	

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	app/nodes/ proposal_evidence_factor.py:179		<code>claim_type: str</code> Internal helper for the materiality step.
<code>_bibtex_key</code>	app/nodes/ proposal_evidence_factor.py:333	3	<code>entry: CitationRegistryEntry</code> Internal helper for the bibtex key step.
<code>ProposalEvidenceFactoryAgent.document</code>	app/nodes/ proposal_evidence_factor.py:907	3	<code>document_id: str, documents: list[FullTextDocument]</code> Internal helper for the document step.
<code>ProposalEvidenceFactoryAgent.relation_from_link</code>	app/nodes/ proposal_evidence_factor.py:974	3	<code>link: ClaimEvidenceLink, document: FullTextDocument</code> Internal helper for the relation from link step.
<code>PythonSnippetAgent.preflight_static_output_values</code>	app/nodes/ python_snippet.py:153	3	<code>cls, config: dict[str, Any]</code> Compute the preflight static output values.
<code>NodeRegistry.register</code>	app/nodes/registry.py:43	3	<code>cls, node_class: Type[NodeType]</code> Register a node class.
<code>RouterConfig.multi_selection_not_supported_in_rule_mode</code>	app/nodes/router.py:174	3	<code>self</code> Compute the multi selection not supported in rule mode.
<code>SQLQueryAgent.failure</code>	app/nodes/ sql_query.py:231	3	<code>self, cfg: SQLQueryConfig, error: MCPToolError</code> Internal helper for the failure step.
<code>_as_field_specs</code>	app/nodes/start.py:264	3	<code>fields: list[InputFieldBinding]</code> Internal helper for the as field specs step.
<code>StructuredDatasetRetrieverAgent.plan_queries</code>	app/nodes/ structured_dataset_retriever.py:350	3	<code>self, cfg: StructuredDatasetRetrieverConfig, graph: Any</code> Internal helper for the plan queries step.
<code>_render_input_value</code>	app/nodes/ transform.py:163	3	<code>value: Any</code> Render one resolved input field's value for embedding in the prompt.
<code>_decode_json_container</code>	app/nodes/ transform.py:228	3	<code>value: Any</code> Decode a JSON object or array returned inside a string.
<code>TransformAgent.run</code>	app/nodes/ transform.py:460	3	<code>self, state, resolved_config: dict[str, Any]</code> Run the result.
<code>WorkflowInputConfig.as_field_specs</code>	app/nodes/ workflow_input.py:225	3	<code>self</code> The declared shape, as the same FieldSpec rows every other node uses.
<code>WorkflowInputAgent.preflight_output_fields</code>	app/nodes/ workflow_input.py:320	3	<code>cls, config: dict[str, Any]</code> Compute the preflight output fields.
<code>WorkflowInputAgent.preflight_static_output_values</code>	app/nodes/ workflow_input.py:341	3	<code>cls, config: dict[str, Any]</code> Compute the preflight static output values.
<code>CostLedger.calculate</code>	app/observability/ cost_ledger.py:172	3	<code>model: str, input_tokens: int, output_tokens: int, *, cache_creation_input_tokens: int=0, cache_read_input_tokens: int=0</code> Compute the result.

Logic unit	Location	Branch score	Contract
CostLedger.session_summary	app/observability/cost_ledger.py:373	3	<code>self, session_id: str</code> Compute the session summary.
_merge_object	app/proposal_graph/graph.py:76	3	<code>left: BaseModel, right: BaseModel</code> Field-by-field merge; right wins only where it has a non-null/non-empty value.
ProposalWorkspaceStore.list_source_versions	app/proposal_graph/workspace_store.py:266	3	<code>self, *, session_id: str, proposal_id: str, source_id: str None=None</code> List the source versions.
_skill_score	app/research/skills.py:495	3	<code>objective: str, skill: SkillDocument</code> Internal helper for the skill score step.
compress_chunks	app/retrieval/compressor.py:83	3	<code>query: str, chunks: list[RetrievedChunk], llm: LLMGateway, model: str='claude-haiku-4-5'</code> Compress all chunks in parallel.
assemble_context	app/retrieval/context.py:67	3	<code>chunks: list[RetrievedChunk]</code> Build the final "[N] (source: ...)" block passed to generation.
metadata_property_name	app/retrieval/filters.py:49	3	<code>key: str</code> Sanitize a user-supplied metadata key into a safe Weaviate property name.
_check_field	app/retrieval/filters.py:82	3	<code>field: str, schema: dict[str, Any]</code> Check the field.
_safe_complete	app/retrieval/query_transform.py:21	3	<code>llm: Any, *, system: str, user: str</code> Internal helper for the safe complete step.
_split_lines	app/retrieval/query_transform.py:41	3	<code>text: str, *, limit: int</code> Split the lines.
_format_candidates	app/retrieval/reranker.py:65	3	<code>candidates: list[RetrievedChunk]</code> Build the numbered candidate block for the user prompt.
RetrievalService.compare	app/retrieval/service.py:644	3	<code>self, requests: list[RetrievalQuery], *, owner_scope_id: str, llm: Any None=None</code> Compare the result.
_get_path	app/runtime/autofix.py:75	3	<code>raw: Any, path: str</code> Return the path.
_set_path	app/runtime/autofix.py:91	3	<code>raw: Any, path: str, value: Any</code> Set the path.
_replace_template_named_root_segment	app/runtime/autofix.py:338	3	<code>text: str, root: str, old_name: str, new_name: str</code> Rewrite a "{root.old_name...}" reference (root is 'inputs' or 'variables') to address 'new_name' instead, keeping any remainder.
format_preflight_feedback	app/runtime/autofix.py:774	3	<code>report: WorkflowPreflightReport</code> Format the preflight feedback.
RedisLease.release	app/runtime/coordination.py:80	3	<code>self</code> Release the result.
FieldSpec.migrate_legacy_item_enum_values	app/runtime/field_schema.py:96	3	<code>cls, data: Any</code> Fold list-of-enum values stored under the field-level 'enum_values' key into 'item_enum_values'.

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<code>_resume_output</code>	<code>app/runtime/hitl.py:42</code>	3	<code>checkpoint: dict[str, Any] None,</code> <code>final_state: dict[str, Any]</code> The same projection a fresh <code>`run_workflow`</code> completion computes (<code>executor._project_output`</code> ) — a resumed completion must produce it too, or every caller reading <code>`result["output"]`</code> (subprocess delivery chief among them — see <code>`app/workflow/subprocess_callback.py`</code> 's <code>`_select_result`</code> , which is exactly how a SubprocessAgent parent learns its child's result) silently gets <code>`None`</code> for any workflow that paused at least once before finishing.
<code>_json_schema_kind</code>	<code>app/runtime/logic_preflight.py:353</code>	3	<code>schema: dict[str, Any]</code> Internal helper for the json schema kind step.
<code>_static_mcp_tool_definition</code>	<code>app/runtime/logic_preflight.py:436</code>	3	<code>server_id: Any, tool_name: Any</code> Internal helper for the static mcp tool definition step.
<code>_static_mcp_output_schema</code>	<code>app/runtime/logic_preflight.py:451</code>	3	<code>server_id: Any, tool_name: Any</code> The declared output_schema for a tool this platform ships itself.
<code>_numeric_bounds</code>	<code>app/runtime/logic_preflight.py:1328</code>	3	<code>described: FieldPath</code> Bounds for a numeric field, when the schema declares them.
<code>_check_mcp_agent</code>	<code>app/runtime/logic_preflight.py:1935</code>	3	<code>node: NodeSpec, issue: Callable[..., None], reviewed_before: Callable[[str], bool]</code> MCPAgent chooses its own tools at runtime, so — unlike MCPToolAgent — there is no single static tool name preflight can classify read/write against.
<code>is_graph_interrupt</code>	<code>app/runtime/node_events.py:39</code>	3	<code>exc: BaseException</code> LangGraph's interrupt class has moved between versions.
<code>_construct_unique_mapping</code>	<code>app/runtime/preflight.py:49</code>	3	<code>loader: _UniqueKeyLoader, node: yaml.MappingNode, deep: bool=False</code> Internal helper for the construct unique mapping step.
<code>_reachable</code>	<code>app/runtime/preflight.py:705</code>	3	<code>start: str, graph: dict[str, set[str]]</code> Internal helper for the reachable step.
<code>_guaranteed_before</code>	<code>app/runtime/preflight.py:755</code>	3	<code>entry: str, forward: dict[str, set[str]], reverse: dict[str, set[str]]</code> For each node reachable from entry, the set of nodes GUARANTEED to have already run by the time it fires.
<code>_selection_deps</code>	<code>app/runtime/preflight.py:1219</code>	3	<code>node_id: str, branch_sets: list[tuple[str, dict[str, set[str]]]]</code> Every (multi_router_id, branch_label) this node's execution is conditioned on — a node exclusively reachable via one branch of a Multi-Route router depends on that branch having been selected.
<code>preflight_workflow_yaml</code>	<code>app/runtime/preflight.py:2208</code>	3	<code>yaml_text: str, *, provided_inputs: dict[str, Any] None=None, services: dict[str, Any] None=None, compile_graph: bool=True</code> Run all deterministic checks.
<code>preflight_workflow_for_run</code>	<code>app/runtime/preflight.py:3009</code>	3	<code>yaml_text: str, *, provided_inputs: dict[str, Any] None, services:</code>

Logic unit	Location	Branch score	Contract
			dict[str, Any], probe_services: bool=True, require_run_history: bool=True, owner_scope_id: str None=None Strict API gate used immediately before a new or retried run.
Action.value_matches_operation	app/runtime/rules.py:347	3	self Compute the value matches operation.
WorkflowInputSpec.effective_max_files	app/runtime/schema.py:91	3	self, platform_limit: int Compute the effective max files.
NodeSpec.selected_model_must_be_allowed	app/runtime/schema.py:284	3	self Compute the selected model must be allowed.
SnippetRunnerClient.probe	app/runtime/snippet_client.py:67	3	self Preflight health check (SNIPPET_RUNNER_UNAVAILABLE) — a real, trivial round trip, not just "did connect() succeed", since a listening-but-wedged daemon is just as unusable.
_rlimit_preexec	app/runtime/snippet_daemon.py:73	3	memory_mb: int, timeout_seconds: float Runs in the child, after fork, before exec — this is real isolation, not the request's own JSON asking nicely.
_resolve_reference	app/runtime/templating.py:90	3	match: 're.Match[str]', state: dict Resolve one matched reference, honouring the optional marker.
_require_session	app/security/audit.py:48	3	session_id: str The isolation boundary, enforced in ONE place.
write_audit_event	app/security/audit.py:75	3	db, run_id: str, session_id: str, node_id: str, event_type: str, actor: str='system', payload: dict[str, Any] None=None Write the audit event.
read_audit_events	app/security/audit.py:112	3	db, session_id: str, run_id: str None=None, limit: int=200 The ONLY read path.
_user_from_token	app/security/dependencies.py:29	3	token: str None Internal helper for the user from token step.
EntityRegistry.register	app/security/entity_registry.py:48	3	self, *, session_id: str, collection_id: str, entity_type: str, value: str, source: str='manual' Register the result.
_overlaps	app/security/entity_tokenizer.py:96	3	start: int, end: int, claimed: list[tuple[int, int]] Internal helper for the overlaps step.
_boundary_pattern	app/security/entity_tokenizer.py:110	3	value: str Internal helper for the boundary pattern step.
_ambiguous_short_ner_matches	app/security/entity_tokenizer.py:125	3	value: str Return True for short acronym-like NER guesses that are too ambiguous.
_redact_matches	app/security/guardrails.py:94	3	text: str, patterns: tuple[tuple[str, re.Pattern[str]], ...], *, path: str Redact the matches.

Logic unit	Location	Branch score	Contract
effective_data_class	app/security/ llm_policy.py:56	3	<code>declared: DataClass, *, entities_detected: bool</code> max(declared, detected) — detecting a real entity in the text implies at least CONFIDENTIAL sensitivity, regardless of what the workflow declared.
hash_password	app/security/ passwords.py:11	3	<code>password: str</code> Hash the password.
verify_password	app/security/ passwords.py:27	3	<code>password: str, encoded: str None</code> Verify the password.
content_hash	app/storage/ minio_client.py:33	3	<code>path: Path, chunk_size: int=65536</code> Compute SHA256 hash of file contents.
ObjectStore.object_exists	app/storage/ minio_client.py:203	3	<code>self, key: str</code> Cheap existence check using HEAD.
PublicDatabaseLookupService.close	app/tools/ database_lookup.py:66	3	<code>self</code> Close the result.
PublicDatabaseLookupService.rate_limit	app/tools/ database_lookup.py:179	3	<code>self</code> Internal helper for the rate limit step.
_set_shading	app/tools/ docx_proposal_rendering.py:201	3	<code>element, fill: str</code> Set the shading.
_add_external_hyperlink	app/tools/ docx_proposal_rendering.py:541	3	<code>paragraph, text: str, url: str, *, bold: bool=False, italic: bool=False</code> Add the external hyperlink.
_bookmark_name	app/tools/ docx_proposal_rendering.py:584	3	<code>name: str, bookmark_id: int</code> Internal helper for the bookmark name step.
_HorizonDocxBuilder.configure_footnote_styles	app/tools/ docx_proposal_rendering.py:995	3	<code>self, styles, normal</code> Configure the footnote styles.
_HorizonDocxBuilder.save	app/tools/ docx_proposal_rendering.py:1874	3	<code>self</code> Save the result.
OpenAllImageGenerationService.available	app/tools/image_io.py:83	3	<code>self, provider: str='openai'</code> Compute the available.
extract_text_from_pdf	app/tools/pdf_io.py:17	3	<code>file_bytes: bytes</code> Return a list of {page_num, text} for each page.
_response_usage	app/tools/web_io.py:537	3	<code>response: Any</code> Internal helper for the response usage step.
WorkflowBuilderStore.save_workflow	app/workflow/ builder_store.py:234	3	<code>self, name: str, yaml_text: str</code> Save the workflow.
WorkflowBuilderStore.delete_workflow	app/workflow/ builder_store.py:333	3	<code>self, name: str</code> Remove the executable YAML together with every Builder-owned trace of it — an autosave draft and the whole immutable version history — so nothing orphaned lingers under “.builder” once the workflow itself is gone.
_tokens	app/workflow/ capability_selection.py:62	3	<code>text: str</code> Internal helper for the tokens step.

Logic unit	Location	Branch score	Contract
<code>_reply</code>	<code>app/workflow/chat_workspace_planner.py:182</code>	3	<pre>message: str, *, sources: str None=None, handoff: dict[str, Any] None=None</pre> No summary declared.
<code>safe_filename</code>	<code>app/workflow/file_inputs.py:77</code>	3	<pre>filename: str None</pre> Strip paths/control characters while retaining a readable name.
<code>category_for_extension</code>	<code>app/workflow/file_inputs.py:97</code>	3	<pre>extension: str</pre> Compute the category for extension.
<code>content_type_for</code>	<code>app/workflow/file_inputs.py:115</code>	3	<pre>filename: str, extension: str</pre> Compute the content type for.
<code>accepted_by_spec</code>	<code>app/workflow/file_inputs.py:133</code>	3	<pre>ref: WorkflowFileRef, spec: WorkflowInputSpec</pre> Accept category names or literal extensions in workflow YAML.
<code>record_uploaded_files</code>	<code>app/workflow/file_inputs.py:396</code>	3	<pre>db: Any None, *, session_id: str, refs: list[WorkflowFileRef]</pre> Persist upload metadata for audit/cleanup without storing file bytes.
<code>BackgroundRunManager.launch</code>	<code>app/workflow/orchestration.py:249</code>	3	<pre>self, coro: Awaitable[dict[str, Any]], *, db: Any, run_id: str, session: str, record_rejection_reason: bool=False, services: dict[str, Any] None=None</pre> Launch the result.
<code>BackgroundRunManager.close</code>	<code>app/workflow/orchestration.py:399</code>	3	<pre>self</pre> Cancel owned work and let run_and_finalize persist interruption.
<code>PromptTemplateStore.update</code>	<code>app/workflow/prompt_template_store.py:101</code>	3	<pre>self, owner_scope_id: str, template_id: str, **changes: Any</pre> No summary declared.
<code>_require_session</code>	<code>app/workflow/run_chat_store.py:17</code>	3	<pre>session_id: str</pre> Internal helper for the require session step.
<code>_require_session</code>	<code>app/workflow/run_history.py:151</code>	3	<pre>session_id: str</pre> Enforce the same isolation boundary on every history operation.
<code>record_node_failed</code>	<code>app/workflow/run_history.py:734</code>	3	<pre>db, *, run_id: str, session_id: str, node_id: str, type_name: str, error: str, error_type: str None=None, error_traceback: str None=None, model_selections: list[dict[str, Any]] None=None, ended_at: float, duration_s: float</pre> Persist node attribution and partial run state on failure.
<code>_process_is_alive</code>	<code>app/workflow/run_history.py:1076</code>	3	<pre>pid: int None</pre> Best-effort same-host liveness check for a run's owning process.
<code>expr_name</code>	<code>scripts/build_documentation_pdf.py:197</code>	3	<pre>node: ast.AST None</pre> No summary declared.
v11		3	

Logic unit	Location	Branch score	Contract
	scripts/ build_documentation_pdf. py:530		<code>m: Model</code> No summary declared.
v12	scripts/ build_documentation_pdf. py:538	3	<code>m: Model</code> No summary declared.
v18	scripts/ build_documentation_pdf. py:608	3	<code>m: Model</code> No summary declared.
update_index	scripts/ build_documentation_pdf. py:718	3	<code>results: list[dict[str, Any]], model: Model</code> No summary declared.
safe_document	scripts/ build_workflow_qa_matrix. py:66	3	<code>path: Path</code> No summary declared.
expected_path	scripts/ build_workflow_qa_matrix. py:111	3	<code>document: dict[str, Any]</code> No summary declared.
load_preflight_reports	scripts/ build_workflow_qa_matrix. py:118	3	<code>path: Path None</code> No summary declared.
_annotation_text	scripts/ generate_docstrings.py:299	3	<code>annotation: ast.expr None</code> No summary declared.
_decorator_names	scripts/ generate_docstrings.py:507	3	<code>node</code> No summary declared.
main	scripts/ generate_fixtures.py:103	3	 Compute the main.
main	scripts/ generate_reference_workflows.py:71	3	 No summary declared.
smoke_openai	scripts/ live_provider_smoke.py:16	3	 Make one bounded, low-cost OpenAI completion.
smoke_anthropic	scripts/ live_provider_smoke.py:45	3	 Compute the smoke anthropic.
main	scripts/ live_provider_smoke.py:74	3	 Compute the main.
main	scripts/ new_node_type.py:62	3	 No summary declared.
_price	scripts/ openrouter_health_check. py:237	3	<code>raw: Any</code> OpenRouter prices are per-token USD strings; report per-million.
ModelResult.cost_usd	scripts/ openrouter_health_check. py:403	3	<code>self</code> The cost usd.

Logic unit	Location	Branch score	Contract
<code>_apply_limit</code>	<code>scripts/openrouter_health_check.py:736</code>	3	<code>selected: list[CatalogEntry], excluded: list[tuple[str, str]], limit: int None</code> Apply the limit.
<code>_probe_summary</code>	<code>scripts/openrouter_health_check.py:804</code>	3	<code>result: ModelResult</code> Probe the summary.
<code>main</code>	<code>scripts/seed_collections.py:17</code>	3	<code></code> Compute the main.
<code>Report.record</code>	<code>scripts/verify_knowledge_studio.py:62</code>	3	<code>self, area: str, name: str, ok: bool, detail: str=''</code> Record the result.
<code>Report.summary</code>	<code>scripts/verify_knowledge_studio.py:101</code>	3	<code>self</code> Compute the summary.
<code>wait_for_job</code>	<code>scripts/verify_knowledge_studio.py:391</code>	3	<code>api: Api, job_id: str, timeout: float=180.0</code> Compute the wait for job.
<code>_compare</code>	<code>scripts/verify_knowledge_studio.py:1370</code>	3	<code>api: Api, cid: str, query: str</code> Compare the result.
<code>_trace_detail</code>	<code>scripts/verify_knowledge_studio.py:1434</code>	3	<code>api: Api, traces: list[dict[str, Any]]</code> Internal helper for the trace detail step.
<code>_agent_query</code>	<code>scripts/verify_knowledge_studio.py:1485</code>	3	<code>api: Api, agent: dict[str, Any], query: str</code> Internal helper for the agent query step.
<code>_foreign</code>	<code>scripts/verify_knowledge_studio.py:1556</code>	3	<code>api: Api</code> Internal helper for the foreign step.
<code>_validate</code>	<code>scripts/verify_knowledge_studio.py:1619</code>	3	<code>api: Api, yaml_text: str</code> Validate the result.
<code>_preflight_ok</code>	<code>scripts/verify_knowledge_studio.py:1648</code>	3	<code>api: Api, agent: dict[str, Any], cid: str</code> Internal helper for the preflight ok step.
<code>_services</code>	<code>app/api/builder.py:76</code>	2	<code>request: Request</code> Internal helper for the services step.
<code>mcp_servers</code>	<code>app/api/builder.py:1124</code>	2	<code>request: Request, user: CurrentUser=Depends(require_consultant)</code> Configured MCP servers for the Builder's connection panel.
<code>mcp_tools</code>	<code>app/api/builder.py:1142</code>	2	<code>server_id: str, request: Request, refresh: bool=False, user: CurrentUser=Depends(require_consultant)</code> Discover what a server can do (§5).
<code>mcp_health</code>	<code>app/api/builder.py:1176</code>	2	<code>server_id: str, request: Request, user: CurrentUser=Depends(require_consultant)</code> Live connection check, for the panel's Test Connection button.

Logic unit	Location	Branch score	Contract
<code>_require_mcp</code>	<code>app/api/builder.py:1281</code>	2	<code>request: Request</code> Internal helper for the require mcp step.
<code>email_connections</code>	<code>app/api/builder.py:1300</code>	2	<code>request: Request, user: CurrentUser=Depends(require_consultant)</code> Configured mailboxes for the Email node's connection picker.
<code>_parse_workflow</code>	<code>app/api/builder.py:1342</code>	2	<code>yaml_text: str</code> Parse the workflow.
<code>_require_llm</code>	<code>app/api/builder.py:1359</code>	2	<code>request: Request</code> Internal helper for the require llm step.
<code>_is_external_write</code>	<code>app/api/builder.py:1409</code>	2	<code>type_name: str, config: dict[str, Any]</code> Return whether external write.
<code>_short</code>	<code>app/api/builder.py:1510</code>	2	<code>value: Any, limit: int=80</code> Internal helper for the short step.
<code>_scope</code>	<code>app/api/chat_conversations.py:23</code>	2	<code>user: CurrentUser</code> No summary declared.
<code>_store</code>	<code>app/api/chat_conversations.py:27</code>	2	<code>request: Request</code> No summary declared.
<code>resolve_conversation</code>	<code>app/api/chat_conversations.py:71</code>	2	<code>body: ResolveConversationBody, request: Request, user: CurrentUser=Depends(require_consultant)</code> No summary declared.
<code>append_message</code>	<code>app/api/chat_conversations.py:108</code>	2	<code>conversation_id: str, body: MessageBody, request: Request, user: CurrentUser=Depends(require_consultant)</code> No summary declared.
<code>replace_message</code>	<code>app/api/chat_conversations.py:124</code>	2	<code>conversation_id: str, message_id: str, body: ReplaceMessageBody, request: Request, user: CurrentUser=Depends(require_consultant)</code> No summary declared.
<code>_scope</code>	<code>app/api/chat_workflows.py:309</code>	2	<code>user: CurrentUser</code> No summary declared.
<code>_store</code>	<code>app/api/chat_workflows.py:313</code>	2	<code>request: Request</code> No summary declared.
<code>_slug</code>	<code>app/api/chat_workflows.py:323</code>	2	<code>value: str</code> No summary declared.
<code>_validate</code>	<code>app/api/chat_workflows.py:349</code>	2	<code>yaml_text: str</code> No summary declared.
<code>get_private_chat_workflow</code>	<code>app/api/chat_workflows.py:637</code>	2	<code>chat_workflow_id: str, request: Request, user: CurrentUser=Depends(require_consultant)</code> No summary declared.
<code>archive_private_chat_workflow</code>	<code>app/api/chat_workflows.py:726</code>	2	<code>chat_workflow_id: str, request: Request, user: CurrentUser=Depends(require_consultant)</code> No summary declared.

Logic unit	Location	Branch score	Contract
request_private_workflow_publication	app/api/chat_workflows.py:738		chat_workflow_id: str, request: Request, user: CurrentUser=Depends(require_consultant) No summary declared.
list_workspace_experiences	app/api/chat_workspace.py:135	2	user: CurrentUser=Depends(require_consultant) No summary declared.
clear_pricing_override	app/api/cost_admin.py:165	2	model: str, request: Request, user: CurrentUser=Depends(require_admin) Clear the pricing override.
set_global_budget	app/api/cost_admin.py:467	2	body: BudgetRequest, request: Request, user: CurrentUser=Depends(require_admin) Set the global budget.
set_session_budget	app/api/cost_admin.py:494	2	session_id: str, body: BudgetRequest, request: Request, user: CurrentUser=Depends(require_admin) Set the session budget.
_services	app/api/email_oauth.py:62	2	request: Request Internal helper for the services step.
_require_db	app/api/email_oauth.py:74	2	request: Request Internal helper for the require db step.
_require_email_service	app/api/email_oauth.py:92	2	request: Request Internal helper for the require email service step.
_connection_id	app/api/email_oauth.py:107	2	provider: Provider, address: str Internal helper for the connection id step.
disconnect	app/api/email_oauth.py:303	2	connection_id: str, request: Request, user: CurrentUser=Depends(require_consultant) Compute the disconnect.
_scope	app/api/entity_registry.py:25	2	user: CurrentUser Internal helper for the scope step.
_service	app/api/entity_registry.py:37	2	request: Request Internal helper for the service step.
list_entities	app/api/entity_registry.py:86	2	request: Request, collection_id: str='default', user: CurrentUser=Depends(require_consultant) List the entities.
register_entity	app/api/entity_registry.py:112	2	req: RegisterEntityRequest, request: Request, user: CurrentUser=Depends(require_consultant) Register the entity.
delete_entity	app/api/entity_registry.py:144	2	req: DeleteEntityRequest, request: Request, user: CurrentUser=Depends(require_consultant) Delete the entity.
_load_reference	app/api/eval.py:62	2	 Extract text from the stored reference proposal (gitignored, private).
history	app/api/eval.py:211	2	

Logic unit	Location	Branch score	Contract
			<pre>request: Request, limit: int=20, _user: CurrentUser=Depends(require_permission('e val:run'))</pre> Compute the history.
<code>_call_probe</code>	<code>app/api/health.py:20</code>	2	<pre>probe: Probe</pre> Internal helper for the call probe step.
<code>_mongo_probe</code>	<code>app/api/health.py:78</code>	2	<pre>services: dict[str, Any]</pre> Internal helper for the mongo probe step.
<code>_weaviate_probe</code>	<code>app/api/health.py:93</code>	2	<pre>services: dict[str, Any]</pre> Internal helper for the weaviate probe step.
<code>_minio_probe</code>	<code>app/api/health.py:108</code>	2	<pre>services: dict[str, Any]</pre> Internal helper for the minio probe step.
<code>_redis_probe</code>	<code>app/api/health.py:126</code>	2	<pre>services: dict[str, Any]</pre> Internal helper for the redis probe step.
<code>_checkpointer_probe</code>	<code>app/api/health.py:141</code>	2	<pre>services: dict[str, Any]</pre> Internal helper for the checkpointer probe step.
<code>ready</code>	<code>app/api/health.py:272</code>	2	<pre>request: Request</pre> Return 503 until every required runtime dependency is reachable.
<code>_native_collection</code>	<code>app/api/inspect.py:172</code>	2	<pre>wv</pre> Return the v4 native collection handle for DocumentChunk.
<code>_services</code>	<code>app/api/integration_oauth.py:57</code>	2	<pre>request: Request</pre> Internal helper for the services step.
<code>_require_db</code>	<code>app/api/integration_oauth.py:69</code>	2	<pre>request: Request</pre> Internal helper for the require db step.
<code>_require_integration_service</code>	<code>app/api/integration_oauth.py:87</code>	2	<pre>request: Request</pre> Internal helper for the require integration service step.
<code>_connection_id</code>	<code>app/api/integration_oauth.py:102</code>	2	<pre>provider: Provider, address: str</pre> Internal helper for the connection id step.
<code>list_connections</code>	<code>app/api/integration_oauth.py:249</code>	2	<pre>request: Request, user: CurrentUser=Depends(require_consultant)</pre> List the connections.
<code>disconnect</code>	<code>app/api/integration_oauth.py:270</code>	2	<pre>connection_id: str, request: Request, user: CurrentUser=Depends(require_consultant)</pre> Compute the disconnect.
<code>_scope</code>	<code>app/api/knowledge.py:40</code>	2	<pre>user: CurrentUser</pre> Internal helper for the scope step.
<code>_service</code>	<code>app/api/knowledge.py:52</code>	2	<pre>request: Request</pre> Internal helper for the service step.
<code>_repository</code>	<code>app/api/knowledge.py:64</code>	2	<pre>request: Request</pre> Internal helper for the repository step.
<code>get_collection</code>	<code>app/api/knowledge.py:124</code>	2	<pre>collection_id: str, request: Request, user: CurrentUser=Depends(require_permission('k</pre>

Logic unit	Location	Branch score	Contract
			<code>knowledge:read'))</code> Return the collection.
<code>create_profile</code>	<code>app/api/knowledge.py:164</code>	2	<code>payload: ProfileCreate, request: Request, user:</code> <code>CurrentUser=Depends(require_permission('knowledge:write'))</code> Create the profile.
<code>get_profile</code>	<code>app/api/knowledge.py:201</code>	2	<code>profile_id: str, request: Request, version: int None=Query(None, ge=1), user:</code> <code>CurrentUser=Depends(require_permission('knowledge:read'))</code> Return the profile.
<code>embedding_models</code>	<code>app/api/knowledge.py:288</code>	2	<code>user:</code> <code>CurrentUser=Depends(require_permission('knowledge:read'))</code> Embedding models selectable when building an Index.
<code>create_index</code>	<code>app/api/knowledge.py:323</code>	2	<code>collection_id: str, payload: IndexCreate, request: Request, user:</code> <code>CurrentUser=Depends(require_permission('knowledge:write'))</code> Create the index.
<code>activate_index</code>	<code>app/api/knowledge.py:363</code>	2	<code>collection_id: str, index_id: str, request: Request, user:</code> <code>CurrentUser=Depends(require_permission('knowledge:write'))</code> Compute the activate index.
<code>get_ingestion</code>	<code>app/api/knowledge.py:622</code>	2	<code>ingestion_job_id: str, request: Request, user:</code> <code>CurrentUser=Depends(require_permission('knowledge:read'))</code> Return the ingestion.
<code>cancel_ingestion</code>	<code>app/api/knowledge.py:641</code>	2	<code>ingestion_job_id: str, request: Request, user:</code> <code>CurrentUser=Depends(require_permission('knowledge:write'))</code> Cancel the ingestion.
<code>get_document</code>	<code>app/api/knowledge.py:686</code>	2	<code>document_id: str, request: Request, user:</code> <code>CurrentUser=Depends(require_permission('knowledge:read'))</code> Return the document.
<code>document_source_url</code>	<code>app/api/knowledge.py:705</code>	2	<code>document_id: str, request: Request, user:</code> <code>CurrentUser=Depends(require_permission('knowledge:read'))</code> Compute the document source url.
<code>_scope</code>	<code>app/api/prompt_templates.py:16</code>	2	<code>user: CurrentUser</code> No summary declared.
<code>_store</code>	<code>app/api/prompt_templates.py:20</code>	2	<code>request: Request</code> No summary declared.
<code>list_prompt_templates</code>	<code>app/api/prompt_templates.py:39</code>	2	<code>request: Request, user:</code> <code>CurrentUser=Depends(require_consultant)</code> No summary declared.

Logic unit	Location	Branch score	Contract
<code>_scope</code>	<code>app/api/proposals.py:36</code>	2	<code>user: CurrentUser</code> Internal helper for the scope step.
<code>register_source_version</code>	<code>app/api/proposals.py:188</code>	2	<code>proposal_id: str, source_id: str, body: SourceVersionRequest, request: Request, user: CurrentUser=Depends(require_consultant)</code> Register the source version.
<code>concept_alternatives</code>	<code>app/api/proposals.py:341</code>	2	<code>proposal_id: str, body: ConceptRequest, request: Request, user: CurrentUser=Depends(require_consultant)</code> Compute the concept alternatives.
<code>request_approval</code>	<code>app/api/proposals.py:395</code>	2	<code>proposal_id: str, body: ApprovalRequest, request: Request, user: CurrentUser=Depends(require_consultant)</code> Compute the request approval.
<code>decide_approval</code>	<code>app/api/proposals.py:435</code>	2	<code>proposal_id: str, approval_id: str, body: ApprovalDecisionRequest, request: Request, user: CurrentUser=Depends(require_consultant)</code> Compute the decide approval.
<code>_scope</code>	<code>app/api/rag_agents.py:16</code>	2	<code>user: CurrentUser</code> Internal helper for the scope step.
<code>create_agent</code>	<code>app/api/rag_agents.py:48</code>	2	<code>payload: RAGAgentCreate, request: Request, user: CurrentUser=Depends(require_permission('rag:write'))</code> Create the agent.
<code>get_agent</code>	<code>app/api/rag_agents.py:87</code>	2	<code>rag_agent_id: str, request: Request, user: CurrentUser=Depends(require_permission('knowledge:read'))</code> Return the agent.
<code>list_scientific_skills</code>	<code>app/api/research.py:10</code>	2	<code>request: Request, user: CurrentUser=Depends(require_consultant)</code> List the scientific skills.
<code>_scope</code>	<code>app/api/retrieval.py:28</code>	2	<code>user: CurrentUser</code> Internal helper for the scope step.
<code>_service</code>	<code>app/api/retrieval.py:40</code>	2	<code>request: Request</code> Internal helper for the service step.
<code>_safe_query</code>	<code>app/api/retrieval.py:142</code>	2	<code>value: str</code> Internal helper for the safe query step.
<code>search</code>	<code>app/api/retrieval.py:158</code>	2	<code>payload: RetrievalSearchRequest, request: Request, user: CurrentUser=Depends(require_permission('rag:query'))</code> Search the result.
<code>get_trace</code>	<code>app/api/retrieval.py:255</code>	2	<code>retrieval_request_id: str, request: Request, user: CurrentUser=Depends(require_permission('knowledge:read'))</code> Return the trace.

Logic unit	Location	Branch score	Contract
_node_type_catalog	app/api/run_chat.py:107	2	Internal helper for the node type catalog step.
_scope	app/api/run_chat.py:214	2	<code>user: CurrentUser</code> Internal helper for the scope step.
get_run_chat	app/api/run_chat.py:227	2	<code>run_id: str, request: Request, user: CurrentUser=Depends(require_consultant)</code> Return the run chat.
_scope	app/api/runs.py:37	2	<code>user: CurrentUser</code> Internal helper for the scope step.
pending_gate	app/api/runs.py:101	2	<code>run_id: str, request: Request, user: CurrentUser=Depends(require_consultant)</code> The human-review gate a paused run is currently waiting on, if any.
_scope	app/api/workflow_files.py:40	2	<code>user: CurrentUser</code> Internal helper for the scope step.
capabilities	app/api/workflow_files.py:53	2	<code>_user: CurrentUser=Depends(require_consultant)</code> Return the authoritative UI picker limits and accepted extensions.
extract_workflow_input_file	app/api/workflow_files.py:148	2	<code>body: ExtractWorkflowFileRequest, request: Request, user: CurrentUser=Depends(require_consultant)</code> Convert one uploaded document into plain text for the HITL editor.
_node_issues	app/api/workflows.py:394	2	<code>report: WorkflowPreflightReport, node_id: str</code> No summary declared.
get_preflight_stats	app/api/workflows.py:620	2	<code>request: Request, since_hours: int None=None, user: CurrentUser=Depends(require_admin)</code> Platform-wide preflight reliability rollup across every / validate, /generate, and /autofix call — success rate, and which error codes recur most before any fix is attempted.
get_workflow	app/api/workflows.py:984	2	<code>name: str</code> Builder calls this to load an existing workflow for editing.
get_workflow_stats	app/api/workflows.py:1032	2	<code>name: str, request: Request, user: CurrentUser=Depends(require_consultant)</code> Library "Runs and performance" tab.
_builder_name	app/api/workflows.py:1109	2	<code>name: str</code> Internal helper for the builder name step.
_builder_version_id	app/api/workflows.py:1124	2	<code>version_id: str</code> Internal helper for the builder version id step.
save_workflow_draft	app/api/workflows.py:1159	2	<code>name: str, req: SaveDraftRequest</code> Autosave a recoverable draft.
get_workflow_draft	app/api/workflows.py:1170	2	<code>name: str</code> Builder calls this on load to offer 'Resume draft' for a newer autosave.
get_workflow_version		2	

Logic unit	Location	Branch score	Contract
	app/api/workflows.py:1204		name: str, version_id: str Return the workflow version.
_sse_message	app/api/workflows.py:1247	2	*, event: str, data: dict[str, Any], event_id: int None=None Internal helper for the sse message step.
Settings.required_readiness_services	app/config.py:572	2	self The required readiness services.
Settings.resolved_paper_search_mcp_path	app/config.py:581	2	self The resolved paper search mcp path.
_csv_values	app/config.py:607	2	value: str Internal helper for the csv values step.
_redis_url_has_password	app/config.py:619	2	url: str Return whether a Redis URL contains a non-empty password component.
_is_placeholder	app/config.py:653	2	value: str Return whether placeholder.
_backfill_knowledge_resources_v2	app/db/migrations.py:88	2	db: Any Expand legacy "collections" (CollectionConfig) records with the lifecycle fields Knowledge Studio needs.
_renew_lock	app/db/migrations.py:212	2	collection: Any, *, owner: str Renew the lock.
MongoClient.ensure_client	app/db/mongo.py:69	2	self Ensure the client.
MongoClient.close	app/db/mongo.py:92	2	self Close the result.
MongoClient.list_manifests	app/db/mongo.py:149	2	self, status: str None=None, limit: int=100 List manifests, optionally filtered by status, newest first.
MongoClient.delete_manifest	app/db/mongo.py:173	2	self, manifest_id: str Delete a manifest.
build_manifest	app/db/mongo.py:217	2	minio_key: str, original_filename: str, source_format: str, byte_size: int, chunk_ids: list[str], metadata: dict[str, Any] None=None, status: str='completed' Construct a manifest dict in the canonical shape.
get_mongo_client	app/db/mongo.py:252	2	 Lazily-constructed module-level client.
LLMJudge.score_one	app/evaluation/judge.py:101	2	self, *, criterion: str, question: str, answer: str, context: str, reference: str Score the one.
LLMJudge.score_one	app/evaluation/judge.py:151	2	self, criterion: str, *, question: str, answer: str, context: str, reference: str='' Public single-criterion API used by focused evaluation runs.
LLMJudge.score_all	app/evaluation/judge.py:174	2	self, *, question: str, answer: str, context: str, reference: str=''

Logic unit	Location	Branch score	Contract
			Score one answer on all four criteria, one independent call each.
StructuredDatasetQuery.safe_identifier	app/evidence/database_models.py:40	2	<code>cls, value: str</code> Internal helper for the safe identifier step.
ResolvedWork.has_strong_identifier	app/evidence/identifiers.py:93	2	<code>self</code> The has strong identifier.
normalise_title	app/evidence/identifiers.py:281	2	<code>title: str None</code> Punctuation/whitespace-insensitive title key.
Chunk.embedding_content	app/ingestion/chunker.py:59	2	<code>self</code> The text that should actually be embedded for this chunk.
_get_tokenizer	app/ingestion/chunker.py:96	2	<code>name: str</code> Return the tokenizer.
_split_paragraphs	app/ingestion/chunker.py:137	2	<code>text: str</code> Split on blank lines.
_split_sentences	app/ingestion/chunker.py:143	2	<code>text: str</code> Split on sentence boundaries.
_hard_split	app/ingestion/chunker.py:152	2	<code>tokens: list[int], target: int, max_size: int, encoding: tiktoken.Encoding</code> Last-resort: split a token list at fixed boundaries.
main	app/ingestion/cli.py:64	2	<code></code> Compute the main.
CollectionConfig.validate_doc_type	app/ingestion/collections.py:35	2	<code>self, dt: str</code> Raise if `dt` is not in this collection's declared vocabulary.
CollectionConfig.validate_doc_types	app/ingestion/collections.py:47	2	<code>self, dts: list[str]</code> Validate a list (retrieval filters carry a list of doc_types).
CollectionRegistry.list_all	app/ingestion/collections.py:104	2	<code>self</code> List the all.
StageRegistry.get_parser	app/ingestion/contracts.py:96	2	<code>self, name: str</code> Return the parser.
StageRegistry.get_chunker	app/ingestion/contracts.py:110	2	<code>self, name: str</code> Return the chunker.
IngestionCoordinator.recover	app/ingestion/coordinator.py:60	2	<code>self</code> Resubmit every job left non-terminal by an interrupted process.
IngestionCoordinator.materialize_sources	app/ingestion/coordinator.py:75	2	<code>self, job: IngestionJob</code> Download this job's already-uploaded sources to a local temp dir.
get_embedder	app/ingestion/embedder.py:146	2	<code></code> Lazily-constructed module-level embedder.
embedding_model_catalog	app/ingestion/embedding_catalog.py:115	2	<code></code> Compute the embedding model catalog.
ExtractedDocument.full_text	app/ingestion/extractor.py:60	2	<code>self</code> All units joined with double newlines.
_is_number		2	

Logic unit	Location	Branch score	Contract
	app/ingestion/extractor.py:489		<code>value: str</code> Return whether number.
<code>get_extractor</code>	app/ingestion/extractor.py:573	2	<code>path: Path</code> Return the right extractor for the file's extension.
<code>WeaviateSearchIndex.ensure_schema</code>	app/ingestion/indexes.py:36	2	<code>self</code> Ensure the schema.
<code>StructureAwareDocumentParser.parse</code>	app/ingestion/strategies.py:41	2	<code>self, path: Path, *, config: dict[str, Any]</code> Parse the result.
<code>_chunk_config</code>	app/ingestion/strategies.py:213	2	<code>config: dict[str, Any]</code> Chunk the config.
<code>_terms</code>	app/ingestion/strategies.py:430	2	<code>text: str</code> Internal helper for the terms step.
<code>_can_render</code>	app/ingestion/vision.py:239	2	<code></code> Return whether render.
<code>EmailAddress.render</code>	app/integrations/email/base.py:54	2	<code>self</code> Render the result.
<code>GmailAdapter.headers</code>	app/integrations/email/gmail.py:53	2	<code>self, connection: EmailConnection</code> Internal helper for the headers step.
<code>_decode_b64</code>	app/integrations/email/gmail.py:372	2	<code>data: str</code> Decode the b64.
<code>InMemoryEmailAdapter.read</code>	app/integrations/email/memory.py:106	2	<code>self, connection: EmailConnection, message_id: str</code> Read the result.
<code>MicrosoftGraphAdapter.headers</code>	app/integrations/email/msgraph.py:55	2	<code>self, connection: EmailConnection</code> Internal helper for the headers step.
<code>MicrosoftGraphAdapter.create_draft</code>	app/integrations/email/msgraph.py:214	2	<code>self, connection: EmailConnection, draft: EmailDraft</code> Create the draft.
<code>_recipient</code>	app/integrations/email/msgraph.py:313	2	<code>address: EmailAddress</code> Internal helper for the recipient step.
<code>EmailService.adapter</code>	app/integrations/email/service.py:218	2	<code>self, connection: EmailConnection</code> Compute the adapter.
<code>TokenVault.store</code>	app/integrations/email/token_vault.py:125	2	<code>self, connection_id: str, *, access_token: str, refresh_token: str None, expires_in_seconds: int</code> Compute the store.
<code>TokenVault.forget</code>	app/integrations/email/token_vault.py:172	2	<code>self, connection_id: str</code> Compute the forget.
<code>ExternalActionService.default_url_validator</code>	app/integrations/external_action.py:90	2	<code>url: str</code> Apply the SSRF guard, translating violations into node errors.
<code>get_external_action_service</code>	app/integrations/external_action.py:300	2	<code></code> Return the external action service.
<code>build_integration_service</code>	app/integrations/files/__init__.py:124	2	<code>*, db: Any=None, connections: dict[str, IntegrationConnection] None=None</code>

Logic unit	Location	Branch score	Contract
			Assemble the service with every provider adapter registered.
GoogleDriveProvider.headers	app/integrations/files/google_drive.py:63	2	<code>self, connection: IntegrationConnection</code> Internal helper for the headers step.
GoogleDriveProvider.list_folder	app/integrations/files/google_drive.py:130	2	<code>self, connection: IntegrationConnection, *, folder_id: str None, page_size: int, page_token: str None</code> List the folder.
GoogleDriveProvider.search_files	app/integrations/files/google_drive.py:164	2	<code>self, connection: IntegrationConnection, *, query: str, folder_id: str None, page_size: int, page_token: str None</code> Search the files.
_to_page	app/integrations/files/google_drive.py:286	2	<code>raw: dict[str, Any], *, parent_id: str None</code> Internal helper for the to page step.
OneDriveProvider.headers	app/integrations/files/onedrive.py:48	2	<code>self, connection: IntegrationConnection</code> Internal helper for the headers step.
OneDriveProvider.download_file	app/integrations/files/onedrive.py:207	2	<code>self, connection: IntegrationConnection, *, file_id: str</code> Download the file.
_to_page	app/integrations/files/onedrive.py:258	2	<code>raw: dict[str, Any]</code> Internal helper for the to page step.
IntegrationService.provider_for	app/integrations/files/service.py:230	2	<code>self, connection: IntegrationConnection</code> Compute the provider for.
TokenVault.store	app/integrations/files/token_vault.py:131	2	<code>self, connection_id: str, *, access_token: str, refresh_token: str None, expires_in_seconds: int</code> Compute the store.
TokenVault.forget	app/integrations/files/token_vault.py:178	2	<code>self, connection_id: str</code> Compute the forget.
ExternalOperationLedger.find	app/integrations/operations.py:72	2	<code>self, key: str</code> Find the result.
ExternalOperationLedger.complete	app/integrations/operations.py:109	2	<code>self, key: str, outcome: dict[str, Any]</code> Complete the result.
ExternalOperationLedger.mark_ambiguous	app/integrations/operations.py:122	2	<code>self, key: str, error: str</code> Mark the ambiguous.
ExternalOperationLedger.release	app/integrations/operations.py:135	2	<code>self, key: str, error: str</code> Drop the reservation after a *definitive* failure.
_is_forbidden_address	app/integrations/url_guard.py:31	2	<code>ip: ipaddress.IPv4Address ipaddress.IPv6Address</code> No summary declared.
doc_type_catalog	app/knowledge/doc_types.py:67	2	<code></code> Compute the doc type catalog.
_encode	app/knowledge/ids.py:49	2	<code>value: int, length: int</code> Encode the result.
prefix_for	app/knowledge/ids.py:66	2	<code>kind: str</code> Compute the prefix for.

Logic unit	Location	Branch score	Contract
kind_of	app/knowledge/ids.py:108	2	<code>resource_id: str</code> Return the resource kind for ``resource_id``, or ``None`` if unknown.
RetrievalProfileConfig.validate_counts	app/knowledge/models.py:273	2	<code>self</code> Validate the counts.
KnowledgeRepository.get_collection	app/knowledge/repository.py:181	2	<code>self, owner_scope_id: str, collection_id: str</code> Return the collection.
KnowledgeRepository.list_collections	app/knowledge/repository.py:196	2	<code>self, owner_scope_id: str</code> List the collections.
KnowledgeRepository.get_document	app/knowledge/repository.py:226	2	<code>self, owner_scope_id: str, document_id: str</code> Return the document.
KnowledgeRepository.find_document_by_hash	app/knowledge/repository.py:241	2	<code>self, owner_scope_id: str, collection_id: str, content_hash: str</code> Find the document by hash.
KnowledgeRepository.list_documents	app/knowledge/repository.py:259	2	<code>self, owner_scope_id: str, collection_id: str</code> List the documents.
KnowledgeRepository.get_source_version	app/knowledge/repository.py:288	2	<code>self, owner_scope_id: str, source_version_id: str</code> Return the source version.
KnowledgeRepository.next_profile_version	app/knowledge/repository.py:308	2	<code>self, owner_scope_id: str, profile_id: str</code> Compute the next profile version.
KnowledgeRepository.get_index	app/knowledge/repository.py:406	2	<code>self, owner_scope_id: str, index_id: str</code> Return the index.
KnowledgeRepository.list_indexes	app/knowledge/repository.py:421	2	<code>self, owner_scope_id: str, collection_id: str</code> List the indexes.
KnowledgeRepository.get_index_document	app/knowledge/repository.py:470	2	<code>self, owner_scope_id: str, index_id: str, document_id: str</code> Return the index document.
KnowledgeRepository.get_ingestion_job	app/knowledge/repository.py:520	2	<code>self, owner_scope_id: str, ingestion_job_id: str</code> Return the ingestion job.
KnowledgeRepository.list_recoverable_jobs	app/knowledge/repository.py:555	2	<code>self</code> Non-terminal jobs across every owner, for startup recovery only.
KnowledgeRepository.get_rag_agent	app/knowledge/repository.py:585	2	<code>self, owner_scope_id: str, rag_agent_id: str</code> Return the rag agent.
KnowledgeRepository.get_trace	app/knowledge/repository.py:636	2	<code>self, owner_scope_id: str, retrieval_request_id: str</code> Return the trace.
KnowledgeRepository.list_traces	app/knowledge/repository.py:653	2	<code>self, owner_scope_id: str, limit: int=100</code> List the traces.

Logic unit	Location	Branch score	Contract
KnowledgeService.create_profile_version	app/knowledge/service.py:125	2	<pre>self, *, owner_scope_id: str, profile_type: ProfileType, name: str, strategy: str, config: dict[str, Any], profile_id: str None=None, description: str='', based_on_preset: str None=None</pre> Create the profile version.
KnowledgeService.validate_profile_config	app/knowledge/service.py:181	2	<pre>profile_type: ProfileType, config: dict[str, Any]</pre> Validate the profile config.
KnowledgeService.create_index	app/knowledge/service.py:201	2	<pre>self, *, owner_scope_id: str, collection_id: str, parser_profile_id: str, parser_profile_version: int, chunking_profile_id: str, chunking_profile_version: int, embedding_profile_id: str, embedding_profile_version: int</pre> Create the index.
AnthropicBatchService.submit	app/llm/anthropic_batch.py:76	2	<pre>self, items: list[BatchRequestItem]</pre> Submit a batch and return its batch_id.
_cacheable_system	app/llm/anthropic_gw.py:57	2	<pre>system: str</pre> Turn a plain system string into a cache-marked content-block list.
model_definition	app/llm/catalog.py:148	2	<pre>model: str</pre> Compute the model definition.
is_local_model	app/llm/catalog.py:175	2	<pre>model: str</pre> Return whether local model.
local_service_name	app/llm/catalog.py:188	2	<pre>model: str</pre> Compute the local service name.
LocalOpenAICompatibleGateway.generation_options	app/llm/local_openai_gw.py:138	2	<pre>self</pre> Internal helper for the generation options step.
LocalOpenAICompatibleGateway.max_tokens_field	app/llm/local_openai_gw.py:165	2	<pre>self</pre> Internal helper for the max tokens field step.
ModelRouter.describe_manual	app/llm/model_router.py:317	2	<pre>self, *, requested_model: str, selected_model: str, method_name: str, kwargs: Mapping[str, Any], input_tokens: int, node_type: str None=None, mode: Literal['manual', 'cost_protection']='manual'</pre> Compute the describe manual.
_accuracy_priority	app/llm/model_router.py:498	2	<pre>value: Any</pre> Internal helper for the accuracy priority step.
OpenAIBatchService.submit	app/llm/openai_batch.py:76	2	<pre>self, items: list[BatchRequestItem]</pre> Upload the JSONL request file, create the batch, return its id.
_parse_jsonl	app/llm/openai_batch.py:162	2	<pre>text: str</pre> Parse the jsonl.
_completion_tokens_for	app/llm/openai_gw.py:63	2	<pre>model: str, requested: int</pre> Apply provider-specific minimums to max_completion_tokens.

Logic unit	Location	Branch score	Contract
<code>_prompt_cache_retention_kw args</code>	<code>app/llm/openai_gw.py:107</code>	2	<code>model: str</code> Extended (24h) cache retention for pre-GPT-5.6 models, at the same cost as the default in-memory (5-10 min) window per OpenAI's prompt caching guide -- a free win for workflows with gaps between calls.
<code>_system_messages</code>	<code>app/llm/openai_gw.py:138</code>	2	<code>system: str, *, model: str None=None</code> Build the leading system message, omitting it when empty.
<code>_chat_tool_reasoning_effort</code>	<code>app/llm/openai_gw.py:150</code>	2	<code>model: str</code> Return the safe Chat Completions effort for function-tool calls.
<code>models_for_kind</code>	<code>app/llm/openai_registry.py:256</code>	2	<code>kind: OpenAIModelKind</code> Compute the models for kind.
<code>openai_model</code>	<code>app/llm/openai_registry.py:336</code>	2	<code>model: str</code> Compute the openai model.
<code>require_openai_model_kind</code>	<code>app/llm/openai_registry.py:353</code>	2	<code>model: str, kind: OpenAIModelKind</code> Compute the require openai model kind.
<code>strip_openrouter_prefix</code>	<code>app/llm/openrouter_gw.py:59</code>	2	<code>model: str</code> Compute the strip openrouter prefix.
<code>unfence_json</code>	<code>app/llm/openrouter_gw.py:71</code>	2	<code>content: str</code> Return the JSON body of a Markdown-fenced payload, or `content` unchanged.
<code>_safe_json_loads</code>	<code>app/llm/openrouter_gw.py:363</code>	2	<code>value: str</code> Internal helper for the safe json loads step.
<code>is_openrouter_router_model</code>	<code>app/llm/openrouter_ranking.py:90</code>	2	<code>value: str</code> True for OpenRouter's own router meta-models (openrouter/auto, openrouter/free, ...).
<code>get_gateway</code>	<code>app/llm/registry.py:372</code>	2	<code>model_name: str</code> Return (gateway, resolved_model_name).
<code>_provider_name_for</code>	<code>app/llm/registry.py:482</code>	2	<code>model: str</code> Internal helper for the provider name for step.
<code>_safe_task_kind</code>	<code>app/llm/registry.py:497</code>	2	<code>method_name: str, kwargs: dict[str, Any], node_type: str None</code> <code>infer_task_kind()</code> , but cost telemetry must never break a call.
<code>RegistryLLMGateway._publish</code>	<code>app/llm/registry.py:1069</code>	2	<code>self, evt: RunEvent</code> Publish the result.
<code>get_llm_gateway</code>	<code>app/llm/registry.py:1727</code>	2	 Module-level singleton used by main.py's lifespan.
<code>execute</code>	<code>app/mcp/business_records/db.py:66</code>	2	<code>conn: MySQLConnectionAbstract, query: str, params: tuple[Any, ...]=()</code> Run a parameterized SELECT and return every row as a dict.
<code>execute_write</code>	<code>app/mcp/business_records/db.py:78</code>	2	<code>conn: MySQLConnectionAbstract, query: str, params: tuple[Any, ...]=()</code> Run a parameterized INSERT/UPDATE, commit, and return the affected/ inserted row count (or `lastrowid` for an auto-increment insert).

Logic unit	Location	Branch score	Contract
<code>_limit</code>	<code>app/mcp/business_records/handlers.py:34</code>	2	<code>arguments: dict[str, Any], default: int, maximum: int</code> Internal helper for the limit step.
<code>_dt</code>	<code>app/mcp/business_records/seed.py:39</code>	2	<code>value: str None</code> ISO 8601 ("...T...Z") -> MySQL DATETIME literal ("...
<code>_upsert</code>	<code>app/mcp/business_records/seed.py:51</code>	2	<code>conn, table: str, row: dict[str, Any]</code> Upsert the result.
<code>ensure_readonly_user</code>	<code>app/mcp/business_records/seed.py:69</code>	2	<code>app_settings: Any=settings</code> Create (idempotently) the genuinely lower-privileged credential query_readonly runs under (app/mcp/business_records/db.py's connect_readonly) — GRANT SELECT only, no INSERT/UPDATE/DELETE/DDL.
<code>seed_all</code>	<code>app/mcp/business_records/seed.py:449</code>	2	<code>app_settings=settings</code> Seed the all.
<code>list_tools</code>	<code>app/mcp/business_records/server.py:59</code>	2	<code></code> List the tools.
<code>_paper_search_mcp_env</code>	<code>app/mcp/client.py:43</code>	2	<code>app_settings: Settings</code> Research credentials for the paper-search-mcp subprocess.
<code>MCPClient.require</code>	<code>app/mcp/client.py:328</code>	2	<code>self, server: str</code> Internal helper for the require step.
<code>MCPClient.list_tools</code>	<code>app/mcp/client.py:356</code>	2	<code>self, server: str=DEFAULT_SERVER</code> List the tools.
<code>_dynamics_tool_policies</code>	<code>app/mcp/connections.py:54</code>	2	<code></code> State each Dynamics tool's class as deployment policy.
<code>dynamics_connection</code>	<code>app/mcp/connections.py:71</code>	2	<code>app_settings: Settings=settings</code> Compute the dynamics connection.
<code>business_records_connection</code>	<code>app/mcp/connections.py:211</code>	2	<code>app_settings: Settings=settings</code> A real, persistent MySQL database — genuine Lookup/Create/Update tools.
<code>_limit</code>	<code>app/mcp/d365_finance/handlers.py:34</code>	2	<code>arguments: dict[str, Any], default: int, maximum: int</code> Internal helper for the limit step.
<code>backend</code>	<code>app/mcp/d365_finance/server.py:53</code>	2	<code></code> Compute the backend.
<code>list_tools</code>	<code>app/mcp/d365_finance/server.py:67</code>	2	<code></code> List the tools.
<code>DataverseClient.http</code>	<code>app/mcp/dynamics/client.py:219</code>	2	<code>self</code> Internal helper for the http step.
<code>DataverseClient.headers</code>	<code>app/mcp/dynamics/client.py:292</code>	2	<code>self, *, write: bool=False</code> Internal helper for the headers step.
<code>DataverseClient.url</code>	<code>app/mcp/dynamics/client.py:316</code>	2	<code>self, path: str, params: dict[str, str] None=None</code> Internal helper for the url step.

Logic unit	Location	Branch score	Contract
DataverseClient.query	app/mcp/dynamics/client.py:420	2	<code>self, entity_set: str, *, select: list[str], filter_expression: str None=None, order_by: str None=None, top: int None=None, expand: str None=None</code> Query the result.
_limit	app/mcp/dynamics/handlers.py:306	2	<code>arguments: dict[str, Any], default: int, maximum: int</code> Internal helper for the limit step.
find_account	app/mcp/dynamics/handlers.py:345	2	<code>backend: DynamicsBackend, arguments: dict[str, Any]</code> Find the account.
get_account	app/mcp/dynamics/handlers.py:375	2	<code>backend: DynamicsBackend, arguments: dict[str, Any]</code> Return the account.
get_contacts_for_account	app/mcp/dynamics/handlers.py:436	2	<code>backend: DynamicsBackend, arguments: dict[str, Any]</code> Return the contacts for account.
get_open_opportunities	app/mcp/dynamics/handlers.py:461	2	<code>backend: DynamicsBackend, arguments: dict[str, Any]</code> Return the open opportunities.
get_quotations_for_account	app/mcp/dynamics/handlers.py:489	2	<code>backend: DynamicsBackend, arguments: dict[str, Any]</code> List the stored quotations belonging to one CRM account.
find_previous_orders	app/mcp/dynamics/handlers.py:542	2	<code>backend: DynamicsBackend, arguments: dict[str, Any]</code> Find the previous orders.
get_shipments_for_account	app/mcp/dynamics/handlers.py:612	2	<code>backend: DynamicsBackend, arguments: dict[str, Any]</code> List the stored shipment records belonging to one CRM account.
find_product	app/mcp/dynamics/handlers.py:629	2	<code>backend: DynamicsBackend, arguments: dict[str, Any]</code> Find the product.
get_service_cases_for_account	app/mcp/dynamics/handlers.py:740	2	<code>backend: DynamicsBackend, arguments: dict[str, Any]</code> Return the service cases for account.
get_recent_activities	app/mcp/dynamics/handlers.py:765	2	<code>backend: DynamicsBackend, arguments: dict[str, Any]</code> Return the recent activities.
_column	app/mcp/dynamics/odata.py:176	2	<code>name: str</code> Internal helper for the column step.
encode_params	app/mcp/dynamics/odata.py:208	2	<code>params: dict[str, str]</code> Percent-encode query values while leaving OData's own syntax readable.
backend	app/mcp/dynamics/server.py:101	2	<code></code> Compute the backend.
_annotations		2	

Logic unit	Location	Branch score	Contract
	app/mcp/dynamics/server.py:113		<code>definition: dict[str, Any]</code> Declare the tool's nature honestly.
<code>_JsonDefaultsDataKey.json</code>	app/mcp/paper_search_server.py:179	2	<code>self, *args: Any, **kwargs: Any</code> Compute the json.
<code>_wait_for_semantic_scholar_rate_limit</code>	app/mcp/paper_search_server.py:199	2	<code>min_interval_seconds: float</code> Block until at least `min_interval_seconds` has passed since the last call.
<code>_words</code>	app/mcp/policy.py:95	2	<code>tool_name: str</code> Split a tool name into words across the naming styles servers use: `get-associated-opportunities`, `get_associated_opportunities`, `getAssociatedOpportunities`.
<code>MCPServerConnection.id_is_safe</code>	app/mcp/registry.py:122	2	<code>cls, value: str</code> Compute the id is safe.
<code>MCPServerConnection.label</code>	app/mcp/registry.py:166	2	<code>self</code> The label.
<code>MCPServerConnection.describe</code>	app/mcp/registry.py:192	2	<code>self</code> Builder-facing description.
<code>MCPServerRegistry.all</code>	app/mcp/registry.py:314	2	<code>self</code> Compute the all.
<code>MCPServerRegistry.describe_all</code>	app/mcp/registry.py:359	2	<code>self</code> Compute the describe all.
<code>_trim</code>	app/mcp/results.py:219	2	<code>text: str, max_bytes: int</code> Internal helper for the trim step.
<code>_services</code>	app/mcp/server.py:45	2	<code></code> Internal helper for the services step.
<code>MCPIntegrationService.health_check</code>	app/mcp/service.py:244	2	<code>self, server_id: str</code> Compute the health check.
<code>AITaskAgent.run</code>	app/nodes/ai_task.py:524	2	<code>self, state, resolved_config: dict[str, Any]</code> Run the result.
<code>AITaskAgent._run_text</code>	app/nodes/ai_task.py:631	2	<code>self, cfg: AITaskConfig, llm: Any, system: str, user: str</code> Run the text.
<code>AITaskAgent._failure</code>	app/nodes/ai_task.py:793	2	<code>self, cfg: AITaskConfig, status: str, message: str, *, attempts: int</code> Internal helper for the failure step.
<code>_looks_like_refusal</code>	app/nodes/ai_task.py:842	2	<code>error: Exception</code> Distinguish a model declining the task from a malformed response.
<code>CallCoverageMatrixAgent.run</code>	app/nodes/call_coverage.py:41	2	<code>self, state: dict, resolved_config: dict</code> Run the result.
<code>family_for</code>	app/nodes/categories.py:189	2	<code>type_name: str</code> Compute the family for.
<code>data_types_from_schema</code>	app/nodes/contracts.py:98	2	<code>schema: type[BaseModel]</code> Derive broad transport types from a node's existing Pydantic schema.

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<code>DOCXProposalRenderer._add_title_block</code>	<code>app/nodes/docx_renderer.py:231</code>	2	<code>self, doc, title, client, tpl</code> Add the title block.
<code>is_side_effect</code>	<code>app/nodes/email_integration.py:426</code>	2	<code>operation: str</code> Whether this operation changes something outside the platform.
<code>_sanitize_query</code>	<code>app/nodes/evidence_agent.py:60</code>	2	<code>text: str</code> Strip boolean/advanced-search syntax from a generated query.
<code>ExcelTableExtractor.run</code>	<code>app/nodes/excel_tool.py:64</code>	2	<code>self, state, resolved_config: dict[str, Any]</code> Run the result.
<code>_safe_component</code>	<code>app/nodes/horizon_docx_renderer.py:98</code>	2	<code>value: str</code> Internal helper for the safe component step.
<code>HorizonEvaluationAgent.run</code>	<code>app/nodes/horizon_evaluation.py:65</code>	2	<code>self, state: dict, resolved_config: dict</code> Run the result.
<code>_safe_component</code>	<code>app/nodes/html_proposal_renderer.py:85</code>	2	<code>value: str</code> Internal helper for the safe component step.
<code>HumanInLoopAgent.required_services</code>	<code>app/nodes/human_in_loop.py:198</code>	2	<code>cls, config: dict[str, Any]</code> Compute the required services.
<code>_content_format</code>	<code>app/nodes/human_in_loop.py:459</code>	2	<code>value: Any</code> Internal helper for the content format step.
<code>_RichTextSanitizer.handle_starttag</code>	<code>app/nodes/human_in_loop.py:571</code>	2	<code>self, tag: str, attrs: list[tuple[str, str None]]</code> Handle the starttag.
<code>_RichTextSanitizer.handle_startendtag</code>	<code>app/nodes/human_in_loop.py:586</code>	2	<code>self, tag: str, attrs: list[tuple[str, str None]]</code> Handle the startendtag.
<code>sanitize_rich_html</code>	<code>app/nodes/human_in_loop.py:619</code>	2	<code>value: str None</code> Sanitize the rich html.
<code>_meta_out</code>	<code>app/nodes/integration.py:375</code>	2	<code>meta: CloudFileMeta</code> Internal helper for the meta out step.
<code>_to_workflow_file_ref</code>	<code>app/nodes/integration.py:396</code>	2	<code>downloaded: DownloadedFile</code> Turn a downloaded cloud file into the platform's own file contract.
<code>InternalProjectEvidenceRetrieverAgent.required_services</code>	<code>app/nodes/internal_project_evidence_retriever.py:139</code>	2	<code>cls, config: dict[str, Any]</code> Compute the required services.
<code>_source_class</code>	<code>app/nodes/internal_project_evidence_retriever.py:525</code>	2	<code>metadata: dict[str, Any]</code> Internal helper for the source class step.
<code>_source_path</code>	<code>app/nodes/minio_evidence_ingestion.py:305</code>	2	<code>doc: dict[str, Any], number: int</code> Internal helper for the source path step.
<code>OpenAllImageGenerationAgent.required_services</code>	<code>app/nodes/openai_image_generation_agent.py:101</code>	2	<code>cls, config: dict[str, Any]</code> Compute the required services.

Logic unit	Location	Branch score	Contract
<code>_authority</code>	<code>app/nodes/proposal_evidence_factor.py:195</code>	2	<code>value: str</code> Internal helper for the authority step.
<code>ProposalEvidenceFactoryAgent.source_id_for_document</code>	<code>app/nodes/proposal_evidence_factor.py:926</code>	2	<code>document: FullTextDocument</code> Internal helper for the source id for document step.
<code>PythonSnippetAgent.failure</code>	<code>app/nodes/python_snippet.py:230</code>	2	<code>self, cfg: PythonSnippetConfig, code: str, message: str, *, stdout: str='', stderr: str='', duration_s: float=0.0</code> Internal helper for the failure step.
<code>RAGAgent.required_services</code>	<code>app/nodes/rag.py:165</code>	2	<code>cls, config: dict[str, Any]</code> Compute the required services.
<code>NodeRegistry.get</code>	<code>app/nodes/registry.py:66</code>	2	<code>cls, type_name: str</code> Return the result.
<code>NodeRegistry.manifest</code>	<code>app/nodes/registry.py:83</code>	2	<code>cls</code> Return the JSON schemas and presentation metadata used by the Builder.
<code>_uses_llm</code>	<code>app/nodes/registry.py:158</code>	2	<code>klass: Type[NodeType]</code> Whether this node type needs a model, per its own preflight declaration.
<code>RouteCase.case_is_decidable</code>	<code>app/nodes/router.py:79</code>	2	<code>self</code> Compute the case is decidable.
<code>RouterAgent.required_services</code>	<code>app/nodes/router.py:317</code>	2	<code>cls, config: dict[str, Any]</code> Compute the required services.
<code>_json_text</code>	<code>app/nodes/scientific_research_planner.py:350</code>	2	<code>value: Any</code> Internal helper for the json text step.
<code>ScientificSkillAgent.run</code>	<code>app/nodes/scientific_skill_agent.py:90</code>	2	<code>self, state, resolved_config: dict[str, Any]</code> Run the result.
<code>StartFileField.resolved_source</code>	<code>app/nodes/start.py:54</code>	2	<code>self</code> Compute the resolved source.
<code>StructuredDatasetRetrieverAgent.required_services</code>	<code>app/nodes/structured_dataset_retriever.py:152</code>	2	<code>cls, config: dict[str, Any]</code> Compute the required services.
<code>_json_text</code>	<code>app/nodes/structured_dataset_retriever.py:405</code>	2	<code>value: Any</code> Internal helper for the json text step.
<code>_deduplicate_queries</code>	<code>app/nodes/structured_dataset_retriever.py:419</code>	2	<code>queries: list[StructuredDatasetQuery]</code> Internal helper for the deduplicate queries step.
<code>SubprocessAgent.run</code>	<code>app/nodes/subprocess.py:169</code>	2	<code>self, state, resolved_config: dict[str, Any]</code> Run the result.
<code>_load_child_workflow</code>	<code>app/nodes/subprocess.py:364</code>	2	<code>name: str</code> Load the child workflow.
<code>TextAssemblerAgent.run</code>	<code>app/nodes/text_assembler.py:103</code>	2	

Logic unit	Location	Branch score	Contract
			<pre>self, state: Any, resolved_config: dict[str, Any]</pre> Run the result.
<code>_build_response_model</code>	<code>app/nodes/transform.py:275</code>	2	<pre>name: str, schema: dict[str, str]</pre> Build the response model.
<code>TransformAgent.required_services</code>	<code>app/nodes/transform.py:385</code>	2	<pre>cls, config: dict[str, Any]</pre> Compute the required services.
<code>TransformAgent.run_new_style</code>	<code>app/nodes/transform.py:608</code>	2	<pre>self, cfg: TransformConfig, llm: Any</pre> Run the new style.
<code>_friendly_number</code>	<code>app/nodes/workflow_input.py:41</code>	2	<pre>value: float</pre> "1.0" reads as a technical artifact of a float field; a business- friendly error message says "1", not "1.0".
<code>InputFieldBinding.resolved_source</code>	<code>app/nodes/workflow_input.py:194</code>	2	<pre>self</pre> Compute the resolved source.
<code>CostLedger.record</code>	<code>app/observability/cost_ledger.py:206</code>	2	<pre>self, entry: LedgerEntry</pre> Record the result.
<code>CostLedger.record_no_charge</code>	<code>app/observability/cost_ledger.py:256</code>	2	<pre>self, *, run_id: str, session_id: str, node_id: str, stage: str, task_type: str='retrieval', note: str=''</pre> Log a pipeline stage that genuinely has no model charge.
<code>configure_logging</code>	<code>app/observability/logging.py:11</code>	2	<pre>environment: str='development'</pre> Configure the logging.
<code>track_node</code>	<code>app/observability/metrics.py:197</code>	2	<pre>node_type: str</pre> Time a node, record its outcome, and track in-flight count.
<code>_valid_ids</code>	<code>app/proposal_graph/concepts.py:129</code>	2	<pre>values: list[str], known: set[str]</pre> Internal helper for the valid ids step.
<code>_draft_one_concept</code>	<code>app/proposal_graph/concepts.py:260</code>	2	<pre>llm: Any, *, posture: ConceptPosture, graph_payload: dict[str, Any], concept_note: str, model: str</pre> Draft the one concept.
<code>judge_concept</code>	<code>app/proposal_graph/concepts.py:302</code>	2	<pre>llm: Any, *, concept: ConceptAlternative, model: str, skill_guidance: str=''</pre> Independently score the 5 qualitative dimensions and attack the concept.
<code>_known_object_ids</code>	<code>app/proposal_graph/coverage.py:60</code>	2	<pre>graph: ProposalGraph</pre> Internal helper for the known object ids step.
<code>quote_exists_in_source</code>	<code>app/proposal_graph/evidence_verification.py:47</code>	2	<pre>quote: str, source_text: str</pre> Whitespace-tolerant, otherwise verbatim passage validation.
<code>_provider</code>	<code>app/proposal_graph/horizon_evaluator.py:105</code>	2	<pre>model: str</pre> Internal helper for the provider step.
<code>ProposalWorkspaceStore.decide_approval</code>	<code>app/proposal_graph/workspace_store.py:454</code>	2	<pre>self, *, session_id: str, proposal_id: str, approval_id: str, decision: Literal['approved', 'rejected', 'changes_requested'], decided_by: str, comment: str None</pre> Compute the decide approval.

Logic unit	Location	Branch score	Contract
ProposalWorkspaceStore.list_approvals	app/proposal_graph/workspace_store.py:503	2	<code>self, *, session_id: str, proposal_id: str</code> List the approvals.
BoundedToolResearchService.available	app/research/deep_research.py:233	2	<code>self</code> Compute the available.
SkillSelection.names	app/research/skills.py:205	2	<code>self</code> The names.
SkillSelection.versions	app/research/skills.py:210	2	<code>self</code> The versions.
ScientificSkillCatalog.probe	app/research/skills.py:274	2	<code>self</code> Probe the result.
ScientificSkillCatalog.metadata	app/research/skills.py:286	2	<code>self</code> Compute the metadata.
validate_metadata_document	app/retrieval/filters.py:134	2	<code>metadata: dict[str, Any], schema: dict[str, Any]</code> Validate an ingestion-time metadata document against a Collection schema.
validate_metadata_filters	app/retrieval/filters.py:159	2	<code>metadata: MetadataFilterGroup None, schema: dict[str, Any]</code> Validate a caller-supplied (or self-query-generated) filter tree.
RetrievalFilters.user_filter_dump	app/retrieval/models.py:90	2	<code>self</code> Trace-safe view of caller-supplied filters (no security clauses).
RetrievalService.retrieve	app/retrieval/service.py:115	2	<code>self, request: RetrievalQuery, *, owner_scope_id: str, llm: Any None=None</code> Retrieve the result.
dense_search	app/retrieval/strategies.py:63	2	<code>*, client: Any, collection_name: str, embedder: Embedder, query: str, filters: RetrievalFilters, index_id: str None, top_k: int, exclude_parent_chunks: bool=False</code> Compute the dense search.
sparse_search	app/retrieval/strategies.py:106	2	<code>*, client: Any, collection_name: str, query: str, filters: RetrievalFilters, index_id: str None, top_k: int, exclude_parent_chunks: bool=False</code> Compute the sparse search.
native_hybrid_search	app/retrieval/strategies.py:146	2	<code>*, client: Any, collection_name: str, embedder: Embedder, query: str, filters: RetrievalFilters, index_id: str None, top_k: int, alpha: float, exclude_parent_chunks: bool=False</code> Compute the native hybrid search.
ensure_collection_schema_on	app/retrieval/weaviate_client.py:75	2	<code>client: weaviate.WeaviateClient, name: str</code> Create "name" with the standard chunk schema if it doesn't exist yet.
WeaviateClient.close	app/retrieval/weaviate_client.py:191	2	<code>self</code> Close the underlying gRPC channel.

Logic unit	Location	Branch score	Contract
WeaviateClient.ensure_schema	app/retrieval/weaviate_client.py:200	2	<code>self</code> Create the DocumentChunk collection if it doesn't exist.
WeaviateClient.reset_schema	app/retrieval/weaviate_client.py:238	2	<code>self</code> DESTRUCTIVE: delete the collection and recreate it.
get_weaviate_client	app/retrieval/weaviate_client.py:319	2	<code></code> Lazily-constructed module-level client.
RedisLease.renew	app/runtime/coordination.py:62	2	<code>self</code> Renew the result.
RunEvent.to_json	app/runtime/events.py:67	2	<code>self</code> Compute the to json.
RunEventBus._key	app/runtime/events.py:120	2	<code>run_id: str, session_id: str None</code> Internal helper for the key step.
RunEventBus._redis_names	app/runtime/events.py:169	2	<code>run_id: str, session_id: str None</code> Return opaque, tenant-scoped Redis keys without leaking identifiers.
RunEventBus._offer	app/runtime/events.py:307	2	<code>queue: asyncio.Queue[RunEvent], event: RunEvent</code> Internal helper for the offer step.
_scalar_annotation	app/runtime/field_schema.py:240	2	<code>kind: str, enum_values: list[str]</code> Internal helper for the scalar annotation step.
FieldIndex.add	app/runtime/logic_preflight.py:687	2	<code>self, node_id: str, path: FieldPath</code> Add the result.
FieldIndex.paths_for	app/runtime/logic_preflight.py:726	2	<code>self, node_id: str</code> Compute the paths for.
_check_ai_task	app/runtime/logic_preflight.py:1455	2	<code>node: NodeSpec, config: dict[str, Any], issue: Callable[..., None]</code> Contract sanity on a visually built schema.
WorkflowPreflightReport.errors	app/runtime/preflight.py:144	2	<code>self</code> The errors.
WorkflowPreflightReport.warnings	app/runtime/preflight.py:153	2	<code>self</code> The warnings.
_unknown_config_fields	app/runtime/preflight.py:404	2	<code>node_spec: NodeSpec, node_class: type[NodeType]</code> Internal helper for the unknown config fields step.
_validated_node_config	app/runtime/preflight.py:424	2	<code>node_spec: NodeSpec</code> Internal helper for the validated node config step.
require_preflight	app/runtime/preflight.py:3048	2	<code>report: WorkflowPreflightReport</code> Compute the require preflight.
resolve_path	app/runtime/rules.py:168	2	<code>context: dict[str, Any], path: str</code> Read a dotted path out of workflow state, returning “_MISSING” sentinel semantics as “None” for callers that don't care about the distinction.
Action._rename_legacy_append	app/runtime/rules.py:338	2	<code>cls, value: Any</code> `merge` is `append`'s new name — the accumulate-into-a-list behavior is unchanged, only the label.
_truthy	app/runtime/rules.py:619	2	

Logic unit	Location	Branch score	Contract
			<code>value: Any</code> Internal helper for the truthy step.
NodeSpec.effective_config	<code>app/runtime/schema.py:325</code>	2	<code>self</code> Compute the effective config.
_enabled	<code>app/runtime/sleep_guard.py:35</code>	2	<code></code> Internal helper for the enabled step.
main	<code>app/runtime/snippet_daemon.py:208</code>	2	<code></code> Compute the main.
_as_text	<code>app/runtime/templating.py:101</code>	2	<code>value: Any</code> Render a substituted value inside a larger string.
_union_members	<code>app/runtime/templating.py:155</code>	2	<code>annotation: Any</code> Internal helper for the union members step.
get_current_user	<code>app/security/dependencies.py:59</code>	2	<code>token: str None=Depends(oauth2_scheme), access_token: str None=Cookie(default=None)</code> Return the current user.
get_current_user_cookie_or_header	<code>app/security/dependencies.py:75</code>	2	<code>token: str None=Depends(oauth2_scheme), access_token: str None=Cookie(default=None)</code> Header first (API clients), then HttpOnly cookie (browser EventSource).
require_permission	<code>app/security/dependencies.py:82</code>	2	<code>permission: str</code> Compute the require permission.
EntityVault.ensure_indexes	<code>app/security/entity_vault.py:204</code>	2	<code>self</code> Ensure the indexes.
EntityVault.delete_entity	<code>app/security/entity_vault.py:363</code>	2	<code>self, *, session_id: str, collection_id: str, entity_type: str, value: str</code> Delete the entity.
decode_token	<code>app/security/jwt_handler.py:43</code>	2	<code>token: str</code> Decode the token.
processing_mode_to_data_class	<code>app/security/llm_policy.py:44</code>	2	<code>processing_mode: str None</code> Maps Eurskem's per-node ProcessingMode onto the classification lattice.
_parse_csv_env	<code>app/security/llm_policy.py:100</code>	2	<code>name: str</code> Parse the csv env.
get_default_opa_client	<code>app/security/llm_policy.py:217</code>	2	<code></code> Return the default opa client.
_route_label	<code>app/security/middleware.py:229</code>	2	<code>request: Request</code> Internal helper for the route label step.
validate_username	<code>app/security/users.py:24</code>	2	<code>username: str</code> Validate the username.
ObjectStore.put_bytes	<code>app/storage/minio_client.py:173</code>	2	<code>self, data: bytes, key: str, content_type: str None=None</code> Upload raw bytes under a given key.
get_object_store	<code>app/storage/minio_client.py:267</code>	2	<code></code> Lazily-constructed module-level store.
		2	

Logic unit	Location	Branch score	Contract
PublicDatabaseLookupService_client	app/tools/ database_lookup.py:163		<code>self, timeout_seconds: float</code> Internal helper for the client step.
_set_run_shading	app/tools/ docx_proposal_rendering.py:223	2	<code>run, fill: str</code> Set the run shading.
_set_cell_width	app/tools/ docx_proposal_rendering.py:340	2	<code>cell, width_dxa: int</code> Set the cell width.
_set_repeat_header	app/tools/ docx_proposal_rendering.py:356	2	<code>row</code> Set the repeat header.
_set_update_fields	app/tools/ docx_proposal_rendering.py:491	2	<code>doc: DocxDocument</code> Set the update fields.
_add_internal_hyperlink	app/tools/ docx_proposal_rendering.py:602	2	<code>paragraph, text: str, anchor: str, *, bookmark_id: int, bold: bool=False, size: float=10</code> Add the internal hyperlink.
_footnote_body_text	app/tools/ docx_proposal_rendering.py:678	2	<code>number: int, citation: dict[str, Any]</code> Internal helper for the footnote body text step.
_build_footnotes_part_xml	app/tools/ docx_proposal_rendering.py:697	2	<code>footnote_bodies: dict[int, str]</code> Build the footnotes part xml.
_attach_footnotes_part	app/tools/ docx_proposal_rendering.py:765	2	<code>doc: DocxDocument, footnote_bodies: dict[int, str]</code> Internal helper for the attach footnotes part step.
_set_alt_text	app/tools/ docx_proposal_rendering.py:823	2	<code>inline_shape, alt_text: str</code> Set the alt text.
_HorizonDocxBuilder.add_fragment	app/tools/ docx_proposal_rendering.py:1241	2	<code>self, fragment: str</code> Add the fragment.
get_image_generation_service	app/tools/ image_io.py:315	2	<code></code> Return the image generation service.
render_proposal_pdf	app/tools/pdf_io.py:29	2	<code>sections: dict[str, str], template: str, client_name: str, proposal_title: str</code> Render an HTML proposal, then convert to PDF bytes via WeasyPrint.
_slug	app/tools/ proposal_rendering.py:97	2	<code>value: str</code> Internal helper for the slug step.
_unique_id	app/tools/ proposal_rendering.py:110	2	<code>base: str, used: set[str]</code> Internal helper for the unique id step.
_toc	app/tools/ proposal_rendering.py:210	2	<code>fragment: str</code> Internal helper for the toc step.
get_kimi_vision_service	app/tools/ vision_io.py:159	2	<code></code> Return the kimi vision service.

Logic unit	Location	Branch score	Contract
WebSearchService._stub_results	app/tools/web_io.py:460	2	query: str, top_k: int Internal helper for the stub results step.
get_web_search_service	app/tools/web_io.py:564	2	 Return the web search service.
search_web	app/tools/web_io.py:576	2	query: str, top_k: int=8, provider: SearchProvider='auto' MCP-compatible entry point returning JSON-safe candidate records.
WorkflowBuilderStore.validate_name	app/workflow/builder_store.py:73	2	name: str Validate the name.
WorkflowBuilderStore.validate_version_id	app/workflow/builder_store.py:87	2	version_id: str Validate the version id.
WorkflowBuilderStore.delete_draft	app/workflow/builder_store.py:199	2	self, name: str Delete the draft.
WorkflowBuilderStore.record_version	app/workflow/builder_store.py:214	2	self, name: str, yaml_text: str Record the version.
WorkflowBuilderStore.get_version	app/workflow/builder_store.py:254	2	self, name: str, version_id: str Return the version.
ChatConversationStore.get	app/workflow/chat_conversation_store.py:118	2	self, owner_scope_id: str, conversation_id: str No summary declared.
ChatConversationStore.list_messages	app/workflow/chat_conversation_store.py:129	2	self, owner_scope_id: str, conversation_id: str No summary declared.
ChatConversationStore.replace_message	app/workflow/chat_conversation_store.py:179	2	self, owner_scope_id: str, conversation_id: str, message_id: str, *, role: ChatMessageRole, content: dict[str, Any], run_id: str None No summary declared.
ChatWorkflowStore.list_private	app/workflow/chat_workflow_store.py:144	2	self, owner_scope_id: str No summary declared.
ChatWorkflowStore.get	app/workflow/chat_workflow_store.py:151	2	self, owner_scope_id: str, chat_workflow_id: str No summary declared.
ChatWorkflowStore.get_by_slug	app/workflow/chat_workflow_store.py:161	2	self, owner_scope_id: str, slug: str No summary declared.
ChatWorkflowStore.get_by_adapter_key	app/workflow/chat_workflow_store.py:169	2	self, owner_scope_id: str, adapter_key: str No summary declared.
ChatWorkflowStore.archive	app/workflow/chat_workflow_store.py:249	2	self, owner_scope_id: str, chat_workflow_id: str No summary declared.
_experience	app/workflow/chat_workspace_planner.py:119	2	value: str None No summary declared.
build_llm_adapter		2	

Logic unit	Location	Branch score	Contract
	app/workflow/ chat_workspace_planner.py:321		title: str='AI Workspace', previous_result: Any=None No summary declared.
build_scientific_skill_adapter	app/workflow/ chat_workspace_planner.py:337	2	skill_name: str Build a conversational workflow that applies one approved skill.
build_retrieval_adapter	app/workflow/ chat_workspace_planner.py:528	2	title: str, collection_id: str, retrieval_profile_id: str, instructions: str, document_ids: list[str] None=None No summary declared.
build_rag_adapter	app/workflow/ chat_workspace_planner.py:557	2	title: str, rag_agent_id: str, document_ids: list[str] None=None No summary declared.
get_verifications	app/workflow/ claim_verifications.py:38	2	db, run_id: str Return the verifications.
_humanize	app/workflow/ library_metadata.py:35	2	identifier: str Internal helper for the humanize step.
_default_human_review_count	app/workflow/ library_metadata.py:80	2	spec: WorkflowSpec Internal helper for the default human review count step.
record_run_failure	app/workflow/ orchestration.py:90	2	db: Any, *, run_id: str, session: str, error: BaseException str, services: dict[str, Any] None=None Persist an exception raised while executing/resuming a run.
run_and_finalize	app/workflow/ orchestration.py:179	2	coro: Awaitable[dict[str, Any]], *, db: Any, run_id: str, session: str, record_rejection_reason: bool=False, services: dict[str, Any] None=None Await one execution coroutine and persist its outcome end to end.
ensure_indexes	app/workflow/ preflight_stats.py:27	2	db: Any Ensure the indexes.
built_in_templates	app/workflow/ prompt_template_store.py:62	2	 No summary declared.
PromptTemplateStore.list	app/workflow/ prompt_template_store.py:93	2	self, owner_scope_id: str No summary declared.
PromptTemplateStore.get	app/workflow/ prompt_template_store.py:97	2	self, owner_scope_id: str, template_id: str No summary declared.
get_run_chat_turns	app/workflow/ run_chat_store.py:42	2	db, *, session_id: str, run_id: str Return the run chat turns.
_is_document_size_error	app/workflow/ run_history.py:44	2	exc: Exception Return whether document size error.
record_node_paused	app/workflow/ run_history.py:637	2	db, *, run_id: str, session_id: str, node_id: str, paused_at: float, pause_kind: str='hitl_gate' Mark a node and the overall run as paused.

Logic unit	Location	Branch score	Contract
is_pause_requested	app/workflow/ run_history.py:724	2	<code>db, *, run_id: str, session_id: str</code> Cheap per-node check — the executor calls this before every node.
initialize_run_checkpoint	app/workflow/ run_history.py:785	2	<code>db, *, run_id: str, session_id: str, workflow_yaml: str, inputs: dict[str, Any], collection_id: str, retry_of_run_id: str None=None</code> Create the private checkpoint used for zero-token node replay.
record_checkpoint_paused	app/workflow/ run_history.py:912	2	<code>db, *, run_id: str, session_id: str, node_id: str, pause_context: Any, pause_kind: str='hitl_gate'</code> Persist enough information to resume a paused run after process restart.
mark_checkpoint_status	app/workflow/ run_history.py:986	2	<code>db, *, run_id: str, session_id: str, status: str</code> Mark the checkpoint status.
_as_aware_utc	app/workflow/ run_history.py:1064	2	<code>value: datetime</code> Internal helper for the as aware utc step.
delete_run	app/workflow/ run_history.py:1374	2	<code>db, *, run_id: str, session_id: str</code> Permanently remove a run's history, checkpoint, and externalized blobs.
ensure_indexes	app/workflow/ subprocess_launches.py:42	2	<code>db: Any</code> Ensure the indexes.
reserve_launch	app/workflow/ subprocess_launches.py:83	2	<code>db: Any, *, launch_key: str, parent_run_id: str, parent_node_id: str, parent_session_id: str, child_run_id: str, child_workflow: str, result_from: str, result_node: str None, timeout_seconds: float</code> Claim 'launch_key' for one child launch.
_load_legacy_collection_configs	scripts/ backfill_knowledge_resources.py:52	2	<code>db</code> Load the legacy collection configs.
read	scripts/ build_documentation_pdf.py:149	2	<code>path: Path</code> No summary declared.
git_sha	scripts/ build_documentation_pdf.py:164	2	<code></code> No summary declared.
args	scripts/ build_documentation_pdf.py:213	2	<code>node: ast.FunctionDef ast.AsyncFunctionDef</code> No summary declared.
complexity	scripts/ build_documentation_pdf.py:220	2	<code>node: ast.AST</code> No summary declared.
metrics	scripts/ build_documentation_pdf.py:326	2	<code>model: Model</code> No summary declared.

Logic unit	Location	Branch score	Contract
key_files	scripts/ build_documentation_pdf. py:331	2	model: Model, paths: Sequence[str] No summary declared.
symbol_table	scripts/ build_documentation_pdf. py:356	2	symbols: Sequence[Symbol] No summary declared.
route_table	scripts/ build_documentation_pdf. py:360	2	routes: Sequence[Route] No summary declared.
node_table	scripts/ build_documentation_pdf. py:364	2	nodes: Sequence[Node] No summary declared.
workflow_table	scripts/ build_documentation_pdf. py:368	2	model: Model, limit: int=140 No summary declared.
v02	scripts/ build_documentation_pdf. py:435	2	m: Model No summary declared.
v03	scripts/ build_documentation_pdf. py:450	2	m: Model No summary declared.
v08	scripts/ build_documentation_pdf. py:502	2	m: Model No summary declared.
v10	scripts/ build_documentation_pdf. py:520	2	m: Model No summary declared.
v16	scripts/ build_documentation_pdf. py:588	2	m: Model No summary declared.
v21	scripts/ build_documentation_pdf. py:649	2	m: Model No summary declared.
build_one	scripts/ build_documentation_pdf. py:727	2	index: int, directory: Path Render one volume in a fresh process-friendly operation.
workflow_paths	scripts/ build_workflow_qa_matrix. py:41	2	root: Path No summary declared.
_words	scripts/ generate_docstrings.py:2 91	2	name: str No summary declared.
_phrase	scripts/ generate_docstrings.py:2 95	2	name: str No summary declared.
_describe_arg	scripts/ generate_docstrings.py:3 18	2	name: str No summary declared.

Logic unit	Location	Branch score	Contract
parser	scripts/ manage_user.py:25	2	 Compute the parser.
CatalogEntry.ceilings	scripts/ openrouter_health_check. py:200	2	<code>self</code> Per-probe output-token budgets, widened for models that think out loud.
ModelResult.as_dict	scripts/ openrouter_health_check. py:407	2	<code>self</code> Compute the as dict.
_is_fatal	scripts/ openrouter_health_check. py:461	2	<code>error: BaseException</code> Return whether fatal.
main	scripts/ register_biomass.py:11	2	 Compute the main.
request	scripts/ smoke_production.py:10	2	<code>url: str</code> Compute the request.
ensure_physical_class_with_product	scripts/ verder_reingest_producti on.py:53	2	<code>client: weaviate.WeaviateClient, name: str</code> Same as ensure_collection_schema_on, plus a `product` property this collection's metadata_schema declares but the base schema doesn't have.
Report.failed	scripts/ verify_knowledge_studio. py:97	2	<code>self</code> The failed.
_raise	scripts/ verify_knowledge_studio. py:216	2	<code>r: httpx.Response</code> Raise the result.
_multi_file	scripts/ verify_knowledge_studio. py:1081	2	<code>api: Api, cid: str, fixtures: dict[str, Path]</code> Internal helper for the multi file step.
_idempotent	scripts/ verify_knowledge_studio. py:1120	2	<code>api: Api, cid: str, fixtures: dict[str, Path]</code> Internal helper for the idempotent step.
_job_list	scripts/ verify_knowledge_studio. py:1140	2	<code>api: Api, cid: str</code> Internal helper for the job list step.
_cancel	scripts/ verify_knowledge_studio. py:1156	2	<code>api: Api, cid: str, fixtures: dict[str, Path]</code> Cancel the result.
_provenance	scripts/ verify_knowledge_studio. py:1178	2	<code>doc: dict[str, Any]</code> Internal helper for the provenance step.
_index_pins	scripts/ verify_knowledge_studio. py:1210	2	<code>indexes: list[dict[str, Any]]</code> Internal helper for the index pins step.
_ready	scripts/ verify_knowledge_studio. py:1230	2	<code>indexes: list[dict[str, Any]]</code> Internal helper for the ready step.
_chunks		2	

Logic unit	Location	Branch score	Contract
	scripts/ verify_knowledge_studio. py:1281		out: dict[str, Any] Internal helper for the chunks step.
_rerank	scripts/ verify_knowledge_studio. py:1293	2	api: Api, cid: str, query: str Internal helper for the rerank step.
_reserved_filter	scripts/ verify_knowledge_studio. py:1350	2	api: Api, cid: str, query: str Internal helper for the reserved filter step.
_agent_resolution	scripts/ verify_knowledge_studio. py:1504	2	api: Api, agent: dict[str, Any], query: str Internal helper for the agent resolution step.
main	scripts/ verify_knowledge_studio. py:1689	2	 Compute the main.
_set_auth_cookie	app/api/auth.py:55	1	response: Response, token: str Attach the JWT as an HttpOnly cookie.
me	app/api/auth.py:134	1	user: CurrentUser=Depends(require_consultant) Rehydrate identity from the auth cookie after a page refresh.
logout	app/api/auth.py:148	1	response: Response Clear the auth cookie.
_summary	app/api/ chat_workflows.py:362	1	record: ChatWorkflowRecord No summary declared.
ensure_general_chat_workflo w	app/api/ chat_workflows.py:585	1	request: Request, user: CurrentUser=Depends(require_consultant) Return one stable owner-scoped workflow for general Chat requests.
import_private_chat_workflo w	app/api/ chat_workflows.py:652	1	body: ImportChatWorkflowRequest, request: Request, user: CurrentUser=Depends(require_consultant) No summary declared.
_workflow_label_fingerprint	app/api/ chat_workspace.py:116	1	item: WorkflowLabelInput No summary declared.
plan_chat_workspace	app/api/ chat_workspace.py:211	1	body: PlanWorkspaceRequest, user: CurrentUser=Depends(require_consultant) No summary declared.
_db	app/api/cost_admin.py:33	1	request: Request Internal helper for the db step.
_today_start	app/api/ cost_admin.py:406	1	 Internal helper for the today start step.
ensure_indexes	app/api/ email_oauth.py:55	1	db: Any TTL-expires an abandoned connect attempt's pending state — a user who starts connecting a mailbox and never finishes leaves one small, non-secret document (no token, no code) that ages out on its own.
list_entity_types	app/api/ entity_registry.py:76	1	 List the entity types.

Logic unit	Location	Branch score	Contract
health	app/api/health.py:264	1	<code>request: Request</code> Return live dependency status without failing the liveness request.
_get_weaviate	app/api/inspect.py:160	1	<code>request: Request</code> Return the WeaviateClient wrapper.
ensure_indexes	app/api/integration_oauth.py:52	1	<code>db: Any</code> TTL-expires an abandoned connect attempt's pending state.
create_collection	app/api/knowledge.py:92	1	<code>payload: CollectionCreate, request: Request, user: CurrentUser=Depends(require_permission('knowledge:write'))</code> Create the collection.
list_collections	app/api/knowledge.py:110	1	<code>request: Request, user: CurrentUser=Depends(require_permission('knowledge:read'))</code> List the collections.
list_profiles	app/api/knowledge.py:185	1	<code>request: Request, profile_type: ProfileType None=Query(None), user: CurrentUser=Depends(require_permission('knowledge:read'))</code> List the profiles.
create_default_profiles	app/api/knowledge.py:222	1	<code>request: Request, user: CurrentUser=Depends(require_permission('knowledge:write'))</code> Create the default profiles.
ingestion_presets	app/api/knowledge.py:264	1	<code>user: CurrentUser=Depends(require_permission('knowledge:read'))</code> Compute the ingestion presets.
doc_types	app/api/knowledge.py:276	1	<code>user: CurrentUser=Depends(require_permission('knowledge:read'))</code> Document types a Collection may declare.
list_indexes	app/api/knowledge.py:346	1	<code>collection_id: str, request: Request, user: CurrentUser=Depends(require_permission('knowledge:read'))</code> List the indexes.
list_ingestions	app/api/knowledge.py:606	1	<code>request: Request, collection_id: str None=Query(None), user: CurrentUser=Depends(require_permission('knowledge:read'))</code> List the ingestions.
list_documents	app/api/knowledge.py:669	1	<code>collection_id: str, request: Request, user: CurrentUser=Depends(require_permission('knowledge:read'))</code> List the documents.
create_prompt_template	app/api/prompt_templates.py:45	1	<code>body: TemplateBody, request: Request, user:</code>

Logic unit	Location	Branch score	Contract
			<code>CurrentUser=Depends(require_consultant)</code> No summary declared.
<code>_services</code>	<code>app/api/proposals.py:48</code>	1	<code>request: Request</code> Internal helper for the services step.
<code>coverage</code>	<code>app/api/proposals.py:114</code>	1	<code>body: GraphRequest, user: CurrentUser=Depends(require_consultant)</code> Compute the coverage.
<code>list_source_versions</code>	<code>app/api/proposals.py:225</code>	1	<code>proposal_id: str, request: Request, source_id: str None=None, user: CurrentUser=Depends(require_consultant)</code> List the source versions.
<code>list_approvals</code>	<code>app/api/proposals.py:466</code>	1	<code>proposal_id: str, request: Request, user: CurrentUser=Depends(require_consultant)</code> List the approvals.
<code>list_agents</code>	<code>app/api/rag_agents.py:69</code>	1	<code>request: Request, search: str None=Query(default=None), user: CurrentUser=Depends(require_permission('knowledge:read'))</code> List the agents.
<code>presets</code>	<code>app/api/retrieval.py:20</code>	1	<code>user: CurrentUser=Depends(require_permission('knowledge:read'))</code> Return editable starting configurations; saving creates a profile.
<code>RetrievalSearchRequest.to_query</code>	<code>app/api/retrieval.py:110</code>	1	<code>self, owner_scope_id: str</code> Compute the to query.
<code>list_traces</code>	<code>app/api/retrieval.py:239</code>	1	<code>request: Request, limit: int=Query(default=100, ge=1, le=500), user: CurrentUser=Depends(require_permission('knowledge:read'))</code> List the traces.
<code>starter_questions</code>	<code>app/api/run_chat.py:27</code>	1	<code>run_id: str</code> Compute the starter questions.
<code>list_workflow_root</code>	<code>app/api/workflows.py:146</code>	1	<code></code> List the workflows.
<code>list_node_types</code>	<code>app/api/workflows.py:152</code>	1	<code>user: CurrentUser=Depends(require_admin)</code> The Workflow Builder UI calls this to populate the palette and forms.
<code>allowed_models</code>	<code>app/api/workflows.py:158</code>	1	<code>node_type: str</code> Returns the default allowed model list for a node type.
<code>_preflight_http_detail</code>	<code>app/api/workflows.py:212</code>	1	<code>report: WorkflowPreflightReport</code> Internal helper for the preflight http detail step.
<code>delete_workflow_draft</code>	<code>app/api/workflows.py:1180</code>	1	<code>name: str</code> Delete the workflow draft.
<code>list_workflow_versions</code>	<code>app/api/workflows.py:1192</code>	1	<code>name: str</code> List immutable versions without rewriting or preloading their YAML.
	<code>app/config.py:72</code>	1	

Logic unit	Location	Branch score	Contract
Settings.workflow_file_max_bytes			<code>self</code> The workflow file max bytes.
Settings.workflow_file_max_total_bytes	<code>app/config.py:77</code>	1	<code>self</code> The workflow file max total bytes.
Settings.max_request_body_bytes	<code>app/config.py:82</code>	1	<code>self</code> The max request body bytes.
Settings.moonshot_api_key	<code>app/config.py:560</code>	1	<code>self</code> Kimi web search and vision (<code>app/tools/web_io.py</code> , <code>vision_io.py</code>) authenticate as the same Moonshot account as the LLM-gateway route — one credential, <code>LOCAL_KIMI_API_KEY</code> , not a second parallel setting.
Settings.kimi_api_base_url	<code>app/config.py:567</code>	1	<code>self</code> The kimi api base url.
Settings.resolved_scientific_skills_path	<code>app/config.py:587</code>	1	<code>self</code> The resolved scientific skills path.
Settings.allowed_scientific_skills	<code>app/config.py:592</code>	1	<code>self</code> The allowed scientific skills.
Settings.allowed_cors_origins	<code>app/config.py:597</code>	1	<code>self</code> The allowed cors origins.
Settings.allowed_hosts	<code>app/config.py:602</code>	1	<code>self</code> The allowed hosts.
<code>_set_missing</code>	<code>app/db/migrations.py:47</code>	1	<code>collection: Any, field: str, value: Any</code> Set the missing.
MongoClient.manifests	<code>app/db/mongo.py:88</code>	1	<code>self</code> The manifests collection, opened lazily.
MongoClient.scorecards	<code>app/db/mongo.py:100</code>	1	<code>self</code> The eval scorecards collection, opened lazily.
MongoClient.collections	<code>app/db/mongo.py:105</code>	1	<code>self</code> The collections.
MongoClient.ensure_indexes	<code>app/db/mongo.py:111</code>	1	<code>self</code> Create the indexes the pipeline relies on.
MongoClient.get_manifest	<code>app/db/mongo.py:145</code>	1	<code>self, manifest_id: str</code> Fetch a single manifest by <code>_id</code> .
MongoClient.mark_status	<code>app/db/mongo.py:165</code>	1	<code>self, manifest_id: str, status: str</code> Update only the status field of an existing manifest.
MongoClient.save_scorecard	<code>app/db/mongo.py:183</code>	1	<code>self, scorecard: dict[str, Any]</code> Insert one eval scorecard.
<code>_now_iso</code>	<code>app/evaluation/models.py:34</code>	1	<code></code> Internal helper for the now iso step.
Scorecard.per_criteria_mean	<code>app/evaluation/models.py:84</code>	1	<code>self</code> Backward-compatible read alias for earlier API/UI consumers.
<code>_eval_one</code>	<code>app/evaluation/runner.py:30</code>	1	<code>ex: GoldenExample, judge: LLMJudge, produce_answer: ProduceAnswer</code> Internal helper for the eval one step.
		1	

Logic unit	Location	Branch score	Contract
ModelOverrideGateway.with_context	app/evaluation/workflow_golden.py:144		<code>self, **kwargs: Any</code> Compute the with context.
ModelOverrideGateway.complete	app/evaluation/workflow_golden.py:155	1	<code>self, *, model: str None=None, **kwargs: Any</code> Complete the result.
ModelOverrideGateway.complete_structured	app/evaluation/workflow_golden.py:165	1	<code>self, *, model: str None=None, **kwargs: Any</code> Complete the structured.
ModelOverrideGateway.chat_with_tools	app/evaluation/workflow_golden.py:175	1	<code>self, *, model: str None=None, **kwargs: Any</code> Compute the chat with tools.
utc_now	app/evidence/retrieval.py:43	1	<code></code> Compute the utc now.
_normalise_title_for_identity	app/evidence/retrieval.py:330	1	<code>title: str</code> Collapse cosmetic title differences that otherwise defeat dedup.
CollectionRegistry.upsert	app/ingestion/collections.py:90	1	<code>self, cfg: CollectionConfig</code> Upsert the result.
DocumentParser.parse	app/ingestion/contracts.py:19	1	<code>self, path: Path, *, config: dict[str, Any]</code> No summary declared.
ChunkingStrategy.chunk	app/ingestion/contracts.py:24	1	<code>self, document: ExtractedDocument, *, config: dict[str, Any], chunk_id_prefix: str</code> No summary declared.
ChunkEnricher.enrich	app/ingestion/contracts.py:31	1	<code>self, chunks: list[Chunk], *, document: ExtractedDocument, config: dict[str, Any]</code> No summary declared.
EmbeddingProvider.embed	app/ingestion/contracts.py:38	1	<code>self, texts: list[str]</code> No summary declared.
SearchIndex.index	app/ingestion/contracts.py:43	1	<code>self, objects: list[dict[str, Any]], vectors: list[list[float]]</code> Write ``objects`` (with their parallel ``vectors``) and return the number of objects actually inserted, so the caller can detect a partial write.
StageRegistry.parser	app/ingestion/contracts.py:69	1	<code>self, name: str, instance: DocumentParser</code> Compute the parser.
StageRegistry.chunker	app/ingestion/contracts.py:78	1	<code>self, name: str, instance: ChunkingStrategy</code> Compute the chunker.
StageRegistry.enricher	app/ingestion/contracts.py:87	1	<code>self, name: str, instance: ChunkEnricher</code> Compute the enricher.
IngestionCoordinator.submit	app/ingestion/coordinator.py:53	1	<code>self, job: IngestionJob</code> Schedule ``job`` to run on the event loop.
Extractor.extract	app/ingestion/extractor.py:72	1	<code>self, path: Path</code> No summary declared.
supported_extensions		1	

Logic unit	Location	Branch score	Contract
	app/ingestion/extractor.py:583		Sorted list of supported file extensions.
WeaviateSearchIndex.index	app/ingestion/indexes.py:43	1	self, objects: list[dict[str, Any]], vectors: list[list[float]] Compute the index.
StandardDocumentParser.parse	app/ingestion/strategies.py:18	1	self, path: Path, *, config: dict[str, Any] Parse the result.
RecursiveChunkingStrategy.chunk	app/ingestion/strategies.py:228	1	self, document: ExtractedDocument, *, config: dict[str, Any], chunk_id_prefix: str Chunk the result.
build_stage_registry	app/ingestion/strategies.py:512	1	Build the stage registry.
PdfPageVisionDescriber.provider	app/ingestion/vision.py:76	1	self The provider.
EmailAdapter.search	app/integrations/email/base.py:194	1	self, connection: EmailConnection, criteria: EmailSearchCriteria Search the result.
EmailAdapter.read	app/integrations/email/base.py:209	1	self, connection: EmailConnection, message_id: str Read the result.
EmailAdapter.create_draft	app/integrations/email/base.py:224	1	self, connection: EmailConnection, draft: EmailDraft Returns the provider's draft id.
EmailAdapter.send	app/integrations/email/base.py:230	1	self, connection: EmailConnection, draft: EmailDraft Returns the provider's sent-message id.
save_connection_record	app/integrations/email/connections_store.py:23	1	db: Any, *, connection_id: str, provider: str, display_name: str, address: str Save the connection record.
delete_connection_record	app/integrations/email/connections_store.py:55	1	db: Any, connection_id: str Delete the connection record.
set_allow_send	app/integrations/email/connections_store.py:66	1	db: Any, connection_id: str, allow_send: bool Returns True if a record was actually updated (the connection exists).
GmailAdapter.read	app/integrations/email/gmail.py:158	1	self, connection: EmailConnection, message_id: str Read the result.
InMemoryEmailAdapter.add	app/integrations/email/memory.py:45	1	self, message: EmailMessage Add the result.
InMemoryEmailAdapter.create_draft	app/integrations/email/memory.py:124	1	self, connection: EmailConnection, draft: EmailDraft Create the draft.
MicrosoftGraphAdapter.send	app/integrations/email/msgraph.py:235	1	self, connection: EmailConnection, draft: EmailDraft Send the result.

Logic unit	Location	Branch score	Contract
<code>_redirect_uri</code>	<code>app/integrations/email/oauth.py:76</code>	1	<code>provider: Provider</code> Internal helper for the redirect uri step.
<code>generate_pkce_pair</code>	<code>app/integrations/email/oauth.py:92</code>	1	<code></code> (code_verifier, code_challenge) for Google's PKCE requirement.
<code>_now</code>	<code>app/integrations/email/service.py:56</code>	1	<code></code> Internal helper for the now step.
<code>EmailService.add_connection</code>	<code>app/integrations/email/service.py:145</code>	1	<code>self, connection: EmailConnection</code> Adds or replaces a connection at runtime — how a newly completed OAuth connect (app/api/email_oauth.py) becomes usable immediately, with no restart needed.
<code>EmailService.remove_connection</code>	<code>app/integrations/email/service.py:151</code>	1	<code>self, connection_id: str</code> Remove the connection.
<code>_seal</code>	<code>app/integrations/email/token_vault.py:62</code>	1	<code>payload: dict[str, Any], *, connection_id: str</code> Internal helper for the seal step.
<code>_unseal</code>	<code>app/integrations/email/token_vault.py:89</code>	1	<code>doc: dict[str, Any], *, connection_id: str</code> Internal helper for the unseal step.
<code>TokenVault._tokens</code>	<code>app/integrations/email/token_vault.py:121</code>	1	<code>self</code> The tokens.
<code>IntegrationProvider.list_folder</code>	<code>app/integrations/files/base.py:124</code>	1	<code>self, connection: IntegrationConnection, *, folder_id: str None, page_size: int, page_token: str None</code> List the folder.
<code>IntegrationProvider.search_files</code>	<code>app/integrations/files/base.py:146</code>	1	<code>self, connection: IntegrationConnection, *, query: str, folder_id: str None, page_size: int, page_token: str None</code> Search the files.
<code>IntegrationProvider.get_file_meta</code>	<code>app/integrations/files/base.py:170</code>	1	<code>self, connection: IntegrationConnection, *, file_id: str</code> Return the file meta.
<code>IntegrationProvider.download_file</code>	<code>app/integrations/files/base.py:185</code>	1	<code>self, connection: IntegrationConnection, *, file_id: str</code> Download the file.
<code>save_connection_record</code>	<code>app/integrations/files/connections_store.py:23</code>	1	<code>db: Any, *, connection_id: str, provider: str, display_name: str, address: str</code> Save the connection record.
<code>delete_connection_record</code>	<code>app/integrations/files/connections_store.py:58</code>	1	<code>db: Any, connection_id: str</code> Delete the connection record.
<code>set_needs_reauth</code>	<code>app/integrations/files/connections_store.py:69</code>	1	<code>db: Any, connection_id: str, needs_reauth: bool</code> Persist the reauth flag so the Builder can show it without a live provider call — set on a failed refresh, cleared on any success.
<code>GoogleDriveProvider.get_file_meta</code>	<code>app/integrations/files/google_drive.py:203</code>	1	<code>self, connection: IntegrationConnection, *, file_id: str</code> Return the file meta.

Logic unit	Location	Branch score	Contract
<code>_redirect_uri</code>	<code>app/integrations/files/oauth.py:74</code>	1	<code>provider: Provider</code> Internal helper for the redirect uri step.
<code>generate_pkce_pair</code>	<code>app/integrations/files/oauth.py:90</code>	1	<code></code> (code_verifier, code_challenge) for Google's PKCE requirement.
<code>OneDriveProvider.get_file_meta</code>	<code>app/integrations/files/onedrive.py:187</code>	1	<code>self, connection: IntegrationConnection, *, file_id: str</code> Return the file meta.
<code>_now</code>	<code>app/integrations/files/service.py:44</code>	1	<code></code> Internal helper for the now step.
<code>IntegrationService.add_connection</code>	<code>app/integrations/files/service.py:147</code>	1	<code>self, connection: IntegrationConnection</code> Adds or replaces a connection at runtime — how a newly completed OAuth connect (app/api/integration_oauth.py) becomes usable immediately, with no restart needed.
<code>IntegrationService.remove_connection</code>	<code>app/integrations/files/service.py:153</code>	1	<code>self, connection_id: str</code> Remove the connection.
<code>_seal</code>	<code>app/integrations/files/token_vault.py:68</code>	1	<code>payload: dict[str, Any], *, connection_id: str</code> Internal helper for the seal step.
<code>_unseal</code>	<code>app/integrations/files/token_vault.py:95</code>	1	<code>doc: dict[str, Any], *, connection_id: str</code> Internal helper for the unseal step.
<code>TokenVault._tokens</code>	<code>app/integrations/files/token_vault.py:127</code>	1	<code>self</code> The tokens.
<code>_now</code>	<code>app/integrations/operations.py:50</code>	1	<code></code> Internal helper for the now step.
<code>operation_key</code>	<code>app/integrations/operations.py:149</code>	1	<code>*, scope: str, target: str, payload: Any</code> Stable fingerprint of "this exact side effect".
<code>DocTypeChoice.precision_sensitive</code>	<code>app/knowledge/doc_types.py:36</code>	1	<code>self</code> The precision sensitive.
<code>new_resource_id</code>	<code>app/knowledge/ids.py:81</code>	1	<code>kind: str</code> Mint a fresh opaque identifier for "kind".
<code>scoped_legacy_id</code>	<code>app/knowledge/ids.py:90</code>	1	<code>kind: str, owner_scope_id: str, legacy_key: str</code> Derive a stable identifier for a pre-existing record.
<code>is_kind</code>	<code>app/knowledge/ids.py:121</code>	1	<code>resource_id: str, kind: str</code> Return whether kind.
<code>utcnow</code>	<code>app/knowledge/models.py:13</code>	1	<code></code> Compute the utcnow.
<code>CollectionResource.display_name</code>	<code>app/knowledge/models.py:105</code>	1	<code>self</code> The display name.
<code>_scope</code>	<code>app/knowledge/repository.py:49</code>	1	<code>owner_scope_id: str, **extra: Any</code> Internal helper for the scope step.
<code>_strip</code>	<code>app/knowledge/repository.py:62</code>	1	<code>document: dict[str, Any]</code> Internal helper for the strip step.

Logic unit	Location	Branch score	Contract
KnowledgeRepository._collections	app/knowledge/repository.py:88	1	<code>self</code> The collections.
KnowledgeRepository._documents	app/knowledge/repository.py:93	1	<code>self</code> The documents.
KnowledgeRepository._source_versions	app/knowledge/repository.py:98	1	<code>self</code> The source versions.
KnowledgeRepository._profiles	app/knowledge/repository.py:103	1	<code>self</code> The profiles.
KnowledgeRepository._indexes	app/knowledge/repository.py:108	1	<code>self</code> The indexes.
KnowledgeRepository._index_documents	app/knowledge/repository.py:113	1	<code>self</code> The index documents.
KnowledgeRepository._ingestion_jobs	app/knowledge/repository.py:118	1	<code>self</code> The ingestion jobs.
KnowledgeRepository._rag_agents	app/knowledge/repository.py:123	1	<code>self</code> The rag agents.
KnowledgeRepository._traces	app/knowledge/repository.py:128	1	<code>self</code> The traces.
KnowledgeRepository.ensure_indexes	app/knowledge/repository.py:132	1	<code>self</code> Ensure the indexes.
KnowledgeRepository.create_collection	app/knowledge/repository.py:152	1	<code>self, resource: CollectionResource</code> Create the collection.
KnowledgeRepository.save_collection	app/knowledge/repository.py:164	1	<code>self, resource: CollectionResource</code> Save the collection.
KnowledgeRepository.save_document	app/knowledge/repository.py:209	1	<code>self, resource: DocumentResource</code> Save the document.
KnowledgeRepository.save_source_version	app/knowledge/repository.py:272	1	<code>self, resource: SourceVersionResource</code> Save the source version.
KnowledgeRepository.save_profile	app/knowledge/repository.py:323	1	<code>self, profile: ProfileVersion</code> Save the profile.
KnowledgeRepository.save_index	app/knowledge/repository.py:389	1	<code>self, index: IndexVersion</code> Save the index.
KnowledgeRepository.save_index_document	app/knowledge/repository.py:435	1	<code>self, *, owner_scope_id: str, index_id: str, document_id: str, source_version_id: str, ingestion_job_id: str, chunk_ids: list[str]</code> Save the index document.
KnowledgeRepository.save_ingestion_job	app/knowledge/repository.py:503	1	<code>self, job: IngestionJob</code> Save the ingestion job.
KnowledgeRepository.save_rag_agent	app/knowledge/repository.py:568	1	<code>self, agent: RAGAgentDefinition</code> Save the rag agent.
KnowledgeRepository.save_trace	app/knowledge/repository.py:620	1	<code>self, trace: RetrievalTrace</code> Save the trace.
KnowledgeRepository.record_manifest_index_history	app/knowledge/repository.py:667	1	<code>self, *, storage_key: str, manifest: dict[str, Any], index_history: dict[str, Any]</code>

Logic unit	Location	Branch score	Contract
			Upsert the legacy manifest while appending this index's history.
workspace_for_scope	app/knowledge/service.py:71	1	<code>owner_scope_id: str</code> Compute the workspace for scope.
_tool_name_for	app/llm/anthropic_batch.py:28	1	<code>response_model: Type[BaseModel]</code> Internal helper for the tool name for step.
AnthropicBatchService.status	app/llm/anthropic_batch.py:85	1	<code>self, batch_id: str</code> Compute the status.
AnthropicBatchService.cancel	app/llm/anthropic_batch.py:109	1	<code>self, batch_id: str</code> Request cancellation.
_supports_temperature	app/llm/anthropic_gw.py:52	1	<code>model: str</code> Return True if the model accepts a non-default temperature.
AnthropicGateway.probe_model_access	app/llm/anthropic_gw.py:129	1	<code>self, model: str</code> Verify workspace access through model metadata, with no generation.
LLMGateway.complete	app/llm/base.py:77	1	<code>self, *, model: str, system: str, user: str, temperature: float=0.0, max_tokens: int=1024, reasoning_effort: str None=None</code> Plain text completion.
LLMGateway.complete_structured	app/llm/base.py:97	1	<code>self, *, model: str, system: str, user: str, response_model: Type[T], temperature: float=0.0, max_tokens: int=1024, reasoning_effort: str None=None</code> Structured completion.
LLMGateway.chat_with_tools	app/llm/base.py:118	1	<code>self, *, model: str, system: str, messages: list[dict], tools: list[dict], temperature: float=0.0, max_tokens: int=4096</code> Multi-turn chat with tool calling.
ModelDefinition.as_dict	app/llm/catalog.py:59	1	<code>self</code> Compute the as dict.
provider_for_model	app/llm/catalog.py:163	1	<code>model: str</code> Compute the provider for model.
LocalOpenAICompatibleGateway.fixed_sampling_provider	app/llm/local_openai_gw.py:154	1	<code>self</code> Internal helper for the fixed sampling provider step.
LocalOpenAICompatibleGateway.probe	app/llm/local_openai_gw.py:429	1	<code>self</code> Probe the result.
estimate_model_cost	app/llm/model_catalog.py:151	1	<code>model: str, input_tokens: int, output_tokens: int</code> Estimate request cost using the platform's configured price catalog.
ModelSelection.to_event	app/llm/model_router.py:138	1	<code>self, *, actual_model: str, call_id: int, fallback: bool=False, cache_hit: bool=False</code> Compute the to event.
OpenAIBatchService.status		1	

Logic unit	Location	Branch score	Contract
	app/llm/ openai_batch.py:99		<code>self, batch_id: str</code> Compute the status.
<code>OpenAIBatchService.cancel</code>	app/llm/ openai_batch.py:121	1	<code>self, batch_id: str</code> Request cancellation.
<code>_supports_custom_temperature</code>	app/llm/openai_gw.py:58	1	<code>model: str</code> Return True if the model accepts a non-default temperature.
<code>_prompt_cache_key</code>	app/llm/openai_gw.py:90	1	<code>model: str, cacheable_prefix: str</code> Stable prompt_cache_key so requests sharing a static prefix (same model + system prompt) route to the same cache-holding backend -- OpenAI's own recommendation for boosting cache hit rates.
<code>OpenAIGateway.probe_model_access</code>	app/llm/openai_gw.py:180	1	<code>self, model: str</code> Verify project access through model metadata, with no generation.
<code>OpenAIModelDefinition.as_dict</code>	app/llm/ openai_registry.py:52	1	<code>self</code> Compute the as dict.
<code>OpenRouterModelInfo.as_dict</code>	app/llm/ openrouter_catalog.py:57	1	<code>self</code> Compute the as dict.
<code>is_openrouter_model_id</code>	app/llm/ openrouter_catalog.py:77	1	<code>value: str</code> Structural check only (no live lookup) — used by preflight, which must stay synchronous and network-free.
<code>OpenRouterCatalogCache.is_stale</code>	app/llm/ openrouter_catalog.py:164	1	<code>self</code> Return whether stale.
<code>get_default_cache</code>	app/llm/ openrouter_catalog.py:222	1	<code></code> Return the default cache.
<code>OpenRouterGateway._post</code>	app/llm/ openrouter_gw.py:151	1	<code>self, body: dict[str, Any]</code> Internal helper for the post step.
<code>_cost_tier_for_priority</code>	app/llm/ openrouter_ranking.py:42	1	<code>accuracy_priority: str</code> Internal helper for the cost tier for priority step.
<code>_kimi_profile</code>	app/llm/registry.py:432	1	<code></code> Internal helper for the kimi profile step.
<code>_glm_profile</code>	app/llm/registry.py:451	1	<code></code> Internal helper for the glm profile step.
<code>RegistryLLMGateway.with_context</code>	app/llm/registry.py:695	1	<code>self, run_id: str, session_id: str, node_id: str, ledger=None, *, event_bus=None, node_type: str None=None, allowed_models: list[str] None=None, routing_policy: dict[str, Any] None=None, semantic_cache=None, use_cache: bool=False, entity_tokenizer=None, collection_id: str='default', processing_mode: str None=None, workflow_name: str None=None</code> Return a context-bound copy for per-node cost tracking and routing.
	app/llm/registry.py:770	1	

Logic unit	Location	Branch score	Contract
RegistryLLMGateway.with_collection_id			<code>self, collection_id: str</code> Return a clone with only collection_id overridden.
RegistryLLMGateway.selection_history	<code>app/llm/registry.py:1017</code>	1	<code>self</code> Per-call model-selection events recorded on this bound gateway.
RegistryLLMGateway.next_call_id	<code>app/llm/registry.py:1025</code>	1	<code>self</code> Internal helper for the next call id step.
SemanticLLMCache.probe	<code>app/llm/semantic_cache.py:63</code>	1	<code>self</code> Probe the result.
SemanticLLMCache._scope	<code>app/llm/semantic_cache.py:235</code>	1	<code>*, session_id: str, model: str, system: str, temperature: float, max_tokens: int</code> Internal helper for the scope step.
SemanticLLMCache._entry_key	<code>app/llm/semantic_cache.py:267</code>	1	<code>scope: str, user: str</code> Internal helper for the entry key step.
SemanticLLMCache._index_key	<code>app/llm/semantic_cache.py:281</code>	1	<code>scope: str</code> Internal helper for the index key step.
connect	<code>app/mcp/business_records/db.py:21</code>	1	<code>app_settings: Settings=settings</code> Compute the connect.
connect_readonly	<code>app/mcp/business_records/db.py:47</code>	1	<code>app_settings: Settings=settings</code> A genuinely lower-privileged connection for query_readonly (app/nodes/sql_query.py) — GRANT SELECT only (see seed.py's ensure_readonly_user).
_collection	<code>app/mcp/business_records/handlers.py:19</code>	1	<code>rows: list[dict[str, Any]], key: str, limit: int</code> Internal helper for the collection step.
create_case	<code>app/mcp/business_records/handlers.py:269</code>	1	<code>conn: MySQLConnectionAbstract, arguments: dict[str, Any]</code> Create the case.
create_opportunity	<code>app/mcp/business_records/handlers.py:298</code>	1	<code>conn: MySQLConnectionAbstract, arguments: dict[str, Any]</code> Create the opportunity.
create_order	<code>app/mcp/business_records/handlers.py:326</code>	1	<code>conn: MySQLConnectionAbstract, arguments: dict[str, Any]</code> Create the order.
_order_view	<code>app/mcp/business_records/handlers.py:355</code>	1	<code>row: dict[str, Any]</code> Normalizes a raw `crm_salesorders` row to create_order's field names — a workflow mapping `order_number`/`account_id` after an update must not have to switch to `accountid` because this tool happened to read the row back from SQL instead of building it in Python.
_case_view	<code>app/mcp/business_records/handlers.py:406</code>	1	<code>row: dict[str, Any]</code> Normalizes a raw `crm_service_cases` row to create_case's field names — see _order_view for why this matters.
_next_suffix	<code>app/mcp/business_records/handlers.py:455</code>	1	<code>conn: MySQLConnectionAbstract, table: str, column: str, prefix: str</code> A small counter derived from how many rows already start with `prefix` — good enough for demo-scale

Logic unit	Location	Branch score	Contract
			seeded data; a real deployment would use the source system's own id generation instead of this.
<code>_load</code>	<code>app/mcp/business_records/seed.py:27</code>	1	<code>path: Path</code> Load the result.
<code>_error_result</code>	<code>app/mcp/business_records/server.py:165</code>	1	<code>error: dict[str, Any]</code> Internal helper for the error result step.
<code>main</code>	<code>app/mcp/business_records/server.py:182</code>	1	 Compute the main.
<code>_string</code>	<code>app/mcp/business_records/tools.py:18</code>	1	<code>description: str, **extra: Any</code> Internal helper for the string step.
<code>_collection</code>	<code>app/mcp/business_records/tools.py:31</code>	1	<code>key: str, item_schema: dict[str, Any]</code> Every collection-returning tool shares this envelope.
<code>_customer_item</code>	<code>app/mcp/business_records/tools.py:55</code>	1	 Internal helper for the customer item step.
<code>_order_item</code>	<code>app/mcp/business_records/tools.py:85</code>	1	 Internal helper for the order item step.
<code>_inventory_item</code>	<code>app/mcp/business_records/tools.py:104</code>	1	 Internal helper for the inventory item step.
<code>_product_item</code>	<code>app/mcp/business_records/tools.py:124</code>	1	 Internal helper for the product item step.
<code>_case_record</code>	<code>app/mcp/business_records/tools.py:147</code>	1	 Internal helper for the case record step.
<code>_opportunity_record</code>	<code>app/mcp/business_records/tools.py:171</code>	1	 Internal helper for the opportunity record step.
<code>_order_record</code>	<code>app/mcp/business_records/tools.py:192</code>	1	 Internal helper for the order record step.
<code>_business_records_env</code>	<code>app/mcp/client.py:142</code>	1	<code>app_settings: Settings</code> Environment for the Business Records MCP subprocess.
<code>MCPClient.configured_servers</code>	<code>app/mcp/client.py:259</code>	1	<code>self</code> The configured servers.
<code>MCPClient.running_servers</code>	<code>app/mcp/client.py:264</code>	1	<code>self</code> The running servers.
<code>MCPClient.has_server</code>	<code>app/mcp/client.py:345</code>	1	<code>self, server: str</code> Return whether server.
<code>MCPClient.probe</code>	<code>app/mcp/client.py:371</code>	1	

Logic unit	Location	Branch score	Contract
			<code>self, server: str=DEFAULT_SERVER</code> Perform a live MCP request instead of trusting a cached session.
launch_mcp_session	app/mcp/client.py:416	1	<code>servers: list[str] None=None</code> Convenience used by app/main.py at FastAPI startup.
read_only_dynamics_connection	app/mcp/connections.py:113	1	<code>app_settings: Settings=settings</code> A least-privilege variant, for deployments that want CRM context only.
build_mcp_service	app/mcp/connections.py:306	1	<code>*, client: Any, db: Any=None, app_settings: Settings=settings</code> Build the mcp service.
_collection	app/mcp/d365_finance/handlers.py:19	1	<code>rows: list[dict[str, Any]], key: str, limit: int</code> Internal helper for the collection step.
_error_result	app/mcp/d365_finance/server.py:149	1	<code>error: dict[str, Any]</code> Internal helper for the error result step.
main	app/mcp/d365_finance/server.py:166	1	 Compute the main.
_string	app/mcp/d365_finance/tools.py:12	1	<code>description: str, **extra: Any</code> Internal helper for the string step.
_collection	app/mcp/d365_finance/tools.py:25	1	<code>key: str</code> Internal helper for the collection step.
DynamicsError.as_payload	app/mcp/dynamics/client.py:69	1	<code>self</code> Compute the as payload.
DynamicsBackend.whoami	app/mcp/dynamics/client.py:108	1	<code>self</code> Compute the whoami.
DynamicsBackend.query	app/mcp/dynamics/client.py:117	1	<code>self, entity_set: str, *, select: list[str], filter_expression: str None=None, order_by: str None=None, top: int None=None, expand: str None=None</code> Query the result.
DynamicsBackend.get	app/mcp/dynamics/client.py:143	1	<code>self, entity_set: str, record_id: str, *, select: list[str]</code> Return the result.
DynamicsBackend.create	app/mcp/dynamics/client.py:159	1	<code>self, entity_set: str, payload: dict[str, Any]</code> Create the result.
DynamicsBackend.update	app/mcp/dynamics/client.py:172	1	<code>self, entity_set: str, record_id: str, payload: dict[str, Any]</code> Update the result.
DynamicsBackend.close	app/mcp/dynamics/client.py:184	1	<code>self</code> Close the result.
DataverseClient.update	app/mcp/dynamics/client.py:508	1	<code>self, entity_set: str, record_id: str, payload: dict[str, Any]</code> Update the result.
FixtureBackend.from_file	app/mcp/dynamics/client.py:564	1	<code>cls, path: str Path</code> Compute the from file.

Logic unit	Location	Branch score	Contract
FixtureBackend._next_id	app/mcp/dynamics/client.py:576	1	<code>self, prefix: str</code> Internal helper for the next id step.
FixtureBackend.whoami	app/mcp/dynamics/client.py:590	1	<code>self</code> Compute the whoami.
FixtureBackend.create	app/mcp/dynamics/client.py:659	1	<code>self, entity_set: str, payload: dict[str, Any]</code> Create the result.
_collection	app/mcp/dynamics/handlers.py:292	1	<code>rows: list[dict[str, Any]], key: str, limit: int</code> Wrap results in the shared envelope.
get_current_user	app/mcp/dynamics/handlers.py:329	1	<code>backend: DynamicsBackend, arguments: dict[str, Any]</code> Return the current user.
string_filter	app/mcp/dynamics/odata.py:84	1	<code>field: str, value: str</code> `field` eq `value` with the value escaped.
contains_filter	app/mcp/dynamics/odata.py:89	1	<code>field: str, value: str</code> `contains(field, 'value')` with the value escaped.
lookup_filter	app/mcp/dynamics/odata.py:94	1	<code>field: str, record_id: str</code> `_lookup_value` eq <guid> — the case the reference implementation gets wrong.
_error_result	app/mcp/dynamics/server.py:236	1	<code>error: dict[str, Any]</code> Internal helper for the error result step.
main	app/mcp/dynamics/server.py:257	1	<code></code> Compute the main.
_string	app/mcp/dynamics/tools.py:55	1	<code>description: str, **extra: Any</code> Internal helper for the string step.
_account_summary	app/mcp/dynamics/tools.py:68	1	<code></code> Internal helper for the account summary step.
_contact_summary	app/mcp/dynamics/tools.py:106	1	<code></code> Internal helper for the contact summary step.
_opportunity_summary	app/mcp/dynamics/tools.py:126	1	<code></code> Internal helper for the opportunity summary step.
_product_summary	app/mcp/dynamics/tools.py:152	1	<code></code> Internal helper for the product summary step.
_order_summary	app/mcp/dynamics/tools.py:170	1	<code></code> Internal helper for the order summary step.
_activity_summary	app/mcp/dynamics/tools.py:211	1	<code></code> Internal helper for the activity summary step.
_owner_summary	app/mcp/dynamics/tools.py:231	1	<code></code> Internal helper for the owner summary step.
_quote_summary	app/mcp/dynamics/tools.py:249	1	<code></code> Internal helper for the quote summary step.
_shipment_summary	app/mcp/dynamics/tools.py:269	1	<code></code> Internal helper for the shipment summary step.
_service_case_summary		1	

Logic unit	Location	Branch score	Contract
	app/mcp/dynamics/tools.py:290		Internal helper for the service case summary step.
_collection	app/mcp/dynamics/tools.py:315	1	item_schema: dict[str, Any], key: str Every collection-returning tool shares this envelope.
_JsonNullSafeOpenAireResponse.json	app/mcp/paper_search_server.py:110	1	self, *args: Any, **kwargs: Any Compute the json.
is_write	app/mcp/policy.py:177	1	operation: OperationClass Anything not provably a read is treated as a write.
MCPServerConnection.policy_for	app/mcp/registry.py:230	1	self, tool_name: str Compute the policy for.
MCPServerRegistry.ids	app/mcp/registry.py:274	1	self The ids.
MCPServerRegistry.add	app/mcp/registry.py:278	1	self, connection: MCPServerConnection Add the result.
MCPServerRegistry.get	app/mcp/registry.py:286	1	self, server_id: str Return the result.
MCPServerRegistry.record_health	app/mcp/registry.py:322	1	self, server_id: str, *, healthy: bool, tool_count: int=0, error: str None=None Record the health.
MCPServerRegistry.health	app/mcp/registry.py:345	1	self, server_id: str Compute the health.
list_tools	app/mcp/server.py:59	1	 List the tools.
main	app/mcp/server.py:260	1	 Compute the main.
MCPToolError.as_payload	app/mcp/service.py:75	1	self Compute the as payload.
_humanise	app/mcp/service.py:610	1	name: str Internal helper for the humanise step.
node_discovery_errors	app/nodes/__init__.py:46	1	 Return a copy so callers cannot mutate discovery state.
LiteralNode.run	app/nodes/_stubs.py:41	1	self, state, resolved_config Run the result.
EchoNode.run	app/nodes/_stubs.py:78	1	self, state, resolved_config Run the result.
_repo_root	app/nodes/about_synthesis.py:45	1	 Internal helper for the repo root step.
AITaskAgent.required_services	app/nodes/ai_task.py:469	1	cls, config: dict[str, Any] Compute the required services.
NodeType.run	app/nodes/base.py:113	1	self, state: 'WorkflowState', resolved_config: dict[str, Any] Execute the node.
NodeType.required_services	app/nodes/base.py:133	1	cls, config: dict[str, Any] Service names this node needs at runtime, given its resolved config.

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NodeType.preflight_output_fields	app/nodes/base.py:140	1	<code>cls, config: dict[str, Any]</code> Valid dotted template-reference suffixes for this node's output, beyond the static output_schema field names — e.g.
NodeType.preflight_static_output_values	app/nodes/base.py:150	1	<code>cls, config: dict[str, Any]</code> Output fields whose value preflight can already prove, without running the node — e.g.
BoundedDeepResearchAgent.required_services	app/nodes/bounded_deep_research_agent.py:130	1	<code>cls, config: dict[str, Any]</code> Compute the required services.
category_for	app/nodes/categories.py:92	1	<code>type_name: str</code> Compute the category for.
icon_for	app/nodes/categories.py:121	1	<code>type_name: str</code> Compute the icon for.
ClaimEvidenceVerifier.required_services	app/nodes/claim_evidence_verifier.py:74	1	<code>cls, config: dict[str, Any]</code> Compute the required services.
ConceptAlternativesAgent.required_services	app/nodes/concept_alternatives.py:65	1	<code>cls, config: dict[str, Any]</code> Compute the required services.
ConceptFreezeConfig._coerce_alternatives	app/nodes/concept_freeze.py:40	1	<code>cls, value: Any</code> Internal helper for the coerce alternatives step.
DataTransformAgent.required_services	app/nodes/data_transform.py:212	1	<code>cls, config: dict[str, Any]</code> Compute the required services.
DecisionAgent.required_services	app/nodes/decision.py:216	1	<code>cls, config: dict[str, Any]</code> Compute the required services.
DOCXProposalRenderer.required_services	app/nodes/docx_renderer.py:129	1	<code>cls, config: dict[str, Any]</code> Compute the required services.
DOCXProposalRenderer.run	app/nodes/docx_renderer.py:140	1	<code>self, state, resolved_config: dict[str, Any]</code> Run the result.
_sub_captioned	app/nodes/dynamic_figure_agent.py:297	1	<code>content: str</code> Internal helper for the sub captioned step.
_escape	app/nodes/dynamic_figure_agent.py:323	1	<code>text: str</code> Internal helper for the escape step.
EmailAgent.required_services	app/nodes/email_integration.py:272	1	<code>cls, config: dict[str, Any]</code> Compute the required services.
_criteria	app/nodes/email_integration.py:358	1	<code>cfg: EmailNodeConfig</code> Internal helper for the criteria step.
_humanize	app/nodes/end.py:96	1	<code>value: str</code> Internal helper for the humanize step.
EndAgent.required_services	app/nodes/end.py:155	1	<code>cls, config: dict[str, Any]</code> Compute the required services.
_ScholarlyCandidateDiscovery.required_services	app/nodes/evidence_agent.py:168	1	<code>cls, config: dict[str, Any]</code> Compute the required services.

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ExcelTableExtractor.required_services	app/nodes/excel_tool.py:53	1	<code>cls, config: dict[str, Any]</code> Compute the required services.
ExternalActionAgent.required_services	app/nodes/external_action.py:182	1	<code>cls, config: dict[str, Any]</code> Compute the required services.
_marker_pattern	app/nodes/figure_embedder.py:94	1	<code>marker: str</code> Internal helper for the marker pattern step.
GraphNormalizer.required_services	app/nodes/graph_normalizer.py:225	1	<code>cls, config: dict[str, Any]</code> Compute the required services.
HorizonDOCXProposalRenderer.required_services	app/nodes/horizon_docx_renderer.py:124	1	<code>cls, config: dict[str, Any]</code> Compute the required services.
HorizonEvaluationAgent.required_services	app/nodes/horizon_evaluation.py:54	1	<code>cls, config: dict[str, Any]</code> Compute the required services.
HorizonHTMLProposalRenderer.required_services	app/nodes/html_proposal_renderer.py:110	1	<code>cls, config: dict[str, Any]</code> Compute the required services.
_RichTextSanitizer.handle_data	app/nodes/human_in_loop.py:610	1	<code>self, data: str</code> Handle the data.
IntegrationAgent.required_services	app/nodes/integration.py:243	1	<code>cls, config: dict[str, Any]</code> Compute the required services.
_error_output	app/nodes/integration.py:423	1	<code>cfg: IntegrationNodeConfig, *, status: str, error_code: str, error: str, retryable: bool=False</code> Internal helper for the error output step.
KimiVisionAgent.required_services	app/nodes/kimi_vision_agent.py:82	1	<code>cls, config: dict[str, Any]</code> Compute the required services.
KnowledgeRetrieval.required_services	app/nodes/knowledge_retrieval.py:102	1	<code>cls, config: dict[str, Any]</code> Compute the required services.
MCPAgent.required_services	app/nodes/mcp_agent.py:95	1	<code>cls, config: dict[str, Any]</code> Compute the required services.
MCPToolAgent.required_services	app/nodes/mcp_tool.py:196	1	<code>cls, config: dict[str, Any]</code> Compute the required services.
MCPToolAgent.preflight_output_fields	app/nodes/mcp_tool.py:208	1	<code>cls, config: dict[str, Any]</code> Authorise the fixed envelope.
MCPToolAgent._skipped	app/nodes/mcp_tool.py:372	1	<code>self, cfg: MCPToolConfig, missing: list[str]</code> Internal helper for the skipped step.
MethodologyEngineeringAgent.required_services	app/nodes/methodology_engineering.py:143	1	<code>cls, config: dict[str, Any]</code> Compute the required services.
_rfc3339_now	app/nodes/minio_evidence_ingestion.py:359	1	<code></code> Internal helper for the rfc3339 now step.
PaperQAEvidenceSynthesizerConfig._coerce_documents	app/nodes/paperqa_evidence_synthesizer.py:56	1	<code>cls, value: Any</code> Internal helper for the coerce documents step.

Logic unit	Location	Branch score	Contract
PaperQAEvidenceSynthesizerAgent.required_services	app/nodes/paperqa_evidence_synthesizer.py:144	1	<code>cls, config: dict[str, Any]</code> Compute the required services.
PDFTextExtractor.required_services	app/nodes/pdf_tool.py:59	1	<code>cls, config: dict[str, Any]</code> Compute the required services.
PDFTextExtractor.run	app/nodes/pdf_tool.py:70	1	<code>self, state, resolved_config: dict[str, Any]</code> Run the result.
PDFProposalRenderer.required_services	app/nodes/pdf_tool.py:141	1	<code>cls, config: dict[str, Any]</code> Compute the required services.
PDFProposalRenderer.run	app/nodes/pdf_tool.py:152	1	<code>self, state, resolved_config: dict[str, Any]</code> Run the result.
PowerPointProposalSlides.required_services	app/nodes/powerpoint_tool.py:58	1	<code>cls, config: dict[str, Any]</code> Compute the required services.
PowerPointProposalSlides.run	app/nodes/powerpoint_tool.py:69	1	<code>self, state, resolved_config: dict[str, Any]</code> Run the result.
PriorProjectRetrieverAgent.required_services	app/nodes/prior_project_retriever.py:152	1	<code>cls, config: dict[str, Any]</code> Compute the required services.
ProposalEvidenceFactoryConfig.coerce_candidates	app/nodes/proposal_evidence_factor.py:113	1	<code>cls, value: Any</code> Internal helper for the coerce candidates step.
ProposalEvidenceFactoryConfig.coerce_documents	app/nodes/proposal_evidence_factor.py:126	1	<code>cls, value: Any</code> Internal helper for the coerce documents step.
ProposalEvidenceFactoryConfig.coerce_search_audit	app/nodes/proposal_evidence_factor.py:139	1	<code>cls, value: Any</code> Internal helper for the coerce search audit step.
ProposalEvidenceFactoryConfig.coerce_rejected_candidates	app/nodes/proposal_evidence_factor.py:152	1	<code>cls, value: Any</code> Internal helper for the coerce rejected candidates step.
ProposalEvidenceFactoryAgent.required_services	app/nodes/proposal_evidence_factor.py:363	1	<code>cls, config: dict[str, Any]</code> Compute the required services.
ProposalEvidenceFactoryAgent.source_from_document	app/nodes/proposal_evidence_factor.py:942	1	<code>self, document: FullTextDocument, candidate: CandidateSource None, *, passage: str</code> Internal helper for the source from document step.
ProposalTruthGraphConfig.coerce_verified_claims	app/nodes/proposal_truth_graph.py:55	1	<code>cls, value: Any</code> Internal helper for the coerce verified claims step.
ProposalTruthGraphConfig.coerce_evidence_gaps	app/nodes/proposal_truth_graph.py:72	1	<code>cls, value: Any</code> Internal helper for the coerce evidence gaps step.
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ProposalTruthGraphConfig.coerce_structured_data_records	app/nodes/proposal_truth_graph.py:89		<code>cls, value: Any</code> Internal helper for the coerce structured data records step.
ProposalTruthGraphConfig.coerce_internal_evidence_records	app/nodes/proposal_truth_graph.py:106	1	<code>cls, value: Any</code> Internal helper for the coerce internal evidence records step.
PythonSnippetAgent.required_services	app/nodes/python_snippet.py:141	1	<code>cls, config: dict[str, Any]</code> Compute the required services.
ResearchSourceAcquirerConfig.coerce_candidates	app/nodes/research_source_acquirer.py:70	1	<code>cls, value: Any</code> Internal helper for the coerce candidates step.
ResearchSourceAcquirer.required_services	app/nodes/research_source_acquirer.py:112	1	<code>cls, config: dict[str, Any]</code> Compute the required services.
ScientificResearchPlannerAgent.required_services	app/nodes/scientific_research_planner.py:146	1	<code>cls, config: dict[str, Any]</code> Compute the required services.
ScientificSkillAgent.required_services	app/nodes/scientific_skill_agent.py:79	1	<code>cls, config: dict[str, Any]</code> Compute the required services.
SQLQueryAgent.required_services	app/nodes/sql_query.py:147	1	<code>cls, config: dict[str, Any]</code> Compute the required services.
SQLQueryAgent.preflight_output_fields	app/nodes/sql_query.py:159	1	<code>cls, config: dict[str, Any]</code> Compute the preflight output fields.
StartAgent.required_services	app/nodes/start.py:160	1	<code>cls, config: dict[str, Any]</code> Compute the required services.
SubprocessAgent.required_services	app/nodes/subprocess.py:158	1	<code>cls, config: dict[str, Any]</code> Compute the required services.
is_new_style	app/nodes/transform.py:152	1	<code>cfg: TransformConfig</code> Whether this node was authored through the Inputs/Instructions/Outputs editor rather than with a hand-written <code>'prompt_template'</code> .
TransformAgent.failure_result	app/nodes/transform.py:700	1	<code>self, error: Exception</code> The non-raising counterpart to the two <code>'RuntimeError'</code> s above — only reachable when <code>'fail_on_error=False'</code> .
WebSearchAgent.required_services	app/nodes/web_search_agent.py:97	1	<code>cls, config: dict[str, Any]</code> Compute the required services.
WorkflowFileLoader.required_services	app/nodes/workflow_file_loader.py:96	1	<code>cls, config: dict[str, Any]</code> Compute the required services.
WorkflowInputAgent.required_services	app/nodes/workflow_input.py:308	1	<code>cls, config: dict[str, Any]</code> Compute the required services.
WorkflowInputAgent.run	app/nodes/workflow_input.py:354	1	<code>self, state, resolved_config: dict[str, Any]</code> Run the result.
configure_pricing_db	app/observability/cost_ledger.py:136	1	<code>db: Any</code> Called once from app/main.py's lifespan, right after Mongo is available.

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get_logger	app/observability/logging.py:51	1	<code>name: str=__name__</code> Return the logger.
record_paused	app/observability/metrics.py:222	1	<code>node_type: str</code> HITL pause is not an error and not a success — its own outcome.
_normalise_space	app/proposal_graph/evidence_verification.py:35	1	<code>value: str</code> Internal helper for the normalise space step.
relation_from_verdict	app/proposal_graph/evidence_verification.py:109	1	<code>*, relation_id: str, claim_id: str, source_id: str, source_version_id: str None, locator: str, verifier_model: str, verdict: ClaimSupportVerdict</code> Compute the relation from verdict.
empty_graph	app/proposal_graph/graph.py:108	1	<code></code> Compute the empty graph.
proposal_graph_state_update	app/proposal_graph/state.py:24	1	<code>graph: ProposalGraph</code> Compute the proposal graph state update.
_now	app/proposal_graph/workspace_store.py:21	1	<code></code> Internal helper for the now step.
_canonical_hash	app/proposal_graph/workspace_store.py:30	1	<code>payload: Any</code> Internal helper for the canonical hash step.
ProposalWorkspaceStore.ensure_indexes	app/proposal_graph/workspace_store.py:141	1	<code>self</code> Ensure the indexes.
_projected_call_cost	app/research/deep_research.py:39	1	<code>model: str, prompt_text: str, max_tokens: int</code> Worst-case pre-call cost: rough input-token estimate, full max_tokens output.
_brief_prompt	app/research/deep_research.py:616	1	<code>brief: ResearchBrief</code> Internal helper for the brief prompt step.
get_deep_research_service	app/research/deep_research.py:649	1	<code>*, llm: Any, web_search: Any</code> Return the deep research service.
SkillDocument.metadata	app/research/skills.py:179	1	<code>self</code> Compute the metadata.
ScientificSkillCatalog.load_errors	app/research/skills.py:299	1	<code>self</code> The load errors.
ScientificSkillCatalog.loaded_skill_names	app/research/skills.py:304	1	<code>self</code> Allowlisted skills whose SKILL.md actually parsed successfully.
GraphContextProvider.expand	app/retrieval/context.py:22	1	<code>self, chunks: list[RetrievedChunk], *, query: str</code> No summary declared.
rewrite_query	app/retrieval/query_understanding.py:22	1	<code>query: str, llm: LLMGateway</code> Return a search-optimized version of the query.
_safe_candidate	app/retrieval/service.py:49	1	<code>chunk: RetrievedChunk</code> Trace score/provenance, not duplicate raw source text per stage.

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WeaviateClient.ensure_collection_schema	app/retrieval/weaviate_client.py:226	1	<code>self, name: str</code> Like :meth:`ensure_schema`, but for an arbitrary physical collection.
WeaviateClient.upsert_objects	app/retrieval/weaviate_client.py:231	1	<code>self, collection_name: str, objects: list[dict[str, Any]], vectors: list[list[float]]</code> Like :meth:`upsert_chunks`, but targeting an arbitrary physical collection.
WeaviateClient.count_chunks	app/retrieval/weaviate_client.py:303	1	<code>self</code> Total chunks in the collection.
WeaviateClient.add_property	app/retrieval/weaviate_client.py:309	1	<code>self, prop: Property</code> Non-destructive schema evolution.
_fix_template_unknown_input	app/runtime/autofix.py:405	1	<code>raw: dict, issue: PreflightIssue</code> Internal helper for the fix template unknown input step.
_fix_template_unknown_variable	app/runtime/autofix.py:418	1	<code>raw: dict, issue: PreflightIssue</code> Internal helper for the fix template unknown variable step.
RedisLease.acquire	app/runtime/coordination.py:46	1	<code>self</code> Acquire the result.
DomainStateRegistry.reducer_for	app/runtime/domain_state.py:58	1	<code>cls, namespace: str</code> Compute the reducer for.
RunEvent.terminal	app/runtime/events.py:63	1	<code>self</code> The terminal.
RunEventBus.publish_redis	app/runtime/events.py:178	1	<code>self, evt: RunEvent</code> Publish the redis.
build_response_model	app/runtime/field_schema.py:351	1	<code>fields: list[FieldSpec], *, model_name: str='StructuredOutput'</code> Compile the visual schema into a Pydantic model.
json_schema	app/runtime/field_schema.py:361	1	<code>fields: list[FieldSpec], *, model_name: str='StructuredOutput'</code> JSON Schema for the visual schema, via the compiled Pydantic model.
field_paths	app/runtime/field_schema.py:403	1	<code>fields: list[FieldSpec], prefix: str=''</code> Flatten a visual schema into typed dotted paths, parents included.
resume_workflow	app/runtime/hitl.py:62	1	<code>run_id: str, decision: dict[str, Any]</code> Resume a paused workflow with the user's decision payload.
load_workflow	app/runtime/loader.py:12	1	<code>path: str Path</code> Load the workflow.
load_workflow_from_string	app/runtime/loader.py:26	1	<code>yaml_text: str</code> Load the workflow from string.
_static_mcp_tool_registries	app/runtime/logic_preflight.py:413	1	<code></code> Per-server `TOOLS_BY_NAME` registries for the built-in MCP connectors.
_transform_kind	app/runtime/logic_preflight.py:563	1	<code>operation: str</code> Internal helper for the transform kind step.

Logic unit	Location	Branch score	Contract
FieldIndex.get	app/runtime/ logic_preflight.py:697	1	<code>self, reference: str</code> Return the result.
WorkflowPreflightReport.refresh	app/runtime/ preflight.py:161	1	<code>self</code> Refresh the result.
_issue	app/runtime/ preflight.py:188	1	<code>report: WorkflowPreflightReport, code: str, message: str, *, severity: PreflightSeverity=PreflightSeverity.ERROR, path: str None=None, node_id: str None=None, suggestion: str None=None</code> Internal helper for the issue step.
_load_unique_yaml	app/runtime/ preflight.py:249	1	<code>yaml_text: str</code> Load the unique yaml.
preflight_workflow_spec	app/runtime/ preflight.py:2335	1	<code>spec: WorkflowSpec, *, provided_inputs: dict[str, Any] None=None, services: dict[str, Any] None=None, compile_graph: bool=False</code> Validate an already parsed spec without changing or running it.
operators_for_type	app/runtime/rules.py:129	1	<code>field_type: str</code> Compute the operators for type.
path_exists	app/runtime/rules.py:182	1	<code>context: dict[str, Any], path: str</code> Compute the path exists.
SnippetRunnerClient.run	app/runtime/ snippet_client.py:40	1	<code>self, code: str, inputs: dict[str, Any], *, timeout_seconds: float, memory_mb: int</code> Run the result.
_socket_path	app/runtime/ snippet_daemon.py:47	1	<code></code> Internal helper for the socket path step.
merge_node_outputs	app/runtime/state.py:12	1	<code>a: dict[str, Any], b: dict[str, Any]</code> Reducer for node_outputs.
EntityRegistry.list_entities	app/security/ entity_registry.py:84	1	<code>self, *, session_id: str, collection_id: str</code> List the entities.
EntityRegistry.delete	app/security/ entity_registry.py:100	1	<code>self, *, session_id: str, collection_id: str, entity_type: str, value: str</code> Delete the result.
EntityTokenizerService.registry	app/security/ entity_tokenizer.py:149	1	<code>self</code> Exposed for the manual-registration API and proposal-graph sync.
EntityTokenizerService.ensure_indexes	app/security/ entity_tokenizer.py:153	1	<code>self</code> Ensure the indexes.
_normalize	app/security/ entity_vault.py:76	1	<code>value: str</code> Normalize the result.
entity_hash	app/security/ entity_vault.py:88	1	<code>entity_type: str, value: str</code> Deterministic match key for one (type, normalized value) pair.
_mapping_id	app/security/ entity_vault.py:94	1	<code>session_id: str, collection_id: str, ehash: str</code> Internal helper for the mapping id step.

Logic unit	Location	Branch score	Contract
<code>_counter_id</code>	<code>app/security/entity_vault.py:108</code>	1	<code>session_id: str, collection_id: str, entity_type: str</code> Internal helper for the counter id step.
<code>_expiry</code>	<code>app/security/entity_vault.py:122</code>	1	<code></code> Internal helper for the expiry step.
<code>_seal</code>	<code>app/security/entity_vault.py:133</code>	1	<code>real_value: str, *, session_id: str, collection_id: str, placeholder: str</code> Internal helper for the seal step.
<code>_unseal</code>	<code>app/security/entity_vault.py:161</code>	1	<code>doc: dict[str, Any], *, session_id: str, collection_id: str, placeholder: str</code> Internal helper for the unseal step.
<code>EntityVault._mappings</code>	<code>app/security/entity_vault.py:195</code>	1	<code>self</code> The mappings.
<code>EntityVault._counters</code>	<code>app/security/entity_vault.py:200</code>	1	<code>self</code> The counters.
<code>ensure_indexes</code>	<code>app/security/entity_vault.py:386</code>	1	<code>db: Any</code> Module-level convenience matching this codebase's "ensure_*_indexes" convention.
<code>create_access_token</code>	<code>app/security/jwt_handler.py:16</code>	1	<code>data: dict[str, Any]</code> Create the access token.
<code>provider_and_backend_for_gateway</code>	<code>app/security/llm_policy.py:76</code>	1	<code>gateway_class_name: str</code> Compute the provider and backend for gateway.
<code>_map_entity_type</code>	<code>app/security/presidio_client.py:53</code>	1	<code>presidio_type: str</code> Map the entity type.
<code>has_permission</code>	<code>app/security/rbac.py:24</code>	1	<code>role: Role, permission: str</code> Return whether permission.
<code>stub_login</code>	<code>app/security/sso_stub.py:20</code>	1	<code>username: str, role: str, session_id: str None=None</code> Issue a JWT as if Azure AD had authenticated the user.
<code>ensure_user_indexes</code>	<code>app/security/users.py:14</code>	1	<code>db: Any</code> Ensure the user indexes.
<code>upsert_local_user</code>	<code>app/security/users.py:41</code>	1	<code>db: Any, *, username: str, password: str, role: Role, active: bool=True</code> Upsert the local user.
<code>key_for_path</code>	<code>app/storage/minio_client.py:49</code>	1	<code>path: Path</code> The object key a file should be stored under: 'sha256:<hash>.<ext>'.
<code>knowledge_key_for_path</code>	<code>app/storage/minio_client.py:60</code>	1	<code>path: Path, owner_scope_id: str</code> Content-addressed key for a Knowledge Studio upload, scoped by owner.
<code>ObjectStore.get_bytes</code>	<code>app/storage/minio_client.py:196</code>	1	<code>self, key: str</code> Download an object's contents.
<code>ObjectStore.presigned_url</code>	<code>app/storage/minio_client.py:239</code>	1	<code>self, key: str, expires_seconds: int=3600</code> Generate a temporary URL that grants read access to one object.
<code>narrative_char_count</code>		1	

Logic unit	Location	Branch score	Contract
	app/tools/ content_length.py:20		<code>content: str</code> Return `content`'s length with embedded base64 image data URIs excluded.
<code>get_database_lookup_service</code>	app/tools/ database_lookup.py:212	1	<code></code> Return the database lookup service.
<code>_set_cell_shading</code>	app/tools/ docx_proposal_rendering. py:308	1	<code>cell, fill: str</code> Set the cell shading.
<code>_add_field</code>	app/tools/ docx_proposal_rendering. py:505	1	<code>paragraph, instruction: str, fallback: str=''</code> Add the field.
<code>_add_toc_field</code>	app/tools/ docx_proposal_rendering. py:528	1	<code>paragraph</code> Add the toc field.
<code>_add_bookmark</code>	app/tools/ docx_proposal_rendering. py:642	1	<code>paragraph, name: str, bookmark_id: int</code> Add the bookmark.
<code>_add_footnote_reference</code>	app/tools/ docx_proposal_rendering. py:660	1	<code>paragraph, footnote_id: int</code> Add the footnote reference.
<code>_normalise_metadata</code>	app/tools/ docx_proposal_rendering. py:835	1	<code>metadata: dict[str, Any]</code> Internal helper for the normalise metadata step.
<code>_HorizonDocxBUILDER.config ure_document</code>	app/tools/ docx_proposal_rendering. py:893	1	<code>self</code> Configure the document.
<code>_HorizonDocxBUILDER.render _placeholder</code>	app/tools/ docx_proposal_rendering. py:1651	1	<code>self, prompt: str</code> Render the placeholder.
<code>_markdown</code>	app/tools/ proposal_rendering.py:78	1	<code></code> Internal helper for the markdown step.
<code>KimiVisionService.available</code>	app/tools/ vision_io.py:61	1	<code>self</code> Compute the available.
<code>WebSearchService.resolve_pro vider</code>	app/tools/web_io.py:105	1	<code>self, requested: SearchProvider</code> Resolve Auto without making a provider request.
<code>_rank_score</code>	app/tools/web_io.py:484	1	<code>index: int, count: int</code> Rank the score.
<code>_utc_now</code>	app/workflow/ builder_store.py:26	1	<code></code> Internal helper for the utc now step.
<code>_digest</code>	app/workflow/ builder_store.py:35	1	<code>text: str</code> Internal helper for the digest step.
<code>_atomic_write</code>	app/workflow/ builder_store.py:47	1	<code>path: Path, text: str</code> Internal helper for the atomic write step.
<code>WorkflowBuilderStore.workfl ow_path</code>	app/workflow/ builder_store.py:100	1	<code>self, name: str</code> Compute the workflow path.
<code>WorkflowBuilderStore.draft_p ath</code>	app/workflow/ builder_store.py:111	1	<code>self, name: str</code> Draft the path.

Logic unit	Location	Branch score	Contract
WorkflowBuilderStore.versions_dir	app/workflow/ builder_store.py:122	1	<code>self, name: str</code> Compute the versions dir.
WorkflowBuilderStore.restore_version	app/workflow/ builder_store.py:319	1	<code>self, name: str, version_id: str</code> Restore the version.
_split_pascal_case	app/workflow/ capability_selection.py:55	1	<code>name: str</code> "RouterAgent" -> "router agent"; "MCPToolAgent" -> "mcp tool agent" — without this, a type name is one opaque token and never matches a request's ordinary-language words.
ChatConversationRecord.public	app/workflow/ chat_conversation_store.py:31	1	<code>self</code> No summary declared.
ChatConversationMessageRecord.public	app/workflow/ chat_conversation_store.py:52	1	<code>self</code> No summary declared.
ensure_chat_conversation_indexes	app/workflow/ chat_conversation_store.py:63	1	<code>db: Any</code> No summary declared.
_default_output_compatibility	app/workflow/ chat_workflow_store.py:15	1	<code></code> Safe response shape for records created before output analysis existed.
ChatWorkflowRecord.public_summary	app/workflow/ chat_workflow_store.py:55	1	<code>self</code> Return card metadata without owner identity or workflow YAML.
ensure_chat_workflow_indexes	app/workflow/ chat_workflow_store.py:73	1	<code>db: Any</code> No summary declared.
ChatWorkflowStore.replace_managed_adapter	app/workflow/ chat_workflow_store.py:177	1	<code>self, owner_scope_id: str, chat_workflow_id: str, *, display_name: str, yaml_text: str, description: str, source: Literal['generated', 'imported', 'existing'], source_workflow_name: str None, output_compatibility: dict[str, Any], adapter_key: str, adapter_fingerprint: str</code> Claim or refresh a system-managed adapter without changing its stable ID.
ChatWorkflowStore.replace_executable	app/workflow/ chat_workflow_store.py:217	1	<code>self, owner_scope_id: str, chat_workflow_id: str, *, yaml_text: str, description: str, output_compatibility: dict[str, Any]</code> Replace only executable metadata for an owner-scoped private workflow.
ChatWorkflowStore.request_publication	app/workflow/ chat_workflow_store.py:267	1	<code>self, owner_scope_id: str, chat_workflow_id: str</code> No summary declared.
WorkspaceExperience.public	app/workflow/ chat_workspace_planner.py:34	1	<code>self</code> No summary declared.
WorkspacePlan.public		1	<code>self</code> No summary declared.

Logic unit	Location	Branch score	Contract
	app/workflow/ chat_workspace_planner.py:106		
_base	app/workflow/ chat_workspace_planner.py:123	1	title: str, welcome: str, allow_attachments: bool=True No summary declared.
_experience_block	app/workflow/ chat_workspace_planner.py:154	1	name: str, purpose: str, output: str No summary declared.
_answer_node	app/workflow/ chat_workspace_planner.py:162	1	instructions: str, input_fields: list[dict[str, Any]] No summary declared.
build_file_adapter	app/workflow/ chat_workspace_planner.py:368	1	title: str, instructions: str No summary declared.
build_vision_adapter	app/workflow/ chat_workspace_planner.py:402	1	title: str No summary declared.
build_multimodal_file_adapter	app/workflow/ chat_workspace_planner.py:429	1	title: str, instructions: str Analyze mixed uploaded sources, including one image plus text files.
build_web_adapter	app/workflow/ chat_workspace_planner.py:475	1	title: str No summary declared.
build_mcp_adapter	app/workflow/ chat_workspace_planner.py:499	1	title: str, server_id: str, tool: str No summary declared.
build_image_generation_adapter	app/workflow/ chat_workspace_planner.py:675	1	title: str='Image Creator' No summary declared.
build_grounded_diagram_adapter	app/workflow/ chat_workspace_planner.py:705	1	title: str='Architecture Diagram' Read attached sources, explain them, and generate a grounded visual image.
_key	app/workflow/ claim_verifications.py:14	1	run_id: str, record_id: str Internal helper for the key step.
ensure_indexes	app/workflow/ claim_verifications.py:27	1	db Ensure the indexes.
save_verification	app/workflow/ claim_verifications.py:52	1	db, run_id: str, record_id: str, result: dict[str, Any] Save the verification.
scope_token	app/workflow/ file_inputs.py:59	1	session_id: str Return a non-reversible, path-safe tenant/session token.
workflow_input_prefix	app/workflow/ file_inputs.py:65	1	session_id: str Compute the workflow input prefix.
extension_for	app/workflow/ file_inputs.py:85	1	filename: str Compute the extension for.

Logic unit	Location	Branch score	Contract
template_variables	app/workflow/ prompt_template_store.py:19	1	<code>content: str</code> Return unique placeholders in first-appearance order.
PromptTemplateRecord.public	app/workflow/ prompt_template_store.py:36	1	<code>self</code> No summary declared.
ensure_prompt_template_indexes	app/workflow/ prompt_template_store.py:74	1	<code>db: Any</code> No summary declared.
PromptTemplateStore.create	app/workflow/ prompt_template_store.py:84	1	<code>self, owner_scope_id: str, *, title: str, description: str, category: PromptCategory, content: str</code> No summary declared.
PromptTemplateStore.delete	app/workflow/ prompt_template_store.py:114	1	<code>self, owner_scope_id: str, template_id: str</code> No summary declared.
ensure_run_chat_indexes	app/workflow/ run_chat_store.py:31	1	<code>db</code> Ensure the run chat indexes.
append_run_chat_turn	app/workflow/ run_chat_store.py:60	1	<code>db, *, session_id: str, run_id: str, question: str, answer: str, model: str</code> Append one Q&A turn and return the full updated turn list.
_node_key	app/workflow/ run_history.py:159	1	<code>node_id: str</code> Return a Mongo-safe map key while preserving node_id in the value.
ensure_indexes	app/workflow/ run_history.py:218	1	<code>db</code> Create indexes used by run detail and newest-first history views.
request_pause	app/workflow/ run_history.py:682	1	<code>db, *, run_id: str, session_id: str</code> Ask a running workflow to pause at its next node boundary.
clear_pause_request	app/workflow/ run_history.py:704	1	<code>db, *, run_id: str, session_id: str</code> Clear the pause request.
record_checkpoint_approval	app/workflow/ run_history.py:950	1	<code>db, *, run_id: str, session_id: str, node_id: str, decision: dict[str, Any], actor: str</code> Append an immutable human decision before resuming execution.
get_run	app/workflow/ run_history.py:1193	1	<code>db, session_id: str, run_id: str</code> Return one full run if it belongs to the caller's session.
find_by_launch_key	app/workflow/ subprocess_launches.py:70	1	<code>db: Any, launch_key: str</code> Find the by launch key.
find_by_child_run_id	app/workflow/ subprocess_launches.py:132	1	<code>db: Any, child_run_id: str</code> Find the by child run id.
find_by_token	app/workflow/ subprocess_launches.py:145	1	<code>db: Any, token: str</code> Find the by token.

Logic unit	Location	Branch score	Contract
deliver	app/workflow/ subprocess_launches.py:158	1	db: Any, *, token: str, decision: dict[str, Any] Atomically mark the launch "delivered", once.
esc	scripts/ build_documentation_pdf.py:141	1	value: Any No summary declared.
rel	scripts/ build_documentation_pdf.py:145	1	path: Path No summary declared.
markdown	scripts/ build_documentation_pdf.py:178	1	value: str No summary declared.
v00	scripts/ build_documentation_pdf.py:373	1	m: Model No summary declared.
v01	scripts/ build_documentation_pdf.py:404	1	m: Model No summary declared.
v04	scripts/ build_documentation_pdf.py:454	1	m: Model No summary declared.
v06	scripts/ build_documentation_pdf.py:474	1	m: Model No summary declared.
v07	scripts/ build_documentation_pdf.py:487	1	m: Model No summary declared.
v09	scripts/ build_documentation_pdf.py:507	1	m: Model No summary declared.
v13	scripts/ build_documentation_pdf.py:553	1	m: Model No summary declared.
v14	scripts/ build_documentation_pdf.py:566	1	m: Model No summary declared.
v20	scripts/ build_documentation_pdf.py:624	1	m: Model No summary declared.
document	scripts/ build_documentation_pdf.py:681	1	index: int, body: str, model: Model No summary declared.
render	scripts/ build_documentation_pdf.py:688	1	index: int, model: Model, directory: Path No summary declared.
main	scripts/ build_eu_proposal_copy_bundle.py:122	1	 Compute the main.

Logic unit	Location	Branch score	Contract
main	scripts/ build_hitl_editor_bundle.py:89	1	 Compute the main.
main	scripts/ build_workflow_file_inputs_bundle.py:99	1	 Compute the main.
write_json	scripts/ build_workflow_qa_matrix.py:214	1	 <code>path: Path, rows: list[dict[str, Any]]</code> No summary declared.
missing_required_routes	scripts/ check_api_routes.py:60	1	 Compute the missing required routes.
secret	scripts/ generate_production_env.py:16	1	 Compute the secret.
parse_args	scripts/ generate_production_env.py:25	1	 Parse the args.
slug	scripts/ generate_reference_workflows.py:28	1	 <code>value: str</code> No summary declared.
main	scripts/load_test.py:78	1	 Compute the main.
main	scripts/ manage_user.py:73	1	 Compute the main.
snake_case	scripts/ new_node_type.py:56	1	 <code>value: str</code> Convert a PascalCase type name to its module filename.
CatalogEntry.platform_id	scripts/ openrouter_health_check.py:151	1	 <code>self</code> Eurskem's gateway-routed form — the id a workflow YAML would name.
CatalogEntry.emits_text	scripts/ openrouter_health_check.py:161	1	 <code>self</code> The emits text.
CatalogEntry.is_text_only	scripts/ openrouter_health_check.py:166	1	 <code>self</code> Text and nothing else.
CatalogEntry.is_batch_endpoint	scripts/ openrouter_health_check.py:174	1	 <code>self</code> `batch` variants route to OpenRouter's asynchronous Batch API, which the synchronous /chat/completions path this gateway uses does not exercise.
CatalogEntry.claims_structured	scripts/ openrouter_health_check.py:180	1	 <code>self</code> The claims structured.
CatalogEntry.claims_tools	scripts/ openrouter_health_check.py:187	1	 <code>self</code> The claims tools.
CatalogEntry.claims_reasoning	scripts/ openrouter_health_check.py:192	1	 <code>self</code> The claims reasoning.

Logic unit	Location	Branch score	Contract
ProbeResult.as_dict	scripts/ openrouter_health_check. py:370	1	<code>self</code> Compute the as dict.
_verdict_mark	scripts/ openrouter_health_check. py:792	1	<code>verdict: str</code> Internal helper for the verdict mark step.
parse_args	scripts/ openrouter_health_check. py:962	1	<code>argv: list[str] None=None</code> Parse the args.
main	scripts/ openrouter_health_check. py:1124	1	<code></code> Compute the main.
main	scripts/ render_sample_proposal.p y:16	1	<code></code> Compute the main.
main	scripts/ render_sample_proposal_d ocx.py:16	1	<code></code> Compute the main.
_MinimalLLM.complete	scripts/ smoke_events.py:53	1	<code>self, *a, **kw</code> No summary declared.
_MinimalLLM.complete_structured	scripts/ smoke_events.py:54	1	<code>self, *a, **kw</code> No summary declared.
_MinimalLLM.chat_with_tools	scripts/ smoke_events.py:55	1	<code>self, *a, **kw</code> No summary declared.
main	scripts/smoke_llm.py:6	1	<code></code> Compute the main.
main	scripts/ smoke_structured.py:22	1	<code></code> Compute the main.
Api.h	scripts/ verify_knowledge_studio. py:140	1	<code>self</code> The h.
Api.get	scripts/ verify_knowledge_studio. py:159	1	<code>self, path: str, **kw: Any</code> Return the result.
Api.post	scripts/ verify_knowledge_studio. py:171	1	<code>self, path: str, **kw: Any</code> Compute the post.
Api.jpost	scripts/ verify_knowledge_studio. py:183	1	<code>self, path: str, payload: dict[str, Any]</code> Compute the jpost.
Api.jget	scripts/ verify_knowledge_studio. py:197	1	<code>self, path: str, **kw: Any</code> Compute the jget.
Api.close	scripts/ verify_knowledge_studio. py:211	1	<code>self</code> Close the result.
_long		1	<code>times: int=6</code> Internal helper for the long step.

Logic unit	Location	Branch score	Contract
	scripts/ verify_knowledge_studio. py:239		
profile	scripts/ verify_knowledge_studio. py:452	1	api: Api, kind: str, name: str, strategy: str, config: dict[str, Any] Compute the profile.
_assert_eq	scripts/ verify_knowledge_studio. py:996	1	actual: Any, expected: Any Internal helper for the assert eq step.
_assert_in	scripts/ verify_knowledge_studio. py:1010	1	needle: Any, haystack: Any Internal helper for the assert in step.
_assert_true	scripts/ verify_knowledge_studio. py:1024	1	value: bool, message: str Internal helper for the assert true step.
_assert_keys	scripts/ verify_knowledge_studio. py:1038	1	payload: dict[str, Any], keys: set[str] Internal helper for the assert keys step.
_assert_nonempty	scripts/ verify_knowledge_studio. py:1053	1	value: list[Any], label: str Internal helper for the assert nonempty step.
_assert_status	scripts/ verify_knowledge_studio. py:1067	1	r: httpx.Response, expected: int Internal helper for the assert status step.
_dupe_names	scripts/ verify_knowledge_studio. py:1099	1	api: Api, cid: str, fixtures: dict[str, Path] Internal helper for the dupe names step.
_source_url	scripts/ verify_knowledge_studio. py:1192	1	api: Api, doc_id: str Internal helper for the source url step.
_activate	scripts/ verify_knowledge_studio. py:1242	1	api: Api, cid: str, indexes: list[dict[str, Any]] Internal helper for the activate step.
_switch	scripts/ verify_knowledge_studio. py:1262	1	api: Api, cid: str, indexes: list[dict[str, Any]] Internal helper for the switch step.
_filters	scripts/ verify_knowledge_studio. py:1311	1	api: Api, cid: str, query: str Internal helper for the filters step.
_bad_filter	scripts/ verify_knowledge_studio. py:1331	1	api: Api, cid: str, query: str Internal helper for the bad filter step.
_pinned_index	scripts/ verify_knowledge_studio. py:1395	1	api: Api, cid: str, query: str, indexes: list[dict[str, Any]] Internal helper for the pinned index step.
_inspect	scripts/ verify_knowledge_studio. py:1418	1	api: Api, cid: str, query: str Internal helper for the inspect step.
_make_agent		1	

Logic unit	Location	Branch score	Contract
	scripts/ verify_knowledge_studio. py:1457		api: Api, cid: str, defaults: Any, stamp: str, state: dict[str, Any] Build the agent.
_evidence_boundary	scripts/ verify_knowledge_studio. py:1522	1	api: Api, agent: dict[str, Any], query: str Internal helper for the evidence boundary step.
_anon	scripts/ verify_knowledge_studio. py:1540	1	base: str, cid: str Internal helper for the anon step.
_preflight_bad	scripts/ verify_knowledge_studio. py:1668	1	api: Api Internal helper for the preflight bad step.